Design and Analysis of Experiments

Lecture notes, Chapters 1–11

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Chapter 1. Introduction

1 Experiment, Design, and Statistics

- An experiment aims to **draw inferences** or **make a decision** about a hypothesis, which needs **data**.
- *Example (umbrella).* Barometer, sky, forecast, radio: data assimilated into a decision.
- **Statistics**: making decisions in the presence of variability (uncertainty); **how the data are collected** is therefore crucial.
- **Experimental designs** reduce experimental error; **randomization** averages out uncontrolled variables.

Definition 1.1: Basic principles of experimental design

(i) **Randomization.** Allocation of material and run order are random. This usually validates the assumption of independent errors and "averages out" extraneous (lurking) factors.

(ii) **Replication.** An independent repeat run of each factor combination. It (a) allows estimation of the experimental error σ^2, the yardstick for real differences, and (b) sharpens means: n replicates give variance σ^2/n.

(iii) **Blocking.** Improves precision of comparisons by removing variability from nuisance factors; a block is a set of relatively homogeneous conditions.

Remark 1.2: Why replication shrinks variance

If $y_1, \dots, y_n$ are independent with common variance σ^2, then

$$\begin{aligned}\operatorname{Var}(\bar{y}) &= \operatorname{Var}\Big(\frac{1}{n}\sum_{j=1}^{n} y_j\Big) = \frac{1}{n^2}\operatorname{Var}\Big(\sum_{j=1}^{n} y_j\Big) = \frac{1}{n^2}\sum_{j=1}^{n}\operatorname{Var}(y_j) \quad \text{(independence)} \\ &= \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.\end{aligned}$$

Used in every later ANOVA derivation, e.g. $\operatorname{Var}(\bar{\varepsilon}_{i\cdot}) = \sigma^2/n$.

2 Role of Statistics in Experimentation

- **Project planning phase**
  1. What is to be measured? (choice of the response variable)
  2. How large is the likely variation?
  3. What are the influential factors?
- **Experimental design phase**
  1. Control some known sources of variation (blocking).
  2. Allow estimation of the size of the uncontrolled variation (replication).
  3. Permit an investigation of suitable models.

- **Statistical analysis phase**
  1. Make inferences on design factors.
  2. Guide subsequent design.
  3. Suggest more appropriate models.

3 The General Workflow Used in This Course

Derivation

Every design is analyzed in three steps (two-sample model $y_{ij} = \mu_i + \varepsilon_{ij}$, $\varepsilon_{ij} \sim N(0, \sigma^2)$, i.e. $y_{ij} \sim N(\mu_i, \sigma^2)$, $H_0 : \mu_1 = \mu_2$):

① **Estimation**: fit the model, $y_{ij} = \hat{\mu}_i + e_{ij}$ (fitted value + residual);

② **Inference**: test hypotheses / build confidence intervals on the parameters;

③ **Residual analysis**: check the model assumptions (normality, constant variance, independence) using the residuals e_{ij}.

Design	Model	Chapter
Two-sample / paired comparison	$y_{ij} = \mu_i + \varepsilon_{ij}$, $y_{ij} = \mu_i + \beta_j + \varepsilon_{ij}$	2
Completely randomized design (CRD)	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	3
RCBD / BIBD	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	4
Latin / Graeco-Latin square	$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}$	4
Factorial design	$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	5
2^k factorial, blocking & confounding	effects via contrasts	6, 7
Regression approach	$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$	10
Random / mixed effects	$\tau_i \sim N(0, \sigma_\tau^2)$	13
Nested and split-plot	$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}, \ldots$	14
Analysis of covariance	$y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$	15

4 Selected exercises

Motivation. Short calculations behind randomization and replication.

Exercise 1.1 (Randomization makes the comparison unbiased)

$N = n + m$ experimental units have fixed unit effects $u_1, \ldots, u_N$ (unknown). A completely randomized design assigns a simple random sample A of n units to treatment 1 and the other m units (B) to treatment 2, and unit k responds $y_k = u_k + \tau_1$ or $y_k = u_k + \tau_2$ (additivity). With $\bar{u} = \frac{1}{N}\sum_k u_k$ and $S_u^2 = \frac{1}{N-1}\sum_k (u_k - \bar{u})^2$, show that, over the randomization,

$$\mathrm{E}(\bar{y}_A - \bar{y}_B) = \tau_1 - \tau_2, \qquad \mathrm{Var}(\bar{y}_A - \bar{y}_B) = S_u^2\Big(\frac{1}{n} + \frac{1}{m}\Big).$$

Solution [Supplement].

Step 1 (reduction). $\bar{y}_A - \bar{y}_B = \tau_1 - \tau_2 + \bar{u}_A - \bar{u}_B$, and since $n\bar{u}_A + m\bar{u}_B = N\bar{u}$,

$$\bar{u}_A - \bar{u}_B = \bar{u}_A - \frac{N\bar{u} - n\bar{u}_A}{m} = \frac{N}{m}(\bar{u}_A - \bar{u}).$$

Step 2 (indicators). $I_k = \mathbf{1}\{k \in A\}$: $\mathrm{E}I_k = n/N$ and, for $k \neq l$, $\mathrm{E}(I_k I_l) = \frac{n(n-1)}{N(N-1)}$, so

$$\mathrm{Var}I_k = \frac{n}{N}\Big(1 - \frac{n}{N}\Big), \qquad \mathrm{Cov}(I_k, I_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Step 3 (mean). $\bar{u}_A = \frac{1}{n}\sum_k I_k u_k$ gives $\mathrm{E}\bar{u}_A = \frac{1}{n}\sum_k \frac{n}{N}u_k = \bar{u}$, so $\mathrm{E}(\bar{y}_A - \bar{y}_B) = \tau_1 - \tau_2$.

Step 4 (variance). Write $v_k = u_k - \bar{u}$, so $\sum_k v_k = 0$ and $\bar{u}_A - \bar{u} = \frac{1}{n}\sum_k I_k v_k$. Using $\sum_{k\neq l} v_k v_l = (\sum_k v_k)^2 - \sum_k v_k^2 = -\sum_k v_k^2$,

$$\begin{aligned}\mathrm{Var}\Big(\sum_k I_k v_k\Big) &= \frac{n(N-n)}{N^2}\sum_k v_k^2 + \frac{n(N-n)}{N^2(N-1)}\sum_k v_k^2 \\ &= \frac{n(N-n)}{N(N-1)}\sum_k v_k^2 = \frac{n(N-n)}{N}S_u^2.\end{aligned}$$

Hence $\mathrm{Var}(\bar{u}_A) = \frac{N-n}{nN}S_u^2 = \frac{m}{nN}S_u^2$ and, by Step 1,

$$\mathrm{Var}(\bar{y}_A - \bar{y}_B) = \frac{N^2}{m^2}\cdot\frac{m}{nN}S_u^2 = \frac{N}{nm}S_u^2 = S_u^2\Big(\frac{1}{n} + \frac{1}{m}\Big).$$

No distributional assumption was used: randomization itself creates the "error"; S_u^2 plays the role of σ^2 in Ch. 2. ■

Exercise 1.2 (A hidden time trend)

Two treatments are each run n times in $N = 2n$ consecutive runs, and the response drifts linearly with run order k: $y_k = \gamma k + \tau_{(k)} + \varepsilon_k$ (γ unknown, $\tau_{(k)}$ the treatment of run k, ε_k iid with mean 0).
(a) If all treatment-1 runs come first, find the bias of $\bar{y}_1 - \bar{y}_2$ for $\tau_1 - \tau_2$.
(b) If the run order is randomized, show the estimator is unbiased and find the variance contributed by the trend.

Solution [Supplement].

(a) Average run indices are $(n+1)/2$ and $n + (n+1)/2$, so $\mathrm{E}(\bar{y}_1 - \bar{y}_2) = \tau_1 - \tau_2 - \gamma n$: a bias $-\gamma n$ confounded with the treatment difference.

(b) Exercise 1.1 with $u_k = \gamma k$ (errors average out independently of the assignment): unbiased, and with $\sum_{k=1}^{N}(k - \frac{N+1}{2})^2 = \frac{N(N^2-1)}{12}$,

$$S_u^2 = \gamma^2\frac{N(N+1)}{12}, \qquad \mathrm{Var}_{\mathrm{trend}} = S_u^2\cdot\frac{2}{n} = \gamma^2\frac{2n(2n+1)}{12}\cdot\frac{2}{n} = \gamma^2\frac{2n+1}{3}.$$

Bias becomes random error; blocking on time would remove it. ■

Exercise 1.3 (Replicates versus repeated measurements)

Suppose $y_1, \ldots, y_n$ each have variance σ^2, but every pair has correlation $\rho \geq 0$ (e.g. n measurements on one experimental unit instead of n units).
(a) Show $\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}\{1 + (n-1)\rho\}$ and that it tends to $\rho\sigma^2$, not to 0, as $n \to \infty$.

(b) Show that $\mathrm{E}(S^2) = (1-\rho)\sigma^2$, so the sample variance underestimates the variability that matters for comparing units.

Solution [Supplement].

(a) $\mathrm{Var}(\sum_j y_j) = \sum_j \mathrm{Var}(y_j) + \sum_{j \neq l} \mathrm{Cov}(y_j, y_l) = n\sigma^2 + n(n-1)\rho\sigma^2$; divide by n^2. The limit is $\rho\sigma^2 > 0$ when $\rho > 0$.

(b) $\mathrm{E}\sum_j (y_j - \bar{y})^2 = \sum_j \mathrm{E}(y_j - \mu)^2 - n\mathrm{E}(\bar{y} - \mu)^2 = n\sigma^2 - \sigma^2\{1 + (n-1)\rho\} = (n-1)(1-\rho)\sigma^2$.

Repeated measurements shrink the apparent error and stop reducing $\mathrm{Var}(\bar{y})$; only independent replicates give σ^2/n and an honest σ^2. ■

Chapter 2. Simple Comparative Experiments

1 Basic Concepts

Definition 2.1

- **Population**: all possible items under specified experimental or observational conditions.
- **Sample**: a group of observations taken from a population.
- **Parameter**: a numerical characteristic of a population (e.g. μ, σ^2).
- **Statistic**: a numerical characteristic computed from the sample (e.g. $\bar{y}, S^2$).
- **Mathematical model**: derived from theoretical considerations; represents exact, error-free relationships among variables.
- **Statistical model**: derived from data subject to observation, experimental and/or measurement errors (e.g. $y = \mu + \varepsilon$).

2 Descriptive Statistics

2.1 Traditional statistics

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

2.2 Robust statistics

- **Sample median**: with order statistics $x_{(1)} \le \cdots \le x_{(n)}$, (i) if n is odd, the median is $x_{(q)}$ with $q = (n+1)/2$;

  (ii) if n is even, it is $[x_{(q)} + x_{(q+1)}]/2$ with $q = n/2$.

- **Median absolute deviation**:

$$\text{MAD} = \frac{\text{median}(|x_i - \text{median}(x)|)}{0.6745}.$$

  The constant $0.6745 = \Phi^{-1}(0.75)$ makes MAD a consistent estimator of σ under normality: if $X \sim N(\mu, \sigma^2)$ then $\text{median}|X - \mu| = 0.6745\sigma$, because $P(|X - \mu| \le c\sigma) = 0.5 \iff 2\Phi(c) - 1 = 0.5 \iff \Phi(c) = 0.75$.

Example 2.2: Robustness of median/MAD

Data 1: $1, 2, 3, 4, 5$.

$$\bar{x} = \tfrac{15}{5} = 3, \qquad \text{median} = 3,$$
$$s^2 = \tfrac{1}{4}[(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2] = \tfrac{10}{4} = 2.5, \quad s = 1.581,$$
$$|x_i - 3| = 2, 1, 0, 1, 2 \;\Rightarrow\; \text{median} = 1, \quad \text{MAD} = 1/0.6745 = 1.482.$$

Data 2: $1, 2, 3, 4, 1000000$ (one gross outlier).

$$\bar{x} = \tfrac{1000010}{5} = 200002, \qquad \text{median} = 3,$$
$$s = 447212.5 \quad \text{(explodes)},$$
$$|x_i - 3| = 2, 1, 0, 1, 999997 \;\Rightarrow\; \text{median} = 1, \quad \text{MAD} = 1.482 \quad \text{(unchanged)}.$$

The mean and standard deviation are destroyed by a single outlier, while median and MAD are unaffected.

3 Graphical Display of Data

- **Histogram**: displays the empirical distribution of (continuous) data.
- **Boxplot**: shows the median, lower and upper quartiles, whiskers and outliers; good for comparing several samples side by side.
- **Normal Q–Q plot**: checks normality (details in Ch. 3).

4 Simple Comparative Experiments: the Mortar Example

- Compare two conditions (treatments) = two levels of one factor.
- **Example**: tension bond strength of Portland cement mortar for two formulations, *modified* and *unmodified*; 10 observations each.

Modified (y_{1j})	16.85	16.40	17.21	16.35	16.52	17.04	16.96	17.15	16.59	16.57
Unmodified (y_{2j})	17.50	17.63	18.25	18.00	17.86	17.75	18.22	17.90	17.96	18.15

- Modified: mean = 16.764, median = 16.72, $s = 0.3164$, MAD = 0.4151.
- Unmodified: mean = 17.922, median = 17.93, $s = 0.2479$, MAD = 0.2965.
- Question: $H_0 : \mu_{\text{mod}} = \mu_{\text{unmod}}$?

5 Two-Sample t-Test

Motivation. The two-sample t-test is the template: a difference compared with its within-group noise.

5.1 Model and assumptions

$$y_{ij} = \mu_i + \varepsilon_{ij}, \qquad i = 1, 2,\ j = 1, \ldots, n_i,$$

1. $y_{1j} \sim N(\mu_1, \sigma_1^2)$, $y_{2j} \sim N(\mu_2, \sigma_2^2)$, all independent;
2. common variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

Hypotheses: $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 \neq \mu_2$.

5.2 Derivation of the test statistic

Step 1: distribution of $\bar{y}_1 - \bar{y}_2$. Since $\bar{y}_i = \frac{1}{n_i}\sum_j y_{ij}$ is a linear combination of independent normals,

$$\mathrm{E}(\bar{y}_1 - \bar{y}_2) = \mathrm{E}(\bar{y}_1) - \mathrm{E}(\bar{y}_2) = \mu_1 - \mu_2,$$
$$\mathrm{Var}(\bar{y}_1 - \bar{y}_2) = \mathrm{Var}(\bar{y}_1) + \mathrm{Var}(\bar{y}_2) - 2\underbrace{\mathrm{Cov}(\bar{y}_1, \bar{y}_2)}_{=0 \text{ (indep.)}} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Hence $\bar{y}_1 - \bar{y}_2 \sim N\Big(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\Big)$.

Step 2: known variances. If σ_1^2, σ_2^2 were known, under H_0

$$Z_0 = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \overset{\sigma_1=\sigma_2=\sigma}{=} \frac{\bar{y}_1 - \bar{y}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0,1).$$

Step 3: pooled variance. With $\sigma_1^2 = \sigma_2^2 = \sigma^2$ unknown, both S_1^2 and S_2^2 estimate the same σ^2; pool them:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Theorem 2.3: S_p^2 is unbiased and $(n_1 + n_2 - 2)S_p^2/\sigma^2 \sim \chi^2_{n_1+n_2-2}$

Proof [Supplement]. *Unbiasedness.* For one normal sample, $\mathrm{E}(S_i^2) = \sigma^2$:

$$\mathrm{E}\Big[\sum_j (y_{ij} - \bar{y}_i)^2\Big] = \mathrm{E}\Big[\sum_j y_{ij}^2 - n_i\bar{y}_i^2\Big] = \sum_j \mathrm{E}(y_{ij}^2) - n_i\mathrm{E}(\bar{y}_i^2)$$
$$= n_i(\sigma^2 + \mu_i^2) - n_i\Big(\frac{\sigma^2}{n_i} + \mu_i^2\Big) = (n_i - 1)\sigma^2.$$

Hence

$$\mathrm{E}(S_p^2) = \frac{(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2}{n_1 + n_2 - 2} = \sigma^2.$$

Distribution. For a normal sample, $(n_i - 1)S_i^2/\sigma^2 \sim \chi^2_{n_i-1}$ and $S_i^2 \perp\!\!\!\perp \bar{y}_i$. The two samples are independent, and the sum of independent χ^2's is χ^2 with the dfs added:

$$\frac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} = \frac{(n_1 - 1)S_1^2}{\sigma^2} + \frac{(n_2 - 1)S_2^2}{\sigma^2} \sim \chi^2_{n_1-1} + \chi^2_{n_2-1} = \chi^2_{n_1+n_2-2}.$$

■

Step 4: the t statistic. Recall: if $Z \sim N(0,1)$, $V \sim \chi^2_\nu$, $Z \perp\!\!\!\perp V$, then $Z/\sqrt{V/\nu} \sim t_\nu$. With $Z = Z_0$ and $V = (n_1 + n_2 - 2)S_p^2/\sigma^2$ (independent because $(\bar{y}_1, \bar{y}_2) \perp\!\!\!\perp (S_1^2, S_2^2)$):

$$t_0 = \frac{\frac{\bar{y}_1 - \bar{y}_2}{\sigma\sqrt{1/n_1 + 1/n_2}}}{\sqrt{\frac{(n_1 + n_2 - 2)S_p^2/\sigma^2}{n_1 + n_2 - 2}}} = \frac{\bar{y}_1 - \bar{y}_2}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \overset{H_0}{\sim} t_{n_1+n_2-2}.$$

The unknown σ cancels; this is exactly "replace σ by S_p".

- t_0 is a *distance measure*: how far apart the two averages are, in standard-error units.
- Values of t_0 near 0 are consistent with H_0; values far from 0 support H_1.

- **Decision rule**: reject H_0 if $|t_0| > t_{\alpha/2,\, n_1+n_2-2}$.
- **P-value**: the smallest level of significance that would lead to rejection of H_0, $P = P(|t_{n_1+n_2-2}| \geq |t_0|)$.

5.3 Confidence interval

From $\dfrac{(\bar{y}_1 - \bar{y}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2}$,

$$\begin{aligned} 1-\alpha &= P\left(-t_{\alpha/2} \leq \frac{(\bar{y}_1 - \bar{y}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{1/n_1 + 1/n_2}} \leq t_{\alpha/2}\right) \\ &= P\left(\bar{y}_1 - \bar{y}_2 - t_{\alpha/2}S_p\sqrt{\tfrac{1}{n_1}+\tfrac{1}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{y}_1 - \bar{y}_2 + t_{\alpha/2}S_p\sqrt{\tfrac{1}{n_1}+\tfrac{1}{n_2}}\right). \end{aligned}$$

Example 2.4: Mortar data, line by line

$n_1 = n_2 = 10$, $\bar{y}_1 = 16.764$, $\bar{y}_2 = 17.922$, $S_1^2 = 0.3164^2 = 0.1001$, $S_2^2 = 0.2479^2 = 0.0615$.

$$\begin{aligned} S_p^2 &= \frac{9(0.1001) + 9(0.0615)}{18} = \frac{0.1001 + 0.0615}{2} = 0.0808, \qquad S_p = 0.2843, \\ S_p\sqrt{\tfrac{1}{10}+\tfrac{1}{10}} &= 0.2843 \times 0.4472 = 0.1271, \\ t_0 &= \frac{16.764 - 17.922}{0.1271} = \frac{-1.158}{0.1271} = -9.11. \end{aligned}$$

Since $|t_0| = 9.11 > t_{0.025,18} = 2.101$, reject H_0: the mean strengths of the two formulations differ. P-value $= P(|t_{18}| \geq 9.11) = 3.67 \times 10^{-8}$, so H_0 is rejected at any level $\alpha \geq 3.67 \times 10^{-8}$. 95% CI: $-1.158 \pm 2.101(0.1271) = -1.158 \pm 0.267 = [-1.425,\ -0.891]$ (matches the R output `t.test(..., var.equal=TRUE)`).

5.4 Checking assumptions

- **Normality**: normal Q–Q plot of each sample (`qqnorm`, `qqline`).
- **Equal variances**: F-test. Under $H_0 : \sigma_1^2 = \sigma_2^2$,

$$F_0 = \frac{S_1^2}{S_2^2} = \frac{[(n_1-1)S_1^2/\sigma^2]/(n_1-1)}{[(n_2-1)S_2^2/\sigma^2]/(n_2-1)} \sim F_{n_1-1,\, n_2-1}.$$

  Mortar: $F_0 = 0.1001/0.0615 = 1.629$, $P = 0.4785 \Rightarrow$ no evidence against equal variances.

6 The Case $\sigma_1^2 \neq \sigma_2^2$ (Welch)

Use

$$t_0 = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}},$$

whose null distribution is well approximated by t_ν with (Satterthwaite)

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_2^2/n_2)^2}{n_2-1}}.$$

Remark 2.5: Where ν comes from

Let $W = S_1^2/n_1 + S_2^2/n_2$. We approximate $\nu W/\mathrm{E}(W) \approx \chi^2_\nu$ and choose ν by matching variances. A χ^2_ν has variance 2ν, so we need $\mathrm{Var}(\nu W/EW) = 2\nu \iff \nu = 2[\mathrm{E}W]^2/\mathrm{Var}(W)$. Using $\mathrm{Var}(S_i^2) = 2\sigma_i^4/(n_i - 1)$ (since $(n_i - 1)S_i^2/\sigma_i^2 \sim \chi^2_{n_i-1}$):

$$\mathrm{E}W = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \mathrm{Var}W = \frac{2\sigma_1^4}{n_1^2(n_1-1)} + \frac{2\sigma_2^4}{n_2^2(n_2-1)},$$
$$\nu = \frac{2(\sigma_1^2/n_1 + \sigma_2^2/n_2)^2}{2\frac{(\sigma_1^2/n_1)^2}{n_1-1} + 2\frac{(\sigma_2^2/n_2)^2}{n_2-1}},$$

and plugging in S_i^2 for σ_i^2 gives the formula. Mortar: $\nu = 17.03$ (R: Welch $t = -9.11$, df$= 17.025$).

7 Permutation (Randomization) Test

Derivation

Instead of assuming normality, use the randomization itself as the basis of inference.

- Pool the $N = n + m$ observations. Under $H_0 : \mu_x = \mu_y$ (more precisely, under H_0 the two distributions are identical) the labels are exchangeable.
- Form all $N!$ permutations of the data; for each permutation compute the test statistic $T_1, \ldots, T_{N!}$.
- Under H_0 each of these values is equally likely, so the permutation distribution of T puts mass $1/N!$ on each T_j:

$$p\text{-value} = P(T > t_{\mathrm{obs}} \mid H_0) = \frac{1}{N!}\sum_{j=1}^{N!} I(T_j > t_{\mathrm{obs}}).$$

Example 2.6: A tiny permutation test

$(X_1, X_2, Y_1) = (1, 9, 3)$, i.e. x-sample $\{1, 9\}$ $(n = 2)$, y-sample $\{3\}$ $(m = 1)$. Statistic $T = |\bar{x} - \bar{y}|$.

Permutation (X_1, X_2, Y_1)	$\bar{x}$	$T = \lvert\bar{x} - \bar{y}\rvert$	prob.
$(1, 9, 3)$	5	$\lvert 5 - 3\rvert = 2$	1/6
$(1, 3, 9)$	2	$\lvert 2 - 9\rvert = 7$	1/6
$(9, 1, 3)$	5	2	1/6
$(9, 3, 1)$	6	$\lvert 6 - 1\rvert = 5$	1/6
$(3, 9, 1)$	6	5	1/6
$(3, 1, 9)$	2	7	1/6

Observed $t_{\mathrm{obs}} = 2$. Then $P(T \geq 2) = 6/6 = 1$ and $P(T > 2) = 4/6$: no evidence against H_0 (and with $N = 3$ the smallest attainable p-value is $2/6$, so such a test can never reject at $\alpha = 0.05$).

8 Paired Comparison Design

8.1 Hardness testing example

- A hardness testing machine presses a rod with a pointed tip into a metal specimen with a known force. Two tips; are the readings the same?
- **Experiment 1 (CRD)**: 20 specimens, half tested by tip 1 and half by tip 2, randomly assigned. Two-sample t-test. Problem: specimens from different heats are not homogeneous; their variability inflates the error and masks tip differences.
- **Experiment 2 (RCBD / paired)**: divide each of 10 specimens into two halves; randomly assign tip 1 to one half and tip 2 to the other. Each specimen is a *block*.

8.2 Model and the elimination of the block effect

$$y_{ij} = \mu_i + \beta_j + \varepsilon_{ij}, \qquad i = 1, 2;\ j = 1, \ldots, 10,$$

μ_i: true mean hardness of tip i; β_j: effect of specimen j; $\varepsilon_{ij} \overset{\text{iid}}{\sim} (0, \sigma^2)$. Define the paired difference

$$\begin{aligned} d_j = y_{1j} - y_{2j} &= (\mu_1 + \beta_j + \varepsilon_{1j}) - (\mu_2 + \beta_j + \varepsilon_{2j}) \\ &= (\mu_1 - \mu_2) + (\varepsilon_{1j} - \varepsilon_{2j}). \end{aligned}$$

The specimen effect β_j **cancels**. Hence $\mu_d = \mathrm{E}(d_j) = \mu_1 - \mu_2$, $\mathrm{Var}(d_j) = 2\sigma^2$, and $d_1, \ldots, d_n$ is a one-sample problem:

$$t_0 = \frac{\bar{d}}{S_d/\sqrt{n}} \overset{H_0:\mu_d=0}{\sim} t_{n-1}, \qquad \bar{d} = \frac{1}{n}\sum_j d_j, \quad S_d^2 = \frac{1}{n-1}\sum_j (d_j - \bar{d})^2.$$

Remark 2.7: Why pairing gains precision

If instead the 20 specimens were *random* blocks, $\beta_j \sim (0, \sigma_\beta^2)$, then in a CRD $\mathrm{Var}(y) = \sigma_\beta^2 + \sigma^2$ and

$$\mathrm{Var}(\bar{y}_1 - \bar{y}_2)_{\text{CRD}} = \frac{2(\sigma_\beta^2 + \sigma^2)}{n}, \qquad \mathrm{Var}(\bar{d})_{\text{paired}} = \frac{2\sigma^2}{n}.$$

Pairing removes σ_β^2 from the error at the cost of $n - 1$ instead of $2n - 2$ error df. When specimen variability is large, pairing is far more powerful.

Example 2.8: Tip data

Specimen j	1	2	3	4	5	6	7	8	9	10
Tip 1	7	3	3	4	8	3	2	9	5	4
Tip 2	6	3	5	3	8	2	4	9	4	5
d_j	1	0	−2	1	0	1	−2	0	1	−1

$$\begin{aligned} \bar{d} &= \frac{1+0-2+1+0+1-2+0+1-1}{10} = \frac{-1}{10} = -0.1, \\ \sum d_j^2 &= 1+0+4+1+0+1+4+0+1+1 = 13, \\ S_d^2 &= \frac{\sum d_j^2 - n\bar{d}^2}{n-1} = \frac{13 - 10(0.01)}{9} = \frac{12.9}{9} = 1.4333, \quad S_d = 1.1972, \end{aligned}$$

$$t_0 = \frac{-0.1}{1.1972/\sqrt{10}} = \frac{-0.1}{0.3786} = -0.264.$$

$|t_0| = 0.264 < t_{0.025,9} = 2.262$ ($P = 0.7976$): fail to reject $H_0 : \mu_d = 0$. No evidence that the two tips give different readings. 95% CI: $-0.1 \pm 2.262(0.3786) = [-0.956, 0.756]$.

9 Nonparametric Alternatives

- **Wilcoxon rank-sum test** (two independent samples): replaces the data by their ranks in the pooled sample.
- **Wilcoxon signed-rank test** (paired data): ranks $|d_j|$ and attaches the signs.
- These are used when normality is doubtful (see also Kruskal–Wallis in Ch. 3).

10 Selected exercises

Motivation. Allocation of a budget, interval estimates of variances, and sample size for power.

Exercise 2.1 (Optimal allocation; Montgomery 2.40–2.41)

Two normal populations have known variances σ_1^2, σ_2^2, and the total sample size is fixed, $n_1 + n_2 = N$.
(a) Show that the most powerful z-test of $H_0 : \mu_1 = \mu_2$ (equivalently, the shortest confidence interval for $\mu_1 - \mu_2$) uses $n_1/n_2 = \sigma_1/\sigma_2$.
(b) With $\sigma_1 = 4$, $\sigma_2 = 8$ and $N = 30$, compare the lengths of the 95% intervals for $n_1 = n_2 = 15$ and for the optimal allocation.

Solution.

(a)

Step 1. $\bar{y}_1 - \bar{y}_2 \sim N(\mu_1 - \mu_2, V)$ with $V = \sigma_1^2/n_1 + \sigma_2^2/n_2$. The power of the two-sided z-test at a difference δ is $\Phi(-z_{\alpha/2} + \delta/\sqrt{V}) + \Phi(-z_{\alpha/2} - \delta/\sqrt{V})$, which increases as V decreases (Exercise 2.3), and the interval length is $2z_{\alpha/2}\sqrt{V}$. So we minimize V.

Step 2. By the Cauchy–Schwarz inequality,

$$(\sigma_1 + \sigma_2)^2 = \Big(\frac{\sigma_1}{\sqrt{n_1}}\sqrt{n_1} + \frac{\sigma_2}{\sqrt{n_2}}\sqrt{n_2}\Big)^2 \le \Big(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\Big)(n_1 + n_2) = VN,$$

so $V \ge (\sigma_1 + \sigma_2)^2/N$, with equality iff $(\sigma_i/\sqrt{n_i}) \propto \sqrt{n_i}$, i.e. $n_i \propto \sigma_i$. Hence $n_1/n_2 = \sigma_1/\sigma_2$: sample more from the noisier population.

(b) Equal allocation: $V = 16/15 + 64/15 = 5.333$, length $2(1.96)\sqrt{5.333} = 9.05$. Optimal: $n_1 = 10$, $n_2 = 20$, $V = (4 + 8)^2/30 = 4.8$, length $2(1.96)\sqrt{4.8} = 8.59$, about 0.46 (5%) shorter at no extra cost. ■

Exercise 2.2 (Intervals for variances; Montgomery 2.42–2.43)

Let $\chi^2_{\alpha,\nu}$ and $F_{\alpha,u,v}$ denote upper-α points.
(a) For a normal sample of size n, derive the $100(1-\alpha)\%$ interval $[(n-1)S^2/\chi^2_{\alpha/2,n-1},\ (n-1)S^2/\chi^2_{1-\alpha/2,n-1}]$ for σ^2.

(b) For independent normal samples, derive the interval

$$\frac{S_1^2}{S_2^2} F_{1-\alpha/2,\, n_2-1,\, n_1-1} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{S_1^2}{S_2^2} F_{\alpha/2,\, n_2-1,\, n_1-1},$$

and evaluate it for the mortar data ($F_{0.025,9,9} = 4.03$).

Solution.

(a) $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, so $P\{\chi^2_{1-\alpha/2,n-1} \le (n-1)S^2/\sigma^2 \le \chi^2_{\alpha/2,n-1}\} = 1-\alpha$. All quantities are positive, so taking reciprocals and multiplying by $(n-1)S^2$ inverts the inequalities into the stated interval.

(b)

Step 1. $(n_i-1)S_i^2/\sigma_i^2 \sim \chi^2_{n_i-1}$ independently, so

$$W = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1,n_2-1}, \qquad P(F_{1-\alpha/2,n_1-1,n_2-1} \le W \le F_{\alpha/2,n_1-1,n_2-1}) = 1-\alpha.$$

Step 2. $W \le c \iff \sigma_1^2/\sigma_2^2 \ge (S_1^2/S_2^2)/c$, so the event is

$$\frac{S_1^2/S_2^2}{F_{\alpha/2,n_1-1,n_2-1}} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{S_1^2/S_2^2}{F_{1-\alpha/2,n_1-1,n_2-1}}.$$

Step 3. If $W \sim F_{u,v}$ then $1/W \sim F_{v,u}$, so $1/F_{\alpha/2,u,v} = F_{1-\alpha/2,v,u}$ and $1/F_{1-\alpha/2,u,v} = F_{\alpha/2,v,u}$, which gives the stated form.

Mortar: $S_1^2/S_2^2 = 0.1001/0.0615 = 1.628$ and $F_{0.975,9,9} = 1/4.03$, so the interval is

$$[1.628/4.03,\ 1.628 \times 4.03] = [0.40,\ 6.56].$$

It contains 1, in agreement with the F-test ($P = 0.48$); with $n = 10$ per group a variance ratio is estimated only within a factor of about four. ■

Exercise 2.3 (Power and sample size)

Two samples of size n each come from $N(\mu_i, \sigma^2)$ with σ known, and $H_0 : \mu_1 = \mu_2$ is rejected when $|Z_0| > z_{\alpha/2}$, $Z_0 = (\bar{y}_1 - \bar{y}_2)/(\sigma\sqrt{2/n})$.
(a) Show that the power at $\delta = \mu_1 - \mu_2$ is $\Phi(-z_{\alpha/2} + \Delta) + \Phi(-z_{\alpha/2} - \Delta)$, $\Delta = \delta\sqrt{n}/(\sigma\sqrt{2})$, and that it increases in $|\Delta|$.
(b) Deduce that power at least $1-\beta$ is guaranteed by $n \ge 2\sigma^2(z_{\alpha/2} + z_\beta)^2/\delta^2$, and that this is nearly necessary.

Solution [Supplement].

(a)

Step 1. $Z_0 \sim N(\Delta, 1)$, so $P(|Z_0| > z_{\alpha/2}) = P(Z > z_{\alpha/2} - \Delta) + P(Z < -z_{\alpha/2} - \Delta)$ with $Z \sim N(0,1)$, which is the stated formula.

Step 2. The power $\pi(\Delta)$ is even in Δ, and for $\Delta > 0$

$$\pi'(\Delta) = \phi(z_{\alpha/2} - \Delta) - \phi(z_{\alpha/2} + \Delta) > 0,$$

because $|z_{\alpha/2} - \Delta| < z_{\alpha/2} + \Delta$ and ϕ decreases in $|x|$.

(b) Take $\delta > 0$. Since $\pi(\Delta) \ge \Phi(-z_{\alpha/2} + \Delta)$, power $\ge 1-\beta$ holds if $-z_{\alpha/2} + \Delta \ge z_\beta$, i.e. $\sqrt{n}\,\delta/(\sigma\sqrt{2}) \ge z_{\alpha/2} + z_\beta$, which is the stated bound. Conversely the dropped term $\Phi(-z_{\alpha/2} - \Delta) < \alpha/2$ is negligible, so the bound is essentially also necessary. Halving δ quadruples n. For example $\alpha = 0.05$, $\beta = 0.10$, $\delta = \sigma/2$: $n \ge 2(1.96 + 1.282)^2/0.25 = 84.1$, i.e. 85 per group. ■

Chapter 3. Experiments with a Single Factor: The Analysis of Variance

1 What If There Are More Than Two Factor Levels?

- The t-test does not directly apply when there are more than two levels.
- In practice there are often more than two levels of interest, or several factors of simultaneous interest.
- The **analysis of variance (ANOVA)** is the appropriate analysis engine. It was developed by R. A. Fisher in the early 1920s for agricultural experiments and is widely used today in industrial experiments.

2 Experimental Design Terminology

Definition 3.1

- **Block**: group of homogeneous experimental units.
- **Factor**: a controllable experimental variable thought to influence the response.
- **Level**: a specific value of a factor.
- **Repeat tests**: two or more observations with the same levels of all factors.
- **Replication**: repetition of an entire experiment.
- **Response**: outcome or result of an experiment.
- **Unit**: entity on which a measurement is made.

3 Motivating Example: Tensile Strength

A new synthetic fibre for shirts; response = tensile strength. Cotton content (wt %) can vary between 10–40%; some non-linearity is anticipated. Five levels (15, 20, 25, 30, 35%) are chosen and each is replicated 5 times, runs in random order (a CRD with $a = 5$, $n = 5$, $N = 25$).

Cotton %	Observations					Total $y_{i\cdot}$	Average $\bar{y}_{i\cdot}$
15	7	7	15	11	9	49	9.8
20	12	17	12	18	18	77	15.4
25	14	18	18	19	19	88	17.6
30	19	25	22	19	23	108	21.6
35	7	10	11	15	11	54	10.8
						$y_{\cdot\cdot} = 376$	$\bar{y}_{\cdot\cdot} = 15.04$

Questions:

(1) Does cotton weight percent change the mean tensile strength? (2) Is there an optimum level?

4 The Single-Factor ANOVA Model

4.1 Data layout and notation

$$y_{i\cdot} = \sum_{j=1}^{n} y_{ij}, \quad \bar{y}_{i\cdot} = \frac{y_{i\cdot}}{n}, \quad y_{\cdot\cdot} = \sum_{i=1}^{a}\sum_{j=1}^{n} y_{ij}, \quad \bar{y}_{\cdot\cdot} = \frac{y_{\cdot\cdot}}{N}, \quad N = an.$$

4.2 Effects model and means model

Definition 3.2: Single-factor (one-way) model, fixed effects

Effects model: $y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \ldots, a;\ j = 1, \ldots, n,$
μ: overall mean; τ_i: ith treatment effect; $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, so $y_{ij} \overset{\text{ind}}{\sim} N(\mu + \tau_i, \sigma^2)$.
Means model: $y_{ij} = \mu_i + \varepsilon_{ij}$, with $\mu_i = \mu + \tau_i$.

Derivation: where the constraint $\sum \tau_i = 0$ comes from

The overall mean is defined as the average of the treatment means:

$$\mu = \frac{1}{a}\sum_{i=1}^{a} \mu_i = \frac{1}{a}\sum_{i=1}^{a}(\mu + \tau_i) = \mu + \frac{1}{a}\sum_{i=1}^{a}\tau_i \quad \Longrightarrow \quad \sum_{i=1}^{a}\tau_i = 0.$$

So $\tau_i = \mu_i - \mu$ is the deviation of the ith treatment mean from the overall mean.

- **Fixed effects model**: the a treatments are specifically chosen; conclusions apply only to them; τ_i are unknown constants.
- **Random effects model**: the a treatments are a random sample from a larger population; τ_i are random variables (Ch. 9).

4.3 Hypotheses

$$\begin{aligned} &H_0 : \mu_1 = \mu_2 = \cdots = \mu_a \\ &H_1 : \mu_i \neq \mu_j \text{ for at least one pair } (i, j) \end{aligned} \quad \Longleftrightarrow \quad \begin{aligned} &H_0 : \tau_1 = \cdots = \tau_a = 0 \\ &H_1 : \tau_i \neq 0 \text{ for at least one } i. \end{aligned}$$

5 Partitioning the Total Sum of Squares

Motivation. One test of equal means, splitting total variation into treatment and noise, instead of many pairwise t-tests.

Theorem 3.3: Fundamental ANOVA identity

$$\underbrace{\sum_{i=1}^{a}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{\cdot\cdot})^2}_{SS_T} = \underbrace{n\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}_{SS_{\text{Treatments}}} + \underbrace{\sum_{i=1}^{a}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot})^2}_{SS_E}.$$

Proof [Supplement]. Add and subtract $\bar{y}_{i\cdot}$:

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{n}\left[(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) + (y_{ij} - \bar{y}_{i\cdot})\right]^2$$

$$= \sum_i \sum_j (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2 + 2 \sum_i \sum_j (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})(y_{ij} - \bar{y}_{i\cdot}).$$

The first term does not depend on j, so $\sum_i \sum_j (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = n \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$. For the cross product, pull out the factor that does not depend on j:

$$\sum_{i=1}^{a} \sum_{j=1}^{n} (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})(y_{ij} - \bar{y}_{i\cdot}) = \sum_{i=1}^{a} (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \underbrace{\sum_{j=1}^{n} (y_{ij} - \bar{y}_{i\cdot})}_{= y_{i\cdot} - n\bar{y}_{i\cdot} = 0} = 0.$$

Hence $SS_T = SS_{\text{Treatments}} + SS_E$. ■

- $SS_{\text{Treatments}}$ measures differences *between* treatment means; a large value reflects large differences.
- SS_E measures variation *within* treatments (pure error).
- Degrees of freedom: $an - 1 = (a - 1) + a(n - 1)$, i.e. $N - 1 = (a - 1) + (N - a)$.
- Sums of squares cannot be compared directly; **mean squares** can:

$$MS_{\text{Treatments}} = \frac{SS_{\text{Treatments}}}{a - 1}, \qquad MS_E = \frac{SS_E}{a(n-1)} = \frac{SS_E}{N - a}.$$

Derivation: decomposition in terms of estimates

With $\hat{\mu} = \bar{y}_{\cdot\cdot}$, $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, the fitted model is $y_{ij} = \hat{\mu} + \hat{\tau}_i + e_{ij}$, so $e_{ij} = y_{ij} - \bar{y}_{i\cdot}$ and

$$y_{ij} - \hat{\mu} = \hat{\tau}_i + e_{ij} \quad \Longrightarrow \quad SS_T = \sum_i \sum_j \hat{\tau}_i^2 + \sum_i \sum_j e_{ij}^2.$$

6 Expected Mean Squares (Elementwise Proof)

Motivation. Both mean squares estimate σ^2 under H_0; otherwise $MS_{\text{Treatments}}$ gains $\sum n\tau_i^2/(a-1)$.

Theorem 3.4

$$\mathrm{E}(MS_E) = \sigma^2, \qquad \mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \frac{n \sum_{i=1}^{a} \tau_i^2}{a - 1}.$$

Proof of $\mathrm{E}(MS_E) = \sigma^2$ [Supplement]. Write SS_E through the within-treatment sample variances $S_i^2 = \frac{1}{n-1} \sum_j (y_{ij} - \bar{y}_{i\cdot})^2$:

$$MS_E = \frac{SS_E}{a(n-1)} = \frac{1}{a(n-1)} \sum_{i=1}^{a} \Big[(n-1) \frac{\sum_j (y_{ij} - \bar{y}_{i\cdot})^2}{n-1} \Big] = \frac{1}{a(n-1)} \sum_{i=1}^{a} (n-1) S_i^2.$$

Since $\bar{y}_{i\cdot} = \mu_i + \bar{\varepsilon}_{i\cdot}$, we have $\mathrm{E}(\bar{y}_{i\cdot}) = \mu_i$ and $\mathrm{Var}(\bar{y}_{i\cdot}) = \sigma^2/n$, hence $\mathrm{E}(\bar{y}_{i\cdot}^2) = \sigma^2/n + \mu_i^2$. Then

$$\mathrm{E}(S_i^2) = \frac{1}{n-1} \mathrm{E}\Big(\sum_{j=1}^{n} y_{ij}^2 - n\bar{y}_{i\cdot}^2 \Big) = \frac{1}{n-1} \Big(\sum_{j=1}^{n} \mathrm{E}(y_{ij}^2) - n\mathrm{E}(\bar{y}_{i\cdot}^2) \Big)$$

$$= \frac{1}{n-1} \Big(\sum_{j=1}^{n} (\sigma^2 + \mu_i^2) - n\Big(\frac{\sigma^2}{n} + \mu_i^2 \Big) \Big) = \frac{1}{n-1} (n-1)\sigma^2 = \sigma^2.$$

Therefore $\mathrm{E}(MS_E) = \frac{1}{a(n-1)} \sum_{i=1}^{a} (n-1)\sigma^2 = \sigma^2$. ■

Proof of $\mathrm{E}(MS_{\text{Treatments}})$ [Supplement]. Using $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$ and $\bar{y}_{\cdot\cdot} = \mu + \bar{\varepsilon}_{\cdot\cdot}$ (since $\sum \tau_i = 0$),

$$\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}).$$

Then

$$MS_{\text{Treatments}} = \frac{1}{a-1} \sum_{i=1}^{a} \sum_{j=1}^{n} (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = \frac{n}{a-1} \sum_{i=1}^{a} \big(\tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})\big)^2,$$

$$\begin{aligned} \mathrm{E}\Big[\sum_{i=1}^{a} \big(\tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})\big)^2\Big] &= \mathrm{E}\Big[\sum_i \tau_i^2 + \sum_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2 + 2\sum_i \tau_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})\Big] \\ &= \sum_i \tau_i^2 + \mathrm{E}\Big[\sum_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2\Big] + 0. \end{aligned}$$

(The cross term has mean 0 since $\mathrm{E}\bar{\varepsilon}_{i\cdot} = \mathrm{E}\bar{\varepsilon}_{\cdot\cdot} = 0$.) For the remaining term, note $\bar{\varepsilon}_{\cdot\cdot} = \frac{1}{a}\sum_i \bar{\varepsilon}_{i\cdot}$, so

$$\begin{aligned} \mathrm{E}\Big[\sum_{i=1}^{a} (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2\Big] &= \mathrm{E}\Big[\sum_{i=1}^{a} \bar{\varepsilon}_{i\cdot}^2 - a\bar{\varepsilon}_{\cdot\cdot}^2\Big] = \sum_{i=1}^{a} \mathrm{E}(\bar{\varepsilon}_{i\cdot}^2) - a\mathrm{E}(\bar{\varepsilon}_{\cdot\cdot}^2) \\ &= \sum_{i=1}^{a} \frac{\sigma^2}{n} - a \cdot \frac{\sigma^2}{an} = (a-1)\frac{\sigma^2}{n}. \end{aligned}$$

(Equivalently: $\bar{\varepsilon}_{1\cdot}, \ldots, \bar{\varepsilon}_{a\cdot}$ are iid with variance σ^2/n, and the sample variance of them is unbiased.) Therefore

$$\mathrm{E}(MS_{\text{Treatments}}) = \frac{n}{a-1}\Big[\sum_i \tau_i^2 + (a-1)\frac{\sigma^2}{n}\Big] = \sigma^2 + \frac{n\sum_{i=1}^{a} \tau_i^2}{a-1}.$$

■

- MS_E is always an unbiased estimator of σ^2.
- Under H_0 ($\tau_i \equiv 0$), $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2$, so $MS_{\text{Treatments}}$ also estimates σ^2.
- Let $F_0 = MS_{\text{Treatments}}/MS_E$. Heuristically,

$$\mathrm{E}(F_0) \approx \frac{\mathrm{E}(MS_{\text{Treatments}})}{\mathrm{E}(MS_E)} = \frac{\sigma^2 + \frac{n\sum \tau_i^2}{a-1}}{\sigma^2} \overset{H_0}{=} 1.$$

  Under H_0 the ratio is close to 1; under H_1 it tends to exceed 1. Decision: fail to reject if $F_0 \approx 1$, reject if $F_0 \gg 1$.

7 The ANOVA Table and the F-test

Motivation. Under normality the mean-square ratio is F-distributed: the exact test of the ANOVA table.

Source	SS	DF	MS	F_0
Between treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Error (within)	SS_E	$N-a$	MS_E	
Total	SS_T	$N-1$		

Reference distribution: $F_0 \overset{H_0}{\sim} F_{a-1,\,N-a}$ (proved in Section 9). Reject H_0 if $F_0 > F_{\alpha,a-1,N-a}$.

7.1 Hand-computation formulas

Theorem 3.5: Computing formulas

$$SS_T = \sum_i \sum_j y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Treatments}} = \frac{1}{n}\sum_i y_{i\cdot}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_E = SS_T - SS_{\text{Treatments}}.$$

Proof [Supplement]. $\sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum\sum y_{ij}^2 - 2\bar{y}_{\cdot\cdot}\sum\sum y_{ij} + N\bar{y}_{\cdot\cdot}^2 = \sum\sum y_{ij}^2 - 2N\bar{y}_{\cdot\cdot}^2 + N\bar{y}_{\cdot\cdot}^2 = \sum\sum y_{ij}^2 - y_{\cdot\cdot}^2/N$. Similarly, using $\sum_i \bar{y}_{i\cdot} = a\bar{y}_{\cdot\cdot}$, $n\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = n\sum_i \bar{y}_{i\cdot}^2 - 2n\bar{y}_{\cdot\cdot}\sum_i \bar{y}_{i\cdot} + na\bar{y}_{\cdot\cdot}^2 = \frac{1}{n}\sum_i y_{i\cdot}^2 - 2N\bar{y}_{\cdot\cdot}^2 + N\bar{y}_{\cdot\cdot}^2 = \frac{1}{n}\sum_i y_{i\cdot}^2 - \frac{y_{\cdot\cdot}^2}{N}$. ■

Example 3.6: Tensile strength ANOVA, line by line

$$\begin{aligned}
\sum\sum y_{ij}^2 &= 6292, \qquad \frac{y_{\cdot\cdot}^2}{N} = \frac{376^2}{25} = \frac{141376}{25} = 5655.04,\\
SS_T &= 6292 - 5655.04 = 636.96,\\
SS_{\text{Treatments}} &= \frac{49^2 + 77^2 + 88^2 + 108^2 + 54^2}{5} - 5655.04\\
&= \frac{2401 + 5929 + 7744 + 11664 + 2916}{5} - 5655.04\\
&= \frac{30654}{5} - 5655.04 = 6130.8 - 5655.04 = 475.76,\\
SS_E &= 636.96 - 475.76 = 161.20.
\end{aligned}$$

Source	SS	DF	MS	F_0
Cotton %	475.76	4	118.94	118.94/8.06 = 14.76
Error	161.20	20	8.06	
Total	636.96	24		

$F_0 = 14.76 > F_{0.05,4,20} = 2.87$ ($P = 9.13 \times 10^{-6}$). Reject H_0: cotton weight percent significantly affects mean tensile strength.

8 Parameter Estimation by Least Squares

Derivation

Minimize

$$S(\mu, \tau_1, \ldots, \tau_a) = \sum_{i=1}^{a}\sum_{j=1}^{n} \varepsilon_{ij}^2 = \sum_{i=1}^{a}\sum_{j=1}^{n} (y_{ij} - \mu - \tau_i)^2.$$

Setting partial derivatives to zero at $(\hat{\mu}, \hat{\tau}_i)$:

$$\begin{aligned}
\frac{\partial S}{\partial \mu}\Big|_{\hat{\mu},\hat{\tau}_i} &= -2\sum_{i=1}^{a}\sum_{j=1}^{n} (y_{ij} - \hat{\mu} - \hat{\tau}_i) = 0,\\
\frac{\partial S}{\partial \tau_i}\Big|_{\hat{\mu},\hat{\tau}_i} &= -2\sum_{j=1}^{n} (y_{ij} - \hat{\mu} - \hat{\tau}_i) = 0, \qquad i = 1, \ldots, a.
\end{aligned}$$

These give the $a+1$ **normal equations**

$$\begin{cases} N\hat{\mu} + n\hat{\tau}_1 + n\hat{\tau}_2 + \cdots + n\hat{\tau}_a = y_{\cdot\cdot} \\ n\hat{\mu} + n\hat{\tau}_1 = y_{1\cdot} \\ \quad\vdots \\ n\hat{\mu} + n\hat{\tau}_a = y_{a\cdot} \end{cases}$$

The first equation is the sum of the other a, so the system has rank a: one more condition is needed. Imposing $\sum \hat{\tau}_i = 0$, the first equation gives $N\hat{\mu} = y_{\cdot\cdot}$, so

$$\boxed{\hat{\mu} = \bar{y}_{\cdot\cdot}, \qquad \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}, \quad i = 1, \ldots, a.}$$

Note: $\hat{\mu}_i = \hat{\mu} + \hat{\tau}_i = \bar{y}_{i\cdot}$ is the same whatever constraint is used (it is *estimable*), and $\hat{\tau}_i - \hat{\tau}_j = \bar{y}_{i\cdot} - \bar{y}_{j\cdot}$ is also estimable.

Fitted values and residuals: $\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i = \bar{y}_{i\cdot}$, $e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{i\cdot}$.

9 Matrix (Regression) Approach to the One-Way ANOVA

Motivation. ANOVA in matrix form: constraints, the general linear hypothesis, unbalanced designs.

9.1 Design matrix

Derivation

For $a = 3$, $n = 2$ and the constraint $\tau_3 = -\tau_1 - \tau_2$, write $\mathbf{y} = (y_{11}, y_{12}, y_{21}, y_{22}, y_{31}, y_{32})^T$ and

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \mathbf{X} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \mu \\ \tau_1 \\ \tau_2 \end{pmatrix}, \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}).$$

Hence $\mathrm{E}(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$, $\mathrm{Var}(\mathbf{y}) = \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ and $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$.

9.2 Vector differentiation

Theorem 3.7: Two derivative rules

1. If $\mathbf{a}$ is a constant $n \times 1$ vector, $\dfrac{\partial\, \mathbf{a}^T\mathbf{y}}{\partial \mathbf{y}} = \mathbf{a}$.
2. If $\mathbf{A}$ is a constant $n \times n$ matrix, $\dfrac{\partial\, \mathbf{y}^T\mathbf{A}\mathbf{y}}{\partial \mathbf{y}} = \mathbf{A}\mathbf{y} + \mathbf{A}^T\mathbf{y}$ ($= 2\mathbf{A}\mathbf{y}$ if $\mathbf{A}$ is symmetric).

Proof [Supplement]. (1) $\mathbf{a}^T\mathbf{y} = a_1y_1 + \cdots + a_ny_n$, so the kth component of the gradient is $\partial(a_1y_1 + \cdots + a_ny_n)/\partial y_k = a_k$.

(2) $\mathbf{y}^T\mathbf{A}\mathbf{y} = \sum_i \sum_j A_{ij}y_iy_j$. Differentiating w.r.t. y_k, the terms with $i = k$ give $\sum_j A_{kj}y_j = (\mathbf{A}\mathbf{y})_k$ and the terms with $j = k$ give $\sum_i A_{ik}y_i = (\mathbf{A}^T\mathbf{y})_k$. ■

9.3 Least squares in matrix form

$$S(\boldsymbol{\beta}) = \boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon} = (\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^T(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$
$$= \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} \qquad \text{(a scalar equals its transpose).}$$

By rules (1) and (2) (with $\mathbf{a} = \mathbf{X}^T\mathbf{y}$, $\mathbf{A} = \mathbf{X}^T\mathbf{X}$ symmetric),

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0} \quad \Longrightarrow \quad \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y},$$

and if $(\mathbf{X}^T\mathbf{X})^{-1}$ exists, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$.

9.4 The hat matrix

Definition 3.8

Fitted values $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ (**hat matrix**). Residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}-\mathbf{H})\mathbf{y}$.

Theorem 3.9: Properties of H

1. $\mathbf{H}^T = \mathbf{H}$ and $\mathbf{H}\mathbf{H} = \mathbf{H}$ (symmetric idempotent: orthogonal projection onto $\Omega(\mathbf{X})$, the column space of $\mathbf{X}$).
2. $\mathbf{H}\mathbf{X} = \mathbf{X}$; in particular $\mathbf{H}\mathbf{1} = \mathbf{1}$ whenever $\mathbf{1}$ is a column of $\mathbf{X}$.
3. $\mathbf{I}-\mathbf{H}$ is symmetric idempotent and $(\mathbf{I}-\mathbf{H})\mathbf{X} = \mathbf{0}$.
4. $\operatorname{tr}(\mathbf{H}) = \operatorname{rank}(\mathbf{X}) = p$ ($p = a$ here), $\operatorname{tr}(\mathbf{I}-\mathbf{H}) = N - p$.

Proof [Supplement]. (1)

$$\mathbf{H}\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\underbrace{\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}}_{=\mathbf{I}}\mathbf{X}^T = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = \mathbf{H}.$$

(2) $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X} = \mathbf{X}$. Writing $\mathbf{X} = (\mathbf{1}, \mathbf{X}_{-1})$, $\mathbf{H}\mathbf{X} = (\mathbf{H}\mathbf{1}, \mathbf{H}\mathbf{X}_{-1}) = (\mathbf{1}, \mathbf{X}_{-1})$, so $\mathbf{H}\mathbf{1} = \mathbf{1}$.

(3) $(\mathbf{I}-\mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I}-\mathbf{H}$; $(\mathbf{I}-\mathbf{H})\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$.

(4) $\operatorname{tr}(\mathbf{H}) = \operatorname{tr}(\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T) = \operatorname{tr}((\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}) = \operatorname{tr}(\mathbf{I}_p) = p$ (cyclic property of trace). ■

Geometrically, $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\Omega(\mathbf{X})$ and $\mathbf{e} \in \Omega(\mathbf{X})^{\perp}$.

9.5 Sums of squares as quadratic forms

Let $\mathbf{J}_N = \mathbf{1}\mathbf{1}^T$ ($N \times N$ matrix of ones), so $\frac{1}{N}\mathbf{J}_N\mathbf{y} = \bar{y}_{\cdot\cdot}\mathbf{1}$.

$$SS_E = \sum_i\sum_j (y_{ij} - \bar{y}_{i\cdot})^2 = \sum_i\sum_j (y_{ij} - \hat{y}_{ij})^2 = \mathbf{e}^T\mathbf{e} = \mathbf{y}^T(\mathbf{I}-\mathbf{H})^T(\mathbf{I}-\mathbf{H})\mathbf{y} = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y},$$

$$SS_{\text{Treatments}} = \sum_i\sum_j (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = (\hat{\mathbf{y}} - \bar{\mathbf{y}})^T(\hat{\mathbf{y}} - \bar{\mathbf{y}}), \qquad \hat{\mathbf{y}} - \bar{\mathbf{y}} = \left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right)\mathbf{y},$$
$$= \mathbf{y}^T\left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right)\mathbf{y} \quad \text{(once we show } \mathbf{H} - \tfrac{1}{N}\mathbf{J}_N \text{ is idempotent).}$$

Theorem 3.10: $\mathbf{H} - \frac{1}{N}\mathbf{J}_N$ is symmetric idempotent

Proof [Supplement]. First, $\frac{1}{N}\mathbf{J}_N = \mathbf{1}(\mathbf{1}^T\mathbf{1})^{-1}\mathbf{1}^T$ is itself a projection (onto span$\{\mathbf{1}\}$), so $\frac{1}{N^2}\mathbf{J}_N\mathbf{J}_N = \frac{1}{N}\mathbf{J}_N$.

① $\frac{1}{N}\mathbf{J}_N\mathbf{H} = \mathbf{1}(\mathbf{1}^T\mathbf{1})^{-1}\underbrace{\mathbf{1}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T}_{=(\mathbf{H}\mathbf{1})^T=\mathbf{1}^T} = \frac{1}{N}\mathbf{J}_N.$

② $\frac{1}{N}\mathbf{H}\mathbf{J}_N = \underbrace{\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{1}}_{=\mathbf{H}\mathbf{1}=\mathbf{1}}(\mathbf{1}^T\mathbf{1})^{-1}\mathbf{1}^T = \frac{1}{N}\mathbf{J}_N.$

Hence

$$\begin{aligned}\left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right)\left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right) &= \mathbf{H}\mathbf{H} - \underbrace{\tfrac{1}{N}\mathbf{H}\mathbf{J}_N}_{②} - \underbrace{\tfrac{1}{N}\mathbf{J}_N\mathbf{H}}_{①} + \underbrace{\tfrac{1}{N^2}\mathbf{J}_N\mathbf{J}_N}_{=\frac{1}{N}\mathbf{J}_N} \\ &= \mathbf{H} - \tfrac{1}{N}\mathbf{J}_N - \tfrac{1}{N}\mathbf{J}_N + \tfrac{1}{N}\mathbf{J}_N = \mathbf{H} - \tfrac{1}{N}\mathbf{J}_N.\end{aligned}$$

■

Its rank is $\operatorname{tr}(\mathbf{H} - \frac{1}{N}\mathbf{J}_N) = a - \frac{1}{N}\cdot N = a - 1$.

9.6 Distribution theory

Theorem 3.11: chi-square of a quadratic form

Let $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and $\mathbf{A}$ an $n \times n$ symmetric matrix of rank k. If $\sigma^2\mathbf{A}$ is idempotent, then

$$\mathbf{y}^T\mathbf{A}\mathbf{y} \sim \chi^2_{k,\,\lambda}, \qquad \lambda = \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{A}\mathbf{X}\boldsymbol{\beta} \quad \text{(noncentral; central if } \lambda = 0\text{)}.$$

Theorem 3.12: independence of quadratic forms

Let $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, $\mathbf{A}, \mathbf{B}$ symmetric, with $\mathbf{y}^T\mathbf{A}\mathbf{y}$ and $\mathbf{y}^T\mathbf{B}\mathbf{y}$ both chi-squared. If $\mathbf{A}(\sigma^2\mathbf{I})\mathbf{B} = \mathbf{0}$, then $\mathbf{y}^T\mathbf{A}\mathbf{y} \perp\!\!\!\perp \mathbf{y}^T\mathbf{B}\mathbf{y}$.

Theorem 3.13: expectation of a quadratic form

If $\mathbf{y}$ is an $n \times 1$ random vector with $\mathrm{E}(\mathbf{y}) = \boldsymbol{\mu}$, $\mathrm{Var}(\mathbf{y}) = \mathbf{V}$, then $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \operatorname{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$ (no normality needed).

Proof of C [Supplement]. A scalar equals its trace, and trace and expectation commute:

$$\begin{aligned}\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) &= \mathrm{E}(\operatorname{tr}(\mathbf{y}^T\mathbf{A}\mathbf{y})) = \mathrm{E}(\operatorname{tr}(\mathbf{A}\mathbf{y}\mathbf{y}^T)) = \operatorname{tr}(\mathbf{A}\,\mathrm{E}(\mathbf{y}\mathbf{y}^T)) \\ &= \operatorname{tr}(\mathbf{A}(\mathbf{V} + \boldsymbol{\mu}\boldsymbol{\mu}^T)) \qquad (\mathrm{Var}(\mathbf{y}) = \mathrm{E}(\mathbf{y}\mathbf{y}^T) - \boldsymbol{\mu}\boldsymbol{\mu}^T) \\ &= \operatorname{tr}(\mathbf{A}\mathbf{V}) + \operatorname{tr}(\mathbf{A}\boldsymbol{\mu}\boldsymbol{\mu}^T) = \operatorname{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}.\end{aligned}$$

■

Theorem 3.14: $F_0 \sim F_{a-1,N-a}$ under H_0

1. $SS_E/\sigma^2 \sim \chi^2_{N-a}$ (central, always);

2. $SS_{\text{Treatments}}/\sigma^2 \sim \chi^2_{a-1,\lambda}$ with $\lambda = n\sum_i \tau_i^2/\sigma^2$ (central under H_0);
3. $SS_E \perp\!\!\!\perp SS_{\text{Treatments}}$;
4. hence $F_0 = \dfrac{SS_{\text{Treatments}}/(a-1)}{SS_E/(N-a)} \overset{H_0}{\sim} F_{a-1,N-a}$ and, in general, $F_0 \sim F_{a-1,N-a,\lambda}$ (non-central).

Proof [Supplement]. ① $SS_E = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}$. Let $\mathbf{A} = (\mathbf{I}-\mathbf{H})/\sigma^2$; then $\sigma^2\mathbf{A} = \mathbf{I}-\mathbf{H}$ is idempotent. By Theorem 3.11,

$$\frac{\mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}}{\sigma^2} \sim \chi^2_{\operatorname{tr}(\mathbf{I}-\mathbf{H}),\ \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta}/\sigma^2},$$

with $\operatorname{rank}(\mathbf{I}-\mathbf{H}) = \operatorname{tr}(\mathbf{I}) - \operatorname{tr}(\mathbf{H}) = an - a$ and noncentrality 0 since $(\mathbf{I}-\mathbf{H})\mathbf{X} = \mathbf{0}$. So $SS_E/\sigma^2 \sim \chi^2_{an-a}$.

② $SS_{\text{Treatments}} = \mathbf{y}^T(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)\mathbf{y}$; $\mathbf{A} = (\mathbf{H} - \frac{1}{N}\mathbf{J}_N)/\sigma^2$, $\sigma^2\mathbf{A}$ idempotent, $\operatorname{tr} = a-1$. The noncentrality is $\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)\mathbf{X}\boldsymbol{\beta}/\sigma^2$. Compute

$$\Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\Big)\mathbf{X} = \mathbf{H}\mathbf{X} - \tfrac{1}{N}\mathbf{J}_N\mathbf{X} = \mathbf{X} - \mathbf{1}\Big(\tfrac{1}{N}\mathbf{1}^T\mathbf{X}\Big) = \mathbf{X} - (\mathbf{1}, \mathbf{0}, \ldots, \mathbf{0}),$$

because in a balanced design each coded column sums to zero ($\mathbf{1}^T\mathbf{X} = (N, 0, \ldots, 0)$). Therefore

$$\boldsymbol{\xi} := \Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\Big)\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta} - \mu\mathbf{1} = (\underbrace{\tau_1, \ldots, \tau_1}_{n}, \ldots, \underbrace{\tau_a, \ldots, \tau_a}_{n})^T,$$

and $\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\xi}^T\boldsymbol{\xi} = n\sum_i \tau_i^2$ (using idempotence). Under $H_0 : \tau_i = 0$, $\boldsymbol{\xi} = \mathbf{0}$.

③ By Theorem 3.12 with $\mathbf{A} = (\mathbf{I}-\mathbf{H})/\sigma^2$, $\mathbf{B} = (\mathbf{H} - \frac{1}{N}\mathbf{J}_N)/\sigma^2$:

$$\sigma^2\mathbf{A}\mathbf{B} = \sigma^2 \cdot \frac{\mathbf{I}-\mathbf{H}}{\sigma^2} \cdot \frac{\mathbf{H} - \frac{1}{N}\mathbf{J}_N}{\sigma^2} = \frac{1}{\sigma^2}\Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N - \mathbf{H}\mathbf{H} + \tfrac{1}{N}\mathbf{H}\mathbf{J}_N\Big) = \frac{1}{\sigma^2}\Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N - \mathbf{H} + \tfrac{1}{N}\mathbf{J}_N\Big) = \mathbf{0}.$$

④ The ratio of two independent chi-squares, each divided by its df, is F:

$$F_0 = \frac{MS_{\text{Treatments}}}{MS_E} = \frac{\dfrac{\mathbf{y}^T(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)\mathbf{y}}{\sigma^2}\Big/(a-1)}{\dfrac{\mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}}{\sigma^2}\Big/(N-a)} \sim F_{a-1,\,N-a,\,\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)\mathbf{X}\boldsymbol{\beta}/\sigma^2}.$$

■

Derivation: EMS via Theorem 3.13

With $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, $\mathbf{V} = \sigma^2\mathbf{I}$:

$$\begin{aligned}
\mathrm{E}(SS_E) &= \mathrm{E}(\mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}) = \operatorname{tr}((\mathbf{I}-\mathbf{H})\sigma^2\mathbf{I}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta} \\
&= \sigma^2(\operatorname{tr}(\mathbf{I}) - \operatorname{tr}(\mathbf{H})) + 0 = \sigma^2(N-a), \\
&\Rightarrow\ \mathrm{E}(MS_E) = \mathrm{E}\Big(\frac{SS_E}{N-a}\Big) = \sigma^2. \\
\mathrm{E}(SS_{\text{Treatments}}) &= \operatorname{tr}\Big(\big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\big)\sigma^2\mathbf{I}\Big) \\
&\quad + \boldsymbol{\beta}^T\mathbf{X}^T\Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\Big)^T\Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\Big)\mathbf{X}\boldsymbol{\beta}
\end{aligned}$$

$$= \sigma^2(a-1) + \boldsymbol{\xi}^T\boldsymbol{\xi} = \sigma^2(a-1) + n\sum_{i=1}^{a}\tau_i^2,$$

$$\Rightarrow \ \mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \frac{n\sum_{i=1}^{a}\tau_i^2}{a-1}.$$

This reproduces the elementwise result.

9.7 Moments of fitted values and residuals

Theorem 3.15

Using $\mathrm{Cov}(\mathbf{Ax}, \mathbf{By}) = \mathbf{A}\mathrm{Cov}(\mathbf{x}, \mathbf{y})\mathbf{B}^T$:

$$\begin{aligned}
\mathrm{E}(\mathbf{e}) &= (\mathbf{I}-\mathbf{H})\mathrm{E}(\mathbf{y}) = (\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta} = (\mathbf{X}-\mathbf{X})\boldsymbol{\beta} = \mathbf{0},\\
\mathrm{Var}(\mathbf{e}) &= \mathrm{Cov}((\mathbf{I}-\mathbf{H})\mathbf{y}, (\mathbf{I}-\mathbf{H})\mathbf{y}) = (\mathbf{I}-\mathbf{H})\sigma^2\mathbf{I}(\mathbf{I}-\mathbf{H})^T = \sigma^2(\mathbf{I}-\mathbf{H}),\\
\mathrm{Cov}(\mathbf{e}, \hat{\mathbf{y}}) &= \mathrm{Cov}((\mathbf{I}-\mathbf{H})\mathbf{y}, \mathbf{H}\mathbf{y}) = (\mathbf{I}-\mathbf{H})\sigma^2\mathbf{I}\,\mathbf{H}^T = \sigma^2(\mathbf{H}-\mathbf{HH}) = \mathbf{0},\\
\mathrm{Var}(\hat{\mathbf{y}}) &= \mathbf{H}\sigma^2\mathbf{I}\mathbf{H}^T = \sigma^2\mathbf{H}, \qquad \mathrm{E}(\mathbf{e}^T\mathbf{e}) = \mathrm{tr}(\sigma^2(\mathbf{I}-\mathbf{H})) = \sigma^2(N-a).
\end{aligned}$$

Under normality, $\mathrm{Cov}(\mathbf{e}, \hat{\mathbf{y}}) = \mathbf{0}$ means residuals and fitted values are independent — the basis of the "residuals vs fitted" plot. Note also $\sum_{\ell=1}^{N}\mathrm{Var}(\hat{y}_\ell) = \sigma^2\mathrm{tr}(\mathbf{H}) = a\sigma^2$.

Remark 3.16: SS decomposition in matrix form

$SS_T = \mathbf{y}^T(\mathbf{I} - \frac{1}{N}\mathbf{J}_N)\mathbf{y} = \mathbf{y}^T(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)\mathbf{y} + \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y} = SS_{\text{Treatments}} + SS_E$, with $\chi^2_{a-1,\lambda}$ and χ^2_{N-a} components ($\lambda = \boldsymbol{\xi}^T\boldsymbol{\xi}/\sigma^2$).

10 Model Adequacy Checking

Residuals $e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{i\cdot}$.

- **Normality**: normal probability (Q–Q) plot of residuals.
- **Constant variance**: residuals vs fitted values $\hat{y}_{ij}$; or formal tests (Bartlett, Levene).
- **Independence**: residuals vs time order of data collection.

Derivation: constructing a normal probability plot

Given $z_1, \ldots, z_n$; is $z_i \sim N(\mu, \sigma^2)$?

1. Order: $z_{(1)} \le \cdots \le z_{(n)}$.
2. Consider a normal sample of size n with mean 0 and unit variance; let $u_{(1)} \le \cdots \le u_{(n)}$ be the mean values of the order statistics over (many) samples. Approximately
$$u_{(i)} \approx \Phi^{-1}\Big(\frac{i-3/8}{n+1/4}\Big) \quad \text{(Blom)}.$$
3. If z is normal, $z_{(i)} = \mu + \sigma\cdot$(standard normal order stat), so $\mathrm{E}(z_{(i)}) = \mu + \sigma u_{(i)}$: the plot of $z_{(i)}$ vs $u_{(i)}$ is a straight line with intercept μ and slope σ.

11 Statistical Tests for Equality of Variance

$H_0 : \sigma_1^2 = \cdots = \sigma_a^2$ vs H_1: not all equal.

11.1 Bartlett's test

$$\chi_0^2 = 2.3026\,\frac{q}{c}, \qquad q = (N-a)\log_{10} S_p^2 - \sum_{i=1}^{a}(n_i-1)\log_{10} S_i^2,$$

$$c = 1 + \frac{1}{3(a-1)}\Big(\sum_{i=1}^{a}(n_i-1)^{-1} - (N-a)^{-1}\Big),$$

$S_p^2 = \sum_i (n_i-1)S_i^2/(N-a)$. Reject H_0 if $\chi_0^2 > \chi^2_{\alpha,a-1}$.

Derivation: why q measures inequality

Let $A = \frac{1}{\sum_i (n_i-1)}\sum_i (n_i-1)S_i^2 = S_p^2$ (weighted *arithmetic* mean) and $B = \big[(S_1^2)^{n_1-1}\cdots(S_a^2)^{n_a-1}\big]^{1/(N-a)}$ (weighted *geometric* mean). By the AM–GM inequality $A \geq B$, with equality iff all S_i^2 are equal. Then

$$(N-a)\log\frac{A}{B} = (N-a)(\log A - \log B) = (N-a)\log S_p^2 - \sum_i (n_i-1)\log S_i^2 = q \;\geq 0,$$

and q is large when the sample variances are very different. ($2.3026 = \ln 10$ converts $\log_{10}$ to natural logs.)

Example 3.17: Tensile data

$S_i^2 = 11.2,\ 9.8,\ 4.3,\ 6.8,\ 8.2$; $S_p^2 = MS_E = 8.06$.

$$\begin{aligned} q &= 20\log_{10} 8.06 - 4(\log_{10} 11.2 + \log_{10} 9.8 + \log_{10} 4.3 + \log_{10} 6.8 + \log_{10} 8.2) \\ &= 20(0.9063) - 4(1.0492 + 0.9912 + 0.6335 + 0.8325 + 0.9138) = 18.127 - 17.681 = 0.446, \\ c &= 1 + \frac{1}{3\cdot 4}\Big(\frac{5}{4} - \frac{1}{20}\Big) = 1 + \frac{1.2}{12} = 1.10, \qquad \chi_0^2 = 2.3026\frac{0.446}{1.10} = 0.933. \end{aligned}$$

$0.933 < \chi^2_{0.05,4} = 9.49$: fail to reject equal variances.

11.2 Hartley's test (balanced, $n_i = n$)

$H = \max S_i^2 / \min S_i^2$; reject if $H > H(a, n; 1-\alpha)$. Tensile: $H = 11.2/4.3 = 2.60$.

Bartlett and Hartley require normality and are very sensitive to departures from it.

11.3 Levene's test (robust)

$$L = \frac{\sum_{i=1}^{a} n_i(\bar z_{i\cdot} - \bar z_{\cdot\cdot})^2/(a-1)}{\sum_{i=1}^{a}\sum_{j=1}^{n_i}(z_{ij} - \bar z_{i\cdot})^2/(N-a)}, \qquad z_{ij} = |y_{ij} - \bar y_{i\cdot}|,$$

i.e. a one-way ANOVA on z_{ij}; reject if $L > F_{\alpha,a-1,N-a}$. **Modified Levene**: $z_{ij} = |y_{ij} - \text{median}(y_{i\cdot})|$. Implementation:

(1) compute z_{ij};

(2) run a one-way ANOVA. (Tensile: P-value ≈ 1, fail to reject.)

12 Post-ANOVA Comparison of Means

If H_0 is rejected, we do not know *which* means differ: the **multiple comparisons** problem.

12.1 Family-wise error rate

Derivation

$\alpha_{FW} = P(\text{at least one type I error} \mid \text{all } H_{0i} \text{ true})$. If each of r tests is done at level α and the tests are independent,

$$P(\text{no family-wise error}) = (1-\alpha)^r \;\Rightarrow\; \alpha_{FW} = 1-(1-\alpha)^r.$$

E.g. $r=3$, $\alpha=0.05$: $\alpha_{FW} = 1-0.95^3 = 0.1426 < r\alpha = 0.15$. In general (independence not needed, by Bonferroni/Boole) $\alpha_{FW} \le r\alpha$. **Bonferroni**: test each H_{0i} at α/r, i.e. reject H_{0i} if $p_i < \alpha/r$; then $\alpha_{FW} \le \alpha$.

Proof of $1-(1-\alpha)^r \le r\alpha$ **[Supplement].** By induction: $r=1$ is equality. If $(1-\alpha)^r \ge 1-r\alpha$ then $(1-\alpha)^{r+1} \ge (1-r\alpha)(1-\alpha) = 1-(r+1)\alpha + r\alpha^2 \ge 1-(r+1)\alpha$. Without independence, Boole: $P(\cup_i E_i) \le \sum_i P(E_i) = r\alpha$. ■

With 5 treatments there are $\binom{5}{2} = 10$ pairwise intervals; naive 95% intervals have simultaneous confidence only $\ge 1-10(0.05) = 0.5$.

12.2 Contrasts

Definition 3.18

A **contrast** is $\Gamma = \sum_{i=1}^{a} c_i\mu_i$ with $\sum_i c_i = 0$. E.g. $\Gamma = \mu_1+\mu_2-\mu_3-\mu_4$; $\Gamma = \mu_1-\mu_4$ ($H_0: \Gamma = 0 \iff \mu_1 = \mu_4$).

Estimator $C = \sum_i c_i \bar{y}_{i\cdot}$:

$$\mathrm{E}(C) = \sum_i c_i \mathrm{E}(\bar{y}_{i\cdot}) = \sum_i c_i\mu_i = \Gamma, \qquad \mathrm{Var}(C) = \sum_i c_i^2 \mathrm{Var}(\bar{y}_{i\cdot}) = \frac{\sigma^2}{n}\sum_i c_i^2 \quad (n_i = n).$$

With MS_E for σ^2, $\dfrac{C-\Gamma}{\sqrt{MS_E \sum c_i^2/n}} \sim t_{N-a}$, giving the $100(1-\alpha)\%$ CI

$$\sum_i c_i \bar{y}_{i\cdot} \pm t_{\alpha/2,N-a}\sqrt{\frac{MS_E}{n}\sum_i c_i^2}.$$

Note: $\sum c_i = 0$ is what makes μ cancel: $\sum c_i\mu_i = \sum c_i(\mu+\tau_i) = \sum c_i\tau_i$.

12.3 Scheffé's method (all contrasts)

Scheffé's method gives simultaneous $100(1-\alpha)\%$ intervals for *all* contrasts:

$$C \pm S_\alpha, \qquad S_\alpha = \sqrt{(a-1)F_{\alpha,a-1,N-a}}\sqrt{\frac{MS_E}{n}\sum_i c_i^2}.$$

Reject $H_0 : \Gamma = 0$ if $|C| > S_\alpha$. Since S_α tends to be large, Scheffé is **conservative** (tends not to reject).

Example 3.19: Scheffé on tensile data

$\Gamma_1 = \mu_1 + \mu_3 - \mu_4 - \mu_5$, $\Gamma_2 = \mu_1 - \mu_4$.

$$C_1 = 9.8 + 17.6 - 21.6 - 10.8 = -5.0, \quad C_2 = 9.8 - 21.6 = -11.8,$$
$$\sqrt{(a-1)F_{0.05,4,20}} = \sqrt{4(2.87)} = 3.39,$$
$$S_{0.05,1} = 3.39\sqrt{\tfrac{8.06}{5} \cdot 4} = 3.39(2.54) = 8.61,$$
$$S_{0.05,2} = 3.39\sqrt{\tfrac{8.06}{5} \cdot 2} = 3.39(1.80) = 6.09.$$

$|C_1| = 5.0 < 8.61$: Γ_1 not significant. $|C_2| = 11.8 > 6.09$: $\mu_1 \neq \mu_4$.

12.4 Tukey's test (all pairs)

Studentized range $q = \dfrac{\bar{y}_{\max} - \bar{y}_{\min}}{\sqrt{MS_E/n}}$ with distribution $q(a, f)$, f = error df. Two means differ significantly if

$$|\bar{y}_{i\cdot} - \bar{y}_{j\cdot}| > T_\alpha = q_\alpha(a,f)\sqrt{\frac{MS_E}{n}} \qquad \Big(\text{unbalanced: } \frac{q_\alpha(a,f)}{\sqrt{2}}\sqrt{MS_E\Big(\frac{1}{n_i} + \frac{1}{n_j}\Big)}\Big).$$

Overall error rate is exactly α for all pairwise comparisons (balanced case).

12.5 Fisher's least significant difference (LSD)

$$|\bar{y}_{i\cdot} - \bar{y}_{j\cdot}| > LSD = t_{\alpha/2,N-a}\sqrt{\frac{2MS_E}{n}} \qquad \Big(\text{unbalanced: } t_{\alpha/2,N-a}\sqrt{MS_E\Big(\frac{1}{n_i} + \frac{1}{n_j}\Big)}\Big).$$

Controls error per comparison only (least conservative).

12.6 Duncan's multiple range test

Least significant ranges $R_p = r_\alpha(p, f)\sqrt{MS_E/n}$, $p = 2, \ldots, a$. Rank the means ascending $\bar{y}_{(1)} < \cdots < \bar{y}_{(a)}$; compare $\bar{y}_{(a)} - \bar{y}_{(1)}$ with R_a, $\bar{y}_{(a)} - \bar{y}_{(2)}$ with R_{a-1}, $\ldots$, $\bar{y}_{(a)} - \bar{y}_{(a-1)}$ with R_2; then $\bar{y}_{(a-1)} - \bar{y}_{(1)}$ with R_{a-1}, etc. (a difference of means p steps apart in rank is compared with R_p). Significant if the difference exceeds the corresponding R_p.

Example 3.20: All pairwise comparisons for the tensile data

$\sqrt{MS_E/n} = \sqrt{8.06/5} = 1.270$.

- Tukey: $T_{0.05} = q_{0.05}(5, 20)\sqrt{8.06/5} = 4.24(1.270) = 5.38$.
- LSD: $t_{0.025,20}\sqrt{2(8.06)/5} = 2.086(1.796) = 3.75$.
- Duncan: $r_{0.05}(p, 20) = 2.95, 3.10, 3.18, 3.25$ for $p = 2, \ldots, 5 \Rightarrow R_2 = 3.75$, $R_3 = 3.94$, $R_4 = 4.04$, $R_5 = 4.13$.

Ordered means: $\bar{y}_1 = 9.8 < \bar{y}_5 = 10.8 < \bar{y}_2 = 15.4 < \bar{y}_3 = 17.6 < \bar{y}_4 = 21.6$.

Pair	|diff|	Tukey (5.38)	LSD (3.75)	Duncan (R_p)
4 vs 1	11.8	*	*	$> R_5 = 4.13$ *
4 vs 5	10.8	*	*	$> R_4 = 4.04$ *
4 vs 2	6.2	*	*	$> R_3 = 3.94$ *
4 vs 3	4.0	–	*	$> R_2 = 3.75$ *
3 vs 1	7.8	*	*	$> R_4$ *
3 vs 5	6.8	*	*	$> R_3$ *
3 vs 2	2.2	–	–	$< R_2$
2 vs 1	5.6	*	*	$> R_3$ *
2 vs 5	4.6	–	*	$> R_2$ *
5 vs 1	1.0	–	–	$< R_2$

LSD and Duncan agree: only (1,5) and (2,3) are not different. Tukey, being more conservative, also fails to separate (3,4) and (2,5). Level 30% (treatment 4) gives the largest strength.

Remark 3.21: Conservativeness ranking

From hardest to reject H_0 (most conservative) to easiest to reject H_0 (most liberal):

$$\text{Scheffé} \succ \text{Tukey} \succ \text{SNK} \succ \text{Duncan} \succ \text{Fisher's LSD}.$$

13 Unbalanced Data

If treatment i has n_i observations, $N = \sum_i n_i$; only the SS formulas change:

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{\cdot\cdot})^2, \quad SS_{\text{Treatments}} = \sum_{i=1}^{a} n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2, \quad SS_E = \sum_{i=1}^{a}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i\cdot})^2,$$

with hand formulas $SS_T = \sum\sum y_{ij}^2 - y_{\cdot\cdot}^2/N$, $SS_{\text{Treatments}} = \sum_i y_{i\cdot}^2/n_i - y_{\cdot\cdot}^2/N$. (The cross product in the identity still vanishes because $\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$ for each i.) With the weighted constraint $\sum n_i\tau_i = 0$, $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \sum_i n_i\tau_i^2/(a-1)$.

14 Transformations to Stabilize Variance

Derivation: delta-method derivation

Suppose $\mathrm{Var}(y_i) = w(\mu_i)$, $\mu_i = \mathrm{E}(y_i)$. Find h such that $\mathrm{Var}(h(y))$ is (approximately) constant. First-order Taylor expansion about μ:

$$h(y) = h(\mu) + (y-\mu)h'(\mu) + \cdots \;\Rightarrow\; \mathrm{Var}(h(y)) \approx h'(\mu)^2\, w(\mu) \overset{\text{set}}{=} K.$$

Hence $h'(\mu) = \dfrac{\sqrt{K}}{\sqrt{w(\mu)}}$, i.e. $h(\mu) \propto \displaystyle\int \frac{d\mu}{\sqrt{w(\mu)}}$.

- $\sigma_i^2 \propto \mu_i$ ($w(\mu) = \mu$, Poisson-like): $h(\mu) = \int \mu^{-1/2} d\mu = 2\sqrt{\mu} \Rightarrow y^* = \sqrt{y}$.
- $\sigma_i \propto \mu_i$ ($w(\mu) = \mu^2$): $h(\mu) = \int \mu^{-1} d\mu = \log\mu \Rightarrow y^* = \log y$.
- $\sigma_i \propto \mu_i^2$ ($w(\mu) = \mu^4$): $h(\mu) = \int \mu^{-2} d\mu = -1/\mu \Rightarrow y^* = 1/y$.

In general if $\sigma \propto \mu^{\alpha}$, use $y^{*} = y^{1-\alpha}$ ($\log y$ when $\alpha = 1$). Diagnose with a funnel shape in the residuals-vs-fitted plot.

15 Nonparametric Method: Kruskal–Wallis Test

For H_0: the a treatment distributions are identical (location: $\mu_1 = \cdots = \mu_a$) when normality fails.

1. Rank all N observations jointly in ascending order; replace y_{ij} by its rank R_{ij} (average ranks for ties).
2. Compute
$$H = \frac{1}{S^2}\Big[\sum_{i=1}^{a}\frac{R_{i\cdot}^2}{n_i} - \frac{N(N+1)^2}{4}\Big], \qquad S^2 = \frac{1}{N-1}\Big[\sum_{i=1}^{a}\sum_{j=1}^{n_i} R_{ij}^2 - \frac{N(N+1)^2}{4}\Big].$$
3. Reject H_0 if $H > \chi^2_{\alpha,a-1}$ (large-sample approximation).

Derivation: no-tie form

Without ties the ranks are $1,\dots,N$; $\bar{R}_{\cdot\cdot} = \frac{N+1}{2}$ and

$$S^2 = \frac{1}{N-1}\Big[\sum_{k=1}^{N} k^2 - N\Big(\frac{N+1}{2}\Big)^2\Big] = \frac{1}{N-1}\Big[\frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)^2}{4}\Big] = \frac{N(N+1)}{12},$$

so

$$H = \frac{12}{N(N+1)}\sum_{i=1}^{a} n_i\big(\bar{R}_{i\cdot} - \bar{R}_{\cdot\cdot}\big)^2 = \frac{12}{N(N+1)}\sum_{i=1}^{a} n_i\Big(\bar{R}_{i\cdot} - \frac{N+1}{2}\Big)^2.$$

(The bracket: $\frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)^2}{4} = \frac{N(N+1)\,[2(2N+1)-3(N+1)]}{12} = \frac{N(N+1)(N-1)}{12}$.) And $\sum_i \frac{R_{i\cdot}^2}{n_i} - \frac{N(N+1)^2}{4} = \sum_i n_i(\bar{R}_{i\cdot} - \frac{N+1}{2})^2$, which is $SS_{\text{Treatments}}$ computed on the ranks.

Example 3.22: Tensile data

Rank sums $R_{i\cdot} = 27.5,\ 66,\ 85,\ 113,\ 33.5$ ($N = 25$, ties present). $\sum\sum R_{ij}^2 = 5510$, $N(N+1)^2/4 = 4225$:

$$S^2 = \frac{5510 - 4225}{24} = 53.54,$$
$$\sum_i \frac{R_{i\cdot}^2}{5} = \frac{27.5^2 + 66^2 + 85^2 + 113^2 + 33.5^2}{5} = \frac{26228.5}{5} = 5245.7,$$
$$H = \frac{5245.7 - 4225}{53.54} = 19.06 > \chi^2_{0.05,4} = 9.49 \quad (P = 0.00076).$$

Reject H_0 — same conclusion as the F-test.

16 Summary of Chapter 3

Item	Result
LSE	$\hat{\mu} = \bar{y}_{\cdot\cdot}$, $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $\hat{y}_{ij} = \bar{y}_{i\cdot}$
SS	$SS_T = SS_{\text{Trt}} + SS_E$, df $N-1 = (a-1) + (N-a)$
EMS	$\text{E}MS_E = \sigma^2$, $\text{E}MS_{\text{Trt}} = \sigma^2 + n\sum \tau_i^2/(a-1)$
Matrix	$SS_E = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$, $SS_{\text{Trt}} = \mathbf{y}^T(\mathbf{H} - \mathbf{J}/N)\mathbf{y}$
Test	$F_0 = MS_{\text{Trt}}/MS_E \sim F_{a-1,N-a}$ under H_0
Contrasts	$\sum c_i \bar{y}_{i\cdot} \pm t_{\alpha/2,N-a}\sqrt{MS_E \sum c_i^2/n}$

Chapter 4. Randomized Blocks, Latin Squares, and Related Designs

1 Introduction and the Blocking Principle

- Topics: blocking and nuisance factors; the randomized complete block design (RCBD); extension of ANOVA to the RCBD; other blocking scenarios — Latin squares, Graeco-Latin squares, balanced incomplete block designs (BIBD).
- **Blocking** is a technique for dealing with **nuisance factors**: factors that affect the response but are of no interest to the experimenter; the variability they transmit must be minimized.
- Typical nuisance factors: batches of raw material, operators, pieces of test equipment, time (shifts, days), different experimental units. Failure to block is a common flaw in designing an experiment.

Nuisance factor is …	Strategy
known and controllable	**blocking**
known but uncontrollable (measurable)	**analysis of covariance** (Ch. 11)
unknown and uncontrollable (lurking)	**randomization** (hope it balances out)

2 Hardness Testing Example

- Do four different tips produce different mean hardness readings on a Rockwell tester?
- Tips are applied to test coupons. A CRD would assign tips to coupons at random, but coupon hardness is nuisance variability (and tips should be tested across hardnesses).
- **RCBD**: assign *all four tips to each coupon.* Each coupon is a **block**: large variability *between*, small *within.* Each block holds a complete replicate; randomization is restricted to *within* blocks.

Tip	Test coupon (block) 1	2	3	4	$y_{i\cdot}$
1	9.3	9.4	9.6	10.0	38.3
2	9.4	9.3	9.8	9.9	38.4
3	9.2	9.4	9.5	9.7	37.8
4	9.7	9.6	10.0	10.2	39.5
$y_{\cdot j}$	37.6	37.7	38.9	39.8	$y_{\cdot\cdot} = 154.0$

The data have a two-way structure; we test equality of treatment means after removing block variability.

3 The RCBD Model and ANOVA

Motivation. Blocking compares treatments within homogeneous groups, removing nuisance variation from the error.

3.1 Model

Definition 4.1: RCBD effects model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad i = 1, \dots, a;\ j = 1, \dots, b,$$

$\sum_i \tau_i = 0$, $\sum_j \beta_j = 0$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Means model: $y_{ij} = \mu_{ij} + \varepsilon_{ij}$, $\mu_{ij} = \mu + \tau_i + \beta_j$ (additive: no treatment×block interaction).

$H_0 : \mu_1 = \cdots = \mu_a$ $(\mu_i = \frac{1}{b}\sum_j \mu_{ij} = \mu + \tau_i)$ $\iff$ $H_0 : \tau_1 = \cdots = \tau_a = 0$.

3.2 Partition of the total sum of squares

Theorem 4.2

$$\sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{\cdot\cdot})^2 = b\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + a\sum_{j=1}^{b}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 + \sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$$

$$SS_T = SS_{\text{Treatments}} + SS_{\text{Blocks}} + SS_E,$$

$$ab - 1 = (a-1) + (b-1) + (a-1)(b-1).$$

Proof [Supplement]. Write $y_{ij} - \bar{y}_{\cdot\cdot} = A_i + B_j + C_{ij}$ with $A_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $B_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $C_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. Squaring and summing gives the three squares plus three cross products, each of which vanishes:

$$\sum_i\sum_j A_iB_j = \Big(\sum_i A_i\Big)\Big(\sum_j B_j\Big) = 0 \cdot 0 = 0,$$

$$\sum_i\sum_j A_iC_{ij} = \sum_i A_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}) = \sum_i A_i(b\bar{y}_{i\cdot} - b\bar{y}_{i\cdot} - b\bar{y}_{\cdot\cdot} + b\bar{y}_{\cdot\cdot}) = 0,$$

$$\sum_i\sum_j B_jC_{ij} = \sum_j B_j \sum_i (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}) = \sum_j B_j(a\bar{y}_{\cdot j} - a\bar{y}_{\cdot\cdot} - a\bar{y}_{\cdot j} + a\bar{y}_{\cdot\cdot}) = 0.$$

Since A_i does not depend on j, $\sum_i\sum_j A_i^2 = b\sum_i A_i^2$; similarly $\sum\sum B_j^2 = a\sum_j B_j^2$. Degrees of freedom of SS_E: the ab residuals C_{ij} satisfy $\sum_i C_{ij} = 0$ (b constraints) and $\sum_j C_{ij} = 0$ (a constraints, one redundant), so df $= ab - (a + b - 1) = (a-1)(b-1)$. ■

Source	SS	DF	MS	F_0
Treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Blocks	SS_{Blocks}	$b-1$	MS_{Blocks}	
Error	SS_E	$(a-1)(b-1)$	MS_E	
Total	SS_T	$N-1$		

Reject H_0 if $F_0 > F_{\alpha, a-1, (a-1)(b-1)}$.

Derivation: what about $H_0 : \beta_j = 0$?

Testing blocks is generally *not* of interest (they were expected to differ); under the randomization restriction MS_{Blocks}/MS_E is only an approximate indicator, and a large value means blocking was worthwhile.

3.3 Expected mean squares (elementwise)

Theorem 4.3

$$\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \frac{b\sum_i \tau_i^2}{a-1}, \qquad \mathrm{E}(MS_{\text{Blocks}}) = \sigma^2 + \frac{a\sum_j \beta_j^2}{b-1}, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Treatments [Supplement]. $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\beta}_\cdot + \bar{\varepsilon}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$ (since $\sum_j \beta_j = 0$), and $\bar{y}_{\cdot\cdot} = \mu + \bar{\varepsilon}_{\cdot\cdot}$. Hence $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})$ — the block effects cancel. Then

$$\begin{aligned}
\mathrm{E}(MS_{\text{Treatments}}) &= \frac{b}{a-1}\mathrm{E}\Big(\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2\Big) = \frac{b}{a-1}\mathrm{E}\Big(\sum_i \big(\tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})\big)^2\Big) \\
&= \frac{b}{a-1}\Big[\sum_i \tau_i^2 + \mathrm{E}\sum_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2 + 2\underbrace{\mathrm{E}\sum_i \tau_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})}_{=0}\Big] \\
&= \frac{b}{a-1}\sum_i \tau_i^2 + \frac{b}{a-1}\cdot(a-1)\underbrace{\mathrm{E}\Big[\frac{\sum_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2}{a-1}\Big]}_{=\mathrm{Var}(\bar{\varepsilon}_{i\cdot}) = \sigma^2/b} = \sigma^2 + \frac{b}{a-1}\sum_{i=1}^{a}\tau_i^2.
\end{aligned}$$

By symmetry (swap the roles of i, a, τ and j, b, β), $\mathrm{E}(MS_{\text{Blocks}}) = \sigma^2 + \frac{a}{b-1}\sum_j \beta_j^2$. ■

Error [Supplement]. Substituting the model, μ, τ_i, β_j all cancel:

$$y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot j} + \bar{\varepsilon}_{\cdot\cdot} = (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}) - (\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}).$$

So

$$SS_E = \sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2 + \sum\sum(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2 - 2\sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}).$$

① For each i, $\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2/(b-1)$ is a sample variance of b iid errors:

$$\mathrm{E}\Big[\sum_i\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2\Big] = \sum_{i=1}^{a}\mathrm{E}\Big[\frac{\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2}{b-1}\Big](b-1) = a(b-1)\sigma^2.$$

② $\bar{\varepsilon}_{\cdot 1}, \ldots, \bar{\varepsilon}_{\cdot b}$ are iid with variance σ^2/a:

$$\mathrm{E}\Big[\sum_i\sum_j(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2\Big] = a\,\mathrm{E}\Big[\frac{\sum_j(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2}{b-1}\Big](b-1) = a\cdot\frac{\sigma^2}{a}(b-1) = (b-1)\sigma^2.$$

③ Expand the cross product into four pieces:

$$\sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}) = \underbrace{\sum\sum\varepsilon_{ij}\bar{\varepsilon}_{\cdot j}}_{(i)} - \underbrace{\sum\sum\varepsilon_{ij}\bar{\varepsilon}_{\cdot\cdot}}_{(ii)} - \underbrace{\sum\sum\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j}}_{(iii)} + \underbrace{\sum\sum\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot\cdot}}_{(iv)}.$$

(i) $\sum_j \bar{\varepsilon}_{\cdot j}\sum_i \varepsilon_{ij} = a\sum_j \bar{\varepsilon}_{\cdot j}^2$, so $\mathrm{E}(i) = a\cdot b\cdot\frac{\sigma^2}{a} = b\sigma^2$.

(ii) $\bar{\varepsilon}_{\cdot\cdot}\sum\sum\varepsilon_{ij} = ab\,\bar{\varepsilon}_{\cdot\cdot}^2$, so $\mathrm{E}(ii) = ab\cdot\frac{\sigma^2}{ab} = \sigma^2$.

(iii) $\sum_i\sum_j \bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j} = (\sum_i \bar{\varepsilon}_{i\cdot})(\sum_j \bar{\varepsilon}_{\cdot j}) = a\bar{\varepsilon}_{\cdot\cdot}\cdot b\bar{\varepsilon}_{\cdot\cdot}$, so $\mathrm{E}(iii) = ab\frac{\sigma^2}{ab} = \sigma^2$. (Directly: $\mathrm{E}(\bar{\varepsilon}_{1\cdot}\bar{\varepsilon}_{\cdot 1}) = \mathrm{E}\big[\frac{1}{b}(\varepsilon_{11} + \cdots + \varepsilon_{1b})\frac{1}{a}(\varepsilon_{11} + \cdots + \varepsilon_{a1})\big] = \frac{1}{ab}\mathrm{E}(\varepsilon_{11}^2) = \frac{\sigma^2}{ab}$, and there are ab such terms.)

(iv) Same as (ii) by symmetry: σ^2.

Thus E③ $= b\sigma^2 - \sigma^2 - \sigma^2 + \sigma^2 = (b-1)\sigma^2$, and

$$\mathrm{E}(SS_E) = a(b-1)\sigma^2 + (b-1)\sigma^2 - 2(b-1)\sigma^2 = (a-1)(b-1)\sigma^2 \;\Rightarrow\; \mathrm{E}(MS_E) = \sigma^2.$$

■

Remark 4.4: Matrix shortcut

With $\mathbf{H}$ the hat matrix of the additive model ($\operatorname{rank}\mathbf{X} = 1 + (a-1) + (b-1) = a + b - 1$), $SS_E = (\mathbf{y} - \hat{\mathbf{y}})^T(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$ and by Theorem 3.13

$$\mathrm{E}(SS_E) = \sigma^2 \operatorname{tr}(\mathbf{I} - \mathbf{H}) + \underbrace{\boldsymbol{\beta}^T \mathbf{X}^T (\mathbf{I} - \mathbf{H}) \mathbf{X} \boldsymbol{\beta}}_{=0} = \sigma^2 (ab - (a + b - 1)) = (a-1)(b-1)\sigma^2.$$

For $a = b = 2$: $\mathbf{y} = (y_{11}, y_{12}, y_{21}, y_{22})^T$, $\boldsymbol{\beta} = (\mu, \tau_1, \beta_1)^T$, $\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}$; dropping the block column (if $\beta_1 = 0$) gives $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$, $\boldsymbol{\beta} = (\mu, \tau_1)^T$.

3.4 Least squares estimation

Derivation

$S = \sum_i \sum_j (y_{ij} - \mu - \tau_i - \beta_j)^2$. Setting $\partial S / \partial \theta|_{\hat{\theta}} = 0$:

$$\frac{\partial S}{\partial \mu}: \; -2 \sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j) = 0 \;\Rightarrow\; ab\hat{\mu} + b \sum \hat{\tau}_i + a \sum \hat{\beta}_j = y_{\cdot\cdot} \;\Rightarrow\; \hat{\mu} = \bar{y}_{\cdot\cdot},$$

$$\frac{\partial S}{\partial \beta_j}: \; -2 \sum_{i=1}^{a} (y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j) = 0 \;\Rightarrow\; a\hat{\mu} + \textstyle\sum_i \hat{\tau}_i + a\hat{\beta}_j = y_{\cdot j} \;\Rightarrow\; \hat{\beta}_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot},$$

$$\frac{\partial S}{\partial \tau_i}: \; -2 \sum_{j=1}^{b} (y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j) = 0 \;\Rightarrow\; b\hat{\mu} + b\hat{\tau}_i + \textstyle\sum_j \hat{\beta}_j = y_{i\cdot} \;\Rightarrow\; \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot},$$

using $\sum \hat{\tau}_i = \sum \hat{\beta}_j = 0$. Fitted values and residuals:

$$\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j = \bar{y}_{i\cdot} + \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}, \qquad e_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}.$$

Note $E(\hat{\tau}_i) = \mathrm{E}(\bar{y}_{i\cdot}) - \mathrm{E}(\bar{y}_{\cdot\cdot}) = \frac{1}{b}(b\mu + b\tau_i + \sum_j \beta_j) - \frac{1}{ab}(ab\mu + b\sum \tau_i + a \sum \beta_j) = \tau_i$: in a *complete* block design every treatment sees every block, so the block effects average out. (Contrast with the BIBD in Section 6.)

3.5 Extra sum of squares derivation of the F-test

Derivation: extra sum of squares method

Full model (F): $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$; Reduced model (R) under $H_0 : \tau_i = 0 \; \forall i$: $y_{ij} = \mu + \beta_j + \varepsilon_{ij}$.

$$SS_{E(F)} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2, \qquad SS_{E(R)} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot j})^2,$$

$$SS_H = SS_{E(R)} - SS_{E(F)}.$$

Theorem 4.5: $SS_H = SS_{\text{Treatments}}$

Proof [Supplement]. Write $y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{\cdot j}) - (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})$. Then

$$\begin{aligned} SS_{E(F)} &= \sum\sum (y_{ij} - \bar{y}_{\cdot j})^2 + \sum\sum (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2\sum\sum (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})(y_{ij} - \bar{y}_{\cdot j}) \\ &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \underbrace{\frac{\sum_j (y_{ij} - \bar{y}_{\cdot j})}{b}}_{=\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}} \\ &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_{E(R)} - b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2. \end{aligned}$$

Hence $SS_H = SS_{E(R)} - SS_{E(F)} = b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_{\text{Treatments}}$. ■

Matrix form. With $\mathbf{H}_F = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ and $\mathbf{H}_R = \mathbf{X}_R(\mathbf{X}_R^T\mathbf{X}_R)^{-1}\mathbf{X}_R^T$,

$$SS_{E(F)} = \mathbf{y}^T(\mathbf{I} - \mathbf{H}_F)\mathbf{y}, \qquad SS_{E(R)} = \mathbf{y}^T(\mathbf{I} - \mathbf{H}_R)\mathbf{y} \;\Rightarrow\; SS_H = \mathbf{y}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{y}.$$

Because $\Omega(\mathbf{X}_R) \subset \Omega(\mathbf{X})$, $\mathbf{H}_F\mathbf{X}_R = \mathbf{X}_R$, hence $\mathbf{H}_F\mathbf{H}_R = \mathbf{H}_R$ and (transposing) $\mathbf{H}_R\mathbf{H}_F = \mathbf{H}_R$. Then

$$(\mathbf{H}_F - \mathbf{H}_R)(\mathbf{H}_F - \mathbf{H}_R) = \mathbf{H}_F - \mathbf{H}_R - \mathbf{H}_R + \mathbf{H}_R = \mathbf{H}_F - \mathbf{H}_R \quad \text{(idempotent)},$$
$$\operatorname{tr}(\mathbf{H}_F - \mathbf{H}_R) = (a + b - 1) - b = a - 1.$$

- $\mathrm{E}(SS_H) = \operatorname{tr}((\mathbf{H}_F - \mathbf{H}_R)\sigma^2\mathbf{I}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = \sigma^2(a-1) + b\sum_i \tau_i^2$, because $\boldsymbol{\xi} = (\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = (\mathbf{X} - \mathbf{H}_R\mathbf{X})\boldsymbol{\beta} = (\underbrace{\tau_1, \ldots, \tau_1}_{b}, \ldots, \underbrace{\tau_a, \ldots, \tau_a}_{b})^T$ (projecting on the block space removes μ and β_j and averages τ_i within a block, which gives 0). Hence $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + b\sum \tau_i^2/(a-1)$.
- $SS_{E(F)}/\sigma^2 \sim \chi^2_{(a-1)(b-1)}$ (Theorem 3.11 with $\mathbf{A} = (\mathbf{I} - \mathbf{H}_F)/\sigma^2$); $SS_H/\sigma^2 \sim \chi^2_{a-1}$ under H_0 ($\mathbf{B} = (\mathbf{H}_F - \mathbf{H}_R)/\sigma^2$).
- Independence (Theorem 3.12): $\mathbf{A}\sigma^2\mathbf{I}\mathbf{B} = \frac{1}{\sigma^2}(\mathbf{I} - \mathbf{H}_F)(\mathbf{H}_F - \mathbf{H}_R) = \frac{1}{\sigma^2}(\mathbf{H}_F - \mathbf{H}_R - \mathbf{H}_F + \mathbf{H}_R) = \mathbf{0}$.

Therefore

$$F_0 = \frac{SS_H/(a-1)}{SS_{E(F)}/(a-1)(b-1)} = \frac{MS_{\text{Treatments}}}{MS_E} \overset{H_0}{\sim} F_{a-1,(a-1)(b-1)}.$$

Remark 4.6: R^2 never decreases when blocks are added

Let $R^2 = 1 - SS_E/SS_T$. The full RCBD model (1) and the reduced model $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ (2) share the same SS_T. The reduced model's column space is contained in the full model's, so

$$SS_{E(1)} = \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 \le \min_{\boldsymbol{\beta}_1} \|\mathbf{y} - \mathbf{X}_1\boldsymbol{\beta}_1\|^2 = SS_{E(2)}$$

(indeed $SS_{E(2)} - SS_{E(1)} = SS_{\text{Blocks}} \ge 0$ by the same argument as above). Hence $R^2_{(1)} \ge R^2_{(2)}$.

Remark 4.7: Omitting a real block effect biases the residuals

Suppose the truth is $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ (treatments + blocks) but we fit $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$. Then $\mathbf{e} = (\mathbf{I} - \mathbf{H}_1)\mathbf{y}$ and

$$\mathrm{E}(\mathbf{e}) = (\mathbf{I} - \mathbf{H}_1)(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2) = (\mathbf{I} - \mathbf{H}_1)\mathbf{X}_2\boldsymbol{\beta}_2.$$

$\mathbf{H}_1$ replaces each entry by its treatment average; the block effects average to $\frac{1}{b}\sum_j \beta_j = 0$ within every treatment, so $\mathbf{H}_1\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$ and $\mathrm{E}(e_{ij}) = \beta_j$. The block effects leak into the residuals and inflate SS_E (by $\mathrm{E}\,SS_{\text{Blocks}}$), destroying power.

Example 4.8: Hardness testing, line by line

$\sum\sum y_{ij}^2 = 1483.54$, $y_{..}^2/N = 154^2/16 = 1482.25$.

$$
\begin{aligned}
SS_T &= 1483.54 - 1482.25 = 1.29,\\
SS_{\text{Treatments}} &= \frac{38.3^2 + 38.4^2 + 37.8^2 + 39.5^2}{4} - 1482.25\\
&= \frac{5930.54}{4} - 1482.25 = 1482.635 - 1482.25 = 0.385,\\
SS_{\text{Blocks}} &= \frac{37.6^2 + 37.7^2 + 38.9^2 + 39.8^2}{4} - 1482.25 = \frac{5932.30}{4} - 1482.25 = 0.825,\\
SS_E &= 1.29 - 0.385 - 0.825 = 0.080.
\end{aligned}
$$

Source	SS	DF	MS	F_0
Tips	0.385	3	0.1283	14.44
Coupons	0.825	3	0.2750	(30.94)
Error	0.080	9	0.00889	
Total	1.290	15		

$F_0 = 14.44 > F_{0.05,3,9} = 3.86$ ($P = 0.00087$): tips differ.
Consequence of not blocking: a one-way ANOVA puts SS_{Blocks} into error: $SS_E = 0.080 + 0.825 = 0.905$ on $9 + 3 = 12$ df, $MS_E = 0.0754$, $F_0 = 0.1283/0.0754 = 1.70$ ($P = 0.22$): the tip effect is no longer detected!

3.6 Residual analysis

Residuals $e_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$: normal Q–Q plot; residuals vs treatment, vs block, vs fitted values. Levene's test can be applied between treatments ($z_{ij} = |y_{ij} - \tilde{y}_{i\cdot}|$) and between blocks ($z_{ij} = |y_{ij} - \tilde{y}_{\cdot j}|$); for the hardness data $P = 0.65$ and 0.68: no evidence of unequal variance.

4 Latin Square Design

4.1 Idea

- Controls (eliminates) **two** sources of nuisance variability simultaneously: rows and columns.
- Key assumption: the three factors (rows, columns, treatments) do **not interact**; otherwise the Latin square gives invalid results.
- A $p \times p$ Latin square: p treatments (Latin letters), two blocking variables with p levels each; each treatment appears *exactly once in each row and once in each column.*

Example with days (rows) and operators (columns):

	1	2	3
Mon	A	B	C
Tue	B	C	A
Wed	C	A	B

For $p = 3$ there are 12 distinct Latin squares (e.g. ABC/BCA/CAB, ACB/BAC/CBA, BAC/CBA/ACB, CBA/ACB/BAC, ...). In practice one chooses a square at random (randomize rows, columns and letter assignment).

4.2 Model

Definition 4.9

$$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}, \qquad i, j, k = 1, \ldots, p,$$

y_{ijk}: observation in row i, column k, treatment j; $\sum \alpha_i = \sum \tau_j = \sum \beta_k = 0$; $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Only p^2 of the p^3 index triples occur (the treatment index j is determined by (i, k)). The model is completely additive.

4.3 ANOVA decomposition

$$y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot k} + 2\bar{y}_{\cdots}).$$

Squaring and summing over the p^2 cells, all product terms drop out:

$$SS_T = \underbrace{p \sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2}_{SS_{\text{Rows}}} + \underbrace{p \sum_k (\bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots})^2}_{SS_{\text{Columns}}} + \underbrace{p \sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2}_{SS_{\text{Treatments}}} + SS_E,$$

$$p^2 - 1 = (p-1) + (p-1) + (p-1) + (p-2)(p-1).$$

Why the cross products vanish [Supplement]. Take e.g. rows $\times$ treatments: $\sum_{\text{cells}} (\bar{y}_{i\cdot\cdot} - \bar{y})(\bar{y}_{\cdot j\cdot} - \bar{y}) = \sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}) \sum_{\text{cells in row } i} (\bar{y}_{\cdot j\cdot} - \bar{y})$. Row i contains every treatment exactly once, so the inner sum is $\sum_{j=1}^{p} (\bar{y}_{\cdot j\cdot} - \bar{y}) = 0$. The same "each level of one factor meets each level of another exactly once" (**orthogonality**) kills every main-effect cross product. For a main effect times the residual C_{ijk}, note that the residuals sum to zero over every row, every column and every treatment (e.g. in row i: $\sum_k (y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot k} + 2\bar{y}) = p\bar{y}_{i\cdot\cdot} - p\bar{y}_{i\cdot\cdot} - p\bar{y} - p\bar{y} + 2p\bar{y} = 0$). Error df: $p^2 - 1 - 3(p-1) = (p-1)(p+1-3) = (p-1)(p-2)$. ■

Source	SS	DF	MS	F_0
Treatments	$SS_{\text{Treatments}}$	$p-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Rows	SS_{Rows}	$p-1$	MS_{Rows}	
Columns	SS_{Columns}	$p-1$	MS_{Columns}	
Error	SS_E	$(p-2)(p-1)$	MS_E	
Total	SS_T	p^2-1		

Reject $H_0 : \tau_1 = \cdots = \tau_p = 0$ if $F_0 > F_{\alpha, p-1, (p-2)(p-1)}$. As in the RCBD, $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + p\sum_j \tau_j^2/(p-1)$ and $\mathrm{E}(MS_E) = \sigma^2$ (same proof: $\bar{y}_{\cdot j\cdot} - \bar{y} = \tau_j + \bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}$ because each treatment meets every row and column once).

4.4 Least squares estimation

Derivation

$S(\theta) = \sum\sum\sum(y_{ijk} - \mu - \alpha_i - \beta_k - \tau_j)^2$. $\partial S/\partial\mu|_{\hat\theta} = -2\sum\sum\sum(y_{ijk} - \hat\mu - \hat\alpha_i - \hat\beta_k - \hat\tau_j) = 0 \Rightarrow p^2\hat\mu = y_{\cdots} \Rightarrow \hat\mu = \bar y_{\cdots}$. Similarly (using the zero-sum constraints and orthogonality)

$$\hat\alpha_i = \bar y_{i\cdot\cdot} - \bar y_{\cdots}, \quad \hat\tau_j = \bar y_{\cdot j\cdot} - \bar y_{\cdots}, \quad \hat\beta_k = \bar y_{\cdot\cdot k} - \bar y_{\cdots}, \qquad \hat y_{ijk} = \bar y_{i\cdot\cdot} + \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot k} - 2\bar y_{\cdots}.$$

Also $SS_T = SS_{\text{Model}} + SS_E$ with $SS_{\text{Model}} = SS_{\text{Rows}} + SS_{\text{Columns}} + SS_{\text{Treatments}}$.

E.g. for $\partial S/\partial\tau_j$: the p cells with treatment j lie in all p rows and all p columns, so $p\hat\mu + \sum_i \hat\alpha_i + p\hat\tau_j + \sum_k \hat\beta_k = y_{\cdot j\cdot} \Rightarrow \hat\tau_j = \bar y_{\cdot j\cdot} - \hat\mu$.

Example 4.10: Rocket propellant problem (5×5 Latin square)

Rows = batches of raw material, columns = operators, letters = formulations.

Batch	1	2	3	4	5	$y_{i\cdot\cdot}$
1	A=24	B=20	C=19	D=24	E=24	111
2	B=17	C=24	D=30	E=27	A=36	134
3	C=18	D=38	E=26	A=27	B=21	130
4	D=26	E=31	A=26	B=23	C=22	128
5	E=22	A=30	B=20	C=29	D=31	132
$y_{\cdot\cdot k}$	107	143	121	130	134	635

Treatment totals: $A = 143$, $B = 101$, $C = 112$, $D = 149$, $E = 130$.

$$\frac{y_{\cdots}^2}{N} = \frac{635^2}{25} = 16129, \qquad \sum y^2 = 16805, \qquad SS_T = 16805 - 16129 = 676,$$

$$SS_{\text{Batches}} = \frac{111^2 + 134^2 + 130^2 + 128^2 + 132^2}{5} - 16129 = \frac{80985}{5} - 16129 = 68,$$

$$SS_{\text{Operators}} = \frac{107^2 + 143^2 + 121^2 + 130^2 + 134^2}{5} - 16129 = \frac{81395}{5} - 16129 = 150,$$

$$SS_{\text{Formulations}} = \frac{143^2 + 101^2 + 112^2 + 149^2 + 130^2}{5} - 16129 = \frac{82295}{5} - 16129 = 330,$$

$$SS_E = 676 - 68 - 150 - 330 = 128.$$

Source	SS	DF	MS	F_0
Formulations	330	4	82.50	7.73
Batches	68	4	17.00	
Operators	150	4	37.50	
Error	128	12	10.67	
Total	676	24		

$F_0 = 82.5/10.67 = 7.73 > F_{0.05,4,12} = 3.26$ ($P = 0.0025$): formulations differ.
Fisher's LSD: $LSD = t_{0.025,12}\sqrt{2MS_E/p} = 2.179\sqrt{2(10.67)/5} = 2.179(2.066) = 4.50$.
Means: $\bar y_A = 28.6$, $\bar y_B = 20.2$, $\bar y_C = 22.4$, $\bar y_D = 29.8$, $\bar y_E = 26.0$.

Pair	|diff|	Sig.?	Pair	|diff|	Sig.?
A–B	8.4	*	B–D	9.6	*
A–C	6.2	*	B–E	5.8	*
A–D	1.2	–	C–D	7.4	*
A–E	2.6	–	C–E	3.6	–
B–C	2.2	–	D–E	3.8	–

Levene-type checks use $|y_{ijk} - \bar{y}_{i\cdot\cdot}|$ (rows), $|y_{ijk} - \bar{y}_{\cdot\cdot k}|$ (columns), $|y_{ijk} - \bar{y}_{\cdot j\cdot}|$ (treatments).

5 Graeco-Latin Square Design

- Controls **three** sources of nuisance variability: rows, columns and Greek letters.
- Superimpose two *orthogonal* Latin squares: one with Latin letters (treatments), one with Greek letters; each Greek letter appears exactly once in each row, once in each column, and exactly once with each Latin letter.
- Assumption: the four factors do not interact. (Graeco-Latin squares exist for every $p \geq 3$ except $p = 6$.)

Definition 4.11

$$y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}, \qquad i, j, k, l = 1, \ldots, p,$$

θ_i: row, τ_j: Latin treatment, ω_k: Greek letter, Ψ_l: column; all sum to zero.

$$y_{ijkl} - \bar{y} = (\bar{y}_{i\cdots} - \bar{y}) + (\bar{y}_{\cdot j\cdot\cdot} - \bar{y}) + (\bar{y}_{\cdot\cdot k\cdot} - \bar{y}) + (\bar{y}_{\cdots l} - \bar{y}) + (y_{ijkl} - \bar{y}_{i\cdots} - \bar{y}_{\cdot j\cdot\cdot} - \bar{y}_{\cdot\cdot k\cdot} - \bar{y}_{\cdots l} + 3\bar{y}),$$

$$SS_T = SS_{\text{Rows}} + SS_{\text{Columns}} + SS_{\text{Latin}} + SS_{\text{Greek}} + SS_E, \qquad p^2 - 1 = 4(p-1) + (p-3)(p-1).$$

Example 4.12: Rocket propellant with a Greek-letter factor (test assemblies)

Adding test assemblies (Greek letters) as a third blocking factor, with a standard orthogonal assignment

	1	2	3	4	5
1	$A\alpha$=24	$B\gamma$=20	$C\varepsilon$=19	$D\beta$=24	$E\delta$=24
2	$B\beta$=17	$C\delta$=24	$D\alpha$=30	$E\gamma$=27	$A\varepsilon$=36
3	$C\gamma$=18	$D\varepsilon$=38	$E\beta$=26	$A\delta$=27	$B\alpha$=21
4	$D\delta$=26	$E\alpha$=31	$A\gamma$=26	$B\varepsilon$=23	$C\beta$=22
5	$E\varepsilon$=22	$A\beta$=30	$B\delta$=20	$C\alpha$=29	$D\gamma$=31

Greek totals: $\alpha = 135$, $\beta = 119$, $\gamma = 122$, $\delta = 121$, $\varepsilon = 138$, so

$$SS_{\text{Greek}} = \frac{135^2 + 119^2 + 122^2 + 121^2 + 138^2}{5} - 16129 = \frac{80955}{5} - 16129 = 62,$$

and $SS_E = 676 - 68 - 150 - 330 - 62 = 66$ on $(5-3)(5-1) = 8$ df, $MS_E = 8.25$, $F_0 = 82.5/8.25 = 10.0 > F_{0.05,4,8} = 3.84$.
(An R run with $SS_{\text{Greek}} = 33.39$, $SS_E = 83.88$ on 8 df, $F_0 = 7.87$ used a Greek coding not orthogonal to batches/operators, whose SS changed from 68/150; in a genuine Graeco-Latin square all four SS are order-invariant.)

6 Balanced Incomplete Block Design (BIBD)

Motivation. When a block cannot hold every treatment, equal pair counts keep all comparisons equally precise, at the cost of block adjustment.

6.1 Motivation and definitions

- Sometimes not all treatments fit in every block (e.g. each coupon can test only 3 of the 4 tips). An **incomplete block design** has more treatments than can be put in a single block.
- A **BIBD** is an incomplete block design in which *every pair of treatments occurs together the same number of times* (λ). (R. A. Fisher.)

Definition 4.13: BIBD parameters

a = # treatments, b = # blocks, k = # treatments per block, r = # replicates of each treatment, $N = ar = bk$, and

$$\lambda = \frac{r(k-1)}{a-1} = \#\text{ times each pair of treatments appears together.}$$

Counting arguments [Supplement]. (i) Count all observations two ways: by treatments, a treatments $\times$ r each $= ar$; by blocks, b blocks $\times$ k each $= bk$. So $N = ar = bk$.

(ii) Fix one treatment, say 1. It appears in r blocks; in each such block it is paired with $k-1$ other treatments, giving $r(k-1)$ pairs containing treatment 1. Counted the other way, treatment 1 is paired with each of the other $a-1$ treatments exactly λ times: $\lambda(a-1)$. Hence $\lambda(a-1) = r(k-1)$. ■

A BIBD with a treatments and blocks of size k always exists by taking all $\binom{a}{k}$ combinations as blocks (unreduced design), although smaller designs often exist.

Example 4.14: Catalyst experiment

Catalyst	Batch (block)				
	1	2	3	4	$y_{i\cdot}$
1	73	74	–	71	218
2	–	75	67	72	214
3	73	75	68	–	216
4	75	–	72	75	222
$y_{\cdot j}$	221	224	207	218	$y_{\cdot\cdot} = 870$

$a = 4$, $b = 4$, $k = 3$, $r = 3$, $N = 12$, $\lambda = 3(2)/3 = 2$: e.g. catalysts 1 and 2 appear together in blocks 2 and 4.

6.2 Model and why an adjustment is needed

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2), \quad \sum \tau_i = \sum \beta_j = 0,$$

observed only when treatment i is in block j. Let $n_{ij} = 1$ if treatment i appears in block j and 0 otherwise. Then $\sum_j n_{ij} = r$, $\sum_i n_{ij} = k$, $\sum_j n_{ij} n_{lj} = \lambda$ $(i \neq l)$.

Derivation: the naive estimator is biased

RCBD ($a = b = 4$, nothing missing): $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ and

$$\mathrm{E}(\hat{\tau}_i) = \tfrac{1}{4}\mathrm{E}(y_{i1}+y_{i2}+y_{i3}+y_{i4}) - \tfrac{1}{16}\mathrm{E}(y_{11}+\cdots+y_{44}) = \tfrac{1}{4}(4\mu+4\tau_i) - \tfrac{1}{16}(16\mu+\textstyle\sum\tau_i+\sum\beta_j) = \tau_i.$$

BIBD (catalyst layout, treatment 1 in blocks 1, 2, 4):

$$\mathrm{E}(y_{1\cdot}) = \mathrm{E}(y_{11}+y_{12}+y_{14}) = 3\mu + 3\tau_1 + (\beta_1+\beta_2+\beta_4),$$

$$\mathrm{E}(y_{\cdot\cdot}) = N\mu + r\textstyle\sum_i \tau_i + k\sum_j \beta_j = N\mu,$$
$$\mathrm{E}(\bar{y}_{1\cdot} - \bar{y}_{\cdot\cdot}) = \mu + \tau_1 + \tfrac{1}{3}(\beta_1+\beta_2+\beta_4) - \mu = \tau_1 + \tfrac{1}{3}(\beta_1+\beta_2+\beta_4) \neq \tau_1.$$

Because each treatment sees a different set of r blocks, treatment totals are contaminated by block effects: the treatment SS must be **adjusted for blocks**.

6.3 Least squares: normal equations

Derivation

$S(\theta) = \sum_i \sum_j n_{ij}(y_{ij} - \mu - \tau_i - \beta_j)^2$.

$$\frac{\partial S}{\partial \mu} = 0: \; N\hat{\mu} + r\sum_{i=1}^{a}\hat{\tau}_i + k\sum_{j=1}^{b}\hat{\beta}_j = y_{\cdot\cdot} \quad \Rightarrow \; \hat{\mu} = y_{\cdot\cdot}/N = \bar{y}_{\cdot\cdot},$$
$$\frac{\partial S}{\partial \tau_i} = 0: \; r\hat{\mu} + r\hat{\tau}_i + \sum_{j=1}^{b} n_{ij}\hat{\beta}_j = y_{i\cdot}, \qquad i = 1,\dots,a,$$
$$\frac{\partial S}{\partial \beta_j} = 0: \; k\hat{\mu} + \sum_{i=1}^{a} n_{ij}\hat{\tau}_i + k\hat{\beta}_j = y_{\cdot j}, \qquad j = 1,\dots,b \quad (\star).$$

These are $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$.

Theorem 4.15: Adjusted treatment effects

$$\hat{\tau}_i = \frac{k\,Q_i}{\lambda a}, \qquad Q_i = y_{i\cdot} - \frac{1}{k}\sum_{j=1}^{b} n_{ij} y_{\cdot j} \quad \text{(adjusted treatment total)}, \qquad \sum_{i=1}^{a} Q_i = 0.$$

Proof [Supplement]. **Step 1 (eliminate $\hat{\beta}$).** Multiply the τ_i-equation by k:

$$rk\hat{\mu} + rk\hat{\tau}_i + k\sum_j n_{ij}\hat{\beta}_j = k y_{i\cdot}. \tag{a}$$

Multiply $(\star)$ by n_{ij} and sum over j:

$$k\hat{\mu}\sum_j n_{ij} + \sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l + k\sum_j n_{ij}\hat{\beta}_j = \sum_j n_{ij} y_{\cdot j}. \tag{b}$$

Since $\sum_j n_{ij} = r$, the first term of (b) is $rk\hat{\mu}$. Subtract (b) from (a): the $\hat{\mu}$ and $\hat{\beta}$ terms cancel,

$$rk\hat{\tau}_i - \sum_{j=1}^{b}\sum_{l=1}^{a} n_{ij}n_{lj}\hat{\tau}_l = k y_{i\cdot} - \sum_j n_{ij} y_{\cdot j} = kQ_i.$$

Step 2 (use balance). Split the double sum into $l = i$ and $l \neq i$, using $n_{ij}^2 = n_{ij}$:

$$\sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l = \underbrace{\sum_j n_{ij}}_{=r}\hat{\tau}_i + \sum_{l\neq i}\underbrace{\Big(\sum_j n_{ij}n_{lj}\Big)}_{=\lambda}\hat{\tau}_l = r\hat{\tau}_i + \lambda\sum_{l\neq i}\hat{\tau}_l = r\hat{\tau}_i - \lambda\hat{\tau}_i,$$

using $\sum_l \hat{\tau}_l = 0 \Rightarrow \sum_{l\neq i}\hat{\tau}_l = -\hat{\tau}_i$. **Step 3.** Therefore

$$rk\hat{\tau}_i - r\hat{\tau}_i + \lambda\hat{\tau}_i = kQ_i \iff (r(k-1)+\lambda)\hat{\tau}_i = kQ_i \iff (\lambda(a-1)+\lambda)\hat{\tau}_i = kQ_i \iff \hat{\tau}_i = \frac{kQ_i}{\lambda a}.$$

$\sum Q_i = 0$**:** $\sum_i Q_i = \sum_i y_{i\cdot} - \frac{1}{k}\sum_j (\sum_i n_{ij}) y_{\cdot j} = y_{\cdot\cdot} - \frac{1}{k}\cdot k\, y_{\cdot\cdot} = 0.$ ∎

Theorem 4.16: Unbiasedness, $\mathrm{E}(\hat{\tau}_i) = \tau_i$

Proof [Supplement]. $\mathrm{E}(y_{i\cdot}) = r\mu + r\tau_i + \sum_j n_{ij}\beta_j$ and $\mathrm{E}(y_{\cdot j}) = k\mu + \sum_l n_{lj}\tau_l + k\beta_j$. Then

$$\frac{1}{k}\sum_j n_{ij}\mathrm{E}(y_{\cdot j}) = \frac{1}{k}\Big[kr\mu + \sum_j\sum_l n_{ij}n_{lj}\tau_l + k\sum_j n_{ij}\beta_j\Big] = r\mu + \frac{1}{k}\big[r\tau_i + \lambda\textstyle\sum_{l\neq i}\tau_l\big] + \sum_j n_{ij}\beta_j$$
$$= r\mu + \frac{(r-\lambda)\tau_i}{k} + \sum_j n_{ij}\beta_j,$$

so

$$\mathrm{E}(Q_i) = r\tau_i - \frac{(r-\lambda)\tau_i}{k} = \frac{r(k-1)+\lambda}{k}\tau_i = \frac{\lambda a}{k}\tau_i \quad\Rightarrow\quad \mathrm{E}(\hat{\tau}_i) = \frac{k}{\lambda a}\cdot\frac{\lambda a}{k}\tau_i = \tau_i.$$

The block effects $\sum_j n_{ij}\beta_j$ cancel exactly — that is the meaning of "adjusted". ■

6.4 Sums of squares

Theorem 4.17

$$SS_T = \sum_i\sum_j y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Blocks}} = \frac{1}{k}\sum_{j=1}^{b} y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Treatments(adjusted)}} = \frac{k\sum_{i=1}^{a} Q_i^2}{\lambda a},$$
$$SS_E = SS_T - SS_{\text{Treatments(adjusted)}} - SS_{\text{Blocks}}.$$

SS_{Blocks} **[Supplement].** Every block has k observations, so

$$SS_{\text{Blocks}} = k\sum_{j=1}^{b}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 = k\Big[\sum_j \bar{y}_{\cdot j}^2 - b\bar{y}_{\cdot\cdot}^2\Big] = k\Big[\sum_j \frac{y_{\cdot j}^2}{k^2} - b\frac{y_{\cdot\cdot}^2}{b^2k^2}\Big] = \frac{1}{k}\sum_j y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{bk},$$

and $bk = N$. (Blocks are *unadjusted*: fitted first.) ■

$SS_{\text{Treatments(adjusted)}}$ **[Supplement].** **Model SS.** $SS_T = SS_{\text{Model}} + SS_E$ with $SS_T = \mathbf{y}^T\mathbf{y} - \frac{1}{N}y_{\cdot\cdot}^2$ and

$$SS_E = (\mathbf{y}-\hat{\mathbf{y}})^T(\mathbf{y}-\hat{\mathbf{y}}) = \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\hat{\mathbf{y}} - \hat{\mathbf{y}}^T\mathbf{y} + \hat{\mathbf{y}}^T\hat{\mathbf{y}} = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$$

(because $\hat{\mathbf{y}}^T\hat{\mathbf{y}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$ by the normal equations). Hence

$$SS_{\text{Model}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} - \frac{1}{N}y_{\cdot\cdot}^2 = \hat{\mu}y_{\cdot\cdot} + \sum_{i=1}^{a} y_{i\cdot}\hat{\tau}_i + \sum_{j=1}^{b} y_{\cdot j}\hat{\beta}_j - \frac{1}{N}y_{\cdot\cdot}^2.$$

Eliminate $\hat{\beta}_j$. Multiply $(\star)$ by $y_{\cdot j}$: $k\hat{\mu}y_{\cdot j} + \sum_i n_{ij}\hat{\tau}_i y_{\cdot j} + k\hat{\beta}_j y_{\cdot j} = y_{\cdot j}^2$, so

$$\sum_j \hat{\beta}_j y_{\cdot j} = \frac{1}{k}\sum_j y_{\cdot j}^2 - \hat{\mu}\sum_j y_{\cdot j} - \frac{1}{k}\sum_j\sum_i n_{ij}\hat{\tau}_i y_{\cdot j}.$$

Subtract blocks. With $\hat{\mu}\sum_j y_{\cdot j} = \hat{\mu}y_{\cdot\cdot}$,

$$SS_{\text{Model}} - SS_{\text{Blocks}} = \hat{\mu}y_{\cdot\cdot} + \sum_i y_{i\cdot}\hat{\tau}_i + \frac{1}{k}\sum_j y_{\cdot j}^2 - \hat{\mu}y_{\cdot\cdot} - \frac{1}{k}\sum_j\sum_i n_{ij}\hat{\tau}_i y_{\cdot j} - \frac{1}{N}y_{\cdot\cdot}^2 - \frac{1}{k}\sum_j y_{\cdot j}^2 + \frac{1}{N}y_{\cdot\cdot}^2$$
$$= \sum_{i=1}^{a}\Big(y_{i\cdot} - \frac{1}{k}\sum_j n_{ij}y_{\cdot j}\Big)\hat{\tau}_i = \sum_i Q_i\hat{\tau}_i = \sum_i Q_i\cdot\frac{k}{\lambda a}Q_i = \frac{k}{\lambda a}\sum_{i=1}^{a} Q_i^2.$$

So $SS_{\text{Treatments(adjusted)}} = SS_{\text{Model}} - SS_{\text{Blocks}}$ is the extra SS due to treatments after blocks. ■

Source	SS	DF	MS	F_0
Treatments (adjusted)	$\frac{k\sum Q_i^2}{\lambda a}$	$a-1$	$MS_{\text{Trt(adj)}}$	$MS_{\text{Trt(adj)}}/MS_E$
Blocks	$\frac{1}{k}\sum y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{N}$	$b-1$	MS_{Blocks}	
Error	by subtraction	$N-a-b+1$	MS_E	
Total	$\sum\sum y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}$	$N-1$		

Reject $H_0 : \tau_i = 0$ if $F_0 > F_{\alpha,a-1,N-a-b+1}$.

6.5 Variances of the adjusted estimators

Theorem 4.18

$$\text{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2, \qquad \text{Cov}(Q_i, Q_l) = -\frac{\lambda}{k}\sigma^2 \ (i \neq l),$$
$$\text{Var}(\hat{\tau}_i) = \frac{k(a-1)}{\lambda a^2}\sigma^2, \qquad \text{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{2k}{\lambda a}\sigma^2.$$

Proof [Supplement]. Splitting argument for $\text{Var}(Q_i)$. Write

$$kQ_i = ky_{i\cdot} - \sum_j n_{ij}y_{\cdot j} = (k-1)\underbrace{y_{i\cdot}}_{\text{obs. of trt } i} - \underbrace{\Big(\sum_j n_{ij}y_{\cdot j} - y_{i\cdot}\Big)}_{\text{other obs. in those } r \text{ blocks}}.$$

The two pieces involve disjoint sets of observations, hence are independent. $y_{i\cdot}$ is a sum of r observations; the second piece is a sum of $r(k-1)$ observations. So

$$k^2\text{Var}(Q_i) = (k-1)^2 r\sigma^2 + r(k-1)\sigma^2 = r(k-1)k\sigma^2 \ \Rightarrow \ \text{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2.$$

Covariance. For $i \neq l$, using $\text{Cov}(y_{i\cdot}, y_{\cdot j}) = n_{ij}\sigma^2$, $\text{Var}(y_{\cdot j}) = k\sigma^2$, $\text{Cov}(y_{i\cdot}, y_{l\cdot}) = 0$, $\text{Cov}(y_{\cdot j}, y_{\cdot j'}) = 0$ $(j \neq j')$:

$$\begin{aligned}
\text{Cov}(kQ_i, kQ_l) &= \text{Cov}\Big(ky_{i\cdot} - \sum_j n_{ij}y_{\cdot j},\ ky_{l\cdot} - \sum_j n_{lj}y_{\cdot j}\Big) \\
&= 0 - k\sum_j n_{lj}\text{Cov}(y_{i\cdot}, y_{\cdot j}) - k\sum_j n_{ij}\text{Cov}(y_{\cdot j}, y_{l\cdot}) + \sum_j n_{ij}n_{lj}\text{Var}(y_{\cdot j}) \\
&= -k\lambda\sigma^2 - k\lambda\sigma^2 + \lambda k\sigma^2 = -k\lambda\sigma^2 \ \Rightarrow \ \text{Cov}(Q_i, Q_l) = -\frac{\lambda}{k}\sigma^2.
\end{aligned}$$

Variances of $\hat{\tau}$. With $r(k-1) = \lambda(a-1)$:

$$\text{Var}(\hat{\tau}_i) = \frac{k^2}{\lambda^2 a^2}\text{Var}(Q_i) = \frac{k^2}{\lambda^2 a^2}\cdot\frac{\lambda(a-1)}{k}\sigma^2 = \frac{k(a-1)}{\lambda a^2}\sigma^2,$$
$$\text{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{k^2}{\lambda^2 a^2}\big[\text{Var}Q_i + \text{Var}Q_l - 2\text{Cov}(Q_i, Q_l)\big] = \frac{k^2}{\lambda^2 a^2}\Big[\frac{2\lambda(a-1)}{k} + \frac{2\lambda}{k}\Big]\sigma^2 = \frac{2k}{\lambda a}\sigma^2.$$

■

Remark 4.19

Compare with an RCBD with r replicates: $\text{Var}(\hat{\tau}_i - \hat{\tau}_l) = 2\sigma^2/r$. The ratio $\frac{2\sigma^2/r}{2k\sigma^2/(\lambda a)} = \frac{\lambda a}{rk}$ is the **efficiency factor** of the BIBD (< 1). Multiple comparisons on adjusted means use the

standard error $\sqrt{2kMS_E/(\lambda a)}$.

Example 4.20: Catalyst BIBD, line by line

$\sum\sum y_{ij}^2 = 63156$, $y_{\cdot\cdot}^2/N = 870^2/12 = 63075$.

$$SS_T = 63156 - 63075 = 81.00,$$
$$SS_{\text{Blocks}} = \frac{221^2 + 224^2 + 207^2 + 218^2}{3} - 63075 = \frac{189390}{3} - 63075 = 55.00.$$

Adjusted totals (block totals of the blocks containing treatment i):

$$Q_1 = 218 - \tfrac{1}{3}(221 + 224 + 218) = 218 - \tfrac{663}{3} = -3.00,$$
$$Q_2 = 214 - \tfrac{1}{3}(224 + 207 + 218) = 214 - \tfrac{649}{3} = -2.33,$$
$$Q_3 = 216 - \tfrac{1}{3}(221 + 224 + 207) = 216 - \tfrac{652}{3} = -1.33,$$
$$Q_4 = 222 - \tfrac{1}{3}(221 + 207 + 218) = 222 - \tfrac{646}{3} = 6.67 \qquad (\textstyle\sum Q_i = 0 \ \checkmark),$$
$$SS_{\text{Trt(adj)}} = \frac{3\left[(-3)^2 + (-7/3)^2 + (-4/3)^2 + (20/3)^2\right]}{2 \cdot 4}$$
$$= \frac{3(9 + 5.444 + 1.778 + 44.444)}{8} = \frac{3(60.667)}{8} = 22.75,$$
$$SS_E = 81.00 - 22.75 - 55.00 = 3.25.$$

Source	SS	DF	MS	F_0
Treatments (adj.)	22.75	3	7.58	11.67
Blocks	55.00	3	18.33	
Error	3.25	5	0.65	
Total	81.00	11		

$F_0 = 7.58/0.65 = 11.67 > F_{0.05,3,5} = 5.41$ ($P = 0.0107$): catalysts differ. $\hat{\tau}_i = kQ_i/(\lambda a) = 3Q_i/8$: $\hat{\tau} = (-1.125, -0.875, -0.500, 2.500)$. $\widehat{\text{se}}(\hat{\tau}_i - \hat{\tau}_l) = \sqrt{2(3)(0.65)/(2 \cdot 4)} = 0.698$.
R and the order of terms. `lm(y~block+trt)` gives sequential SS: blocks (unadjusted) 55.00, treatments adjusted for blocks 22.75 — the correct BIBD analysis. `lm(y~trt+block)` gives treatments *unadjusted* $= \sum y_{i\cdot}^2/r - y_{\cdot\cdot}^2/N = 11.667$ and blocks adjusted = 66.083; error SS (3.25) the same. Order matters: treatments and blocks are not orthogonal.

Example 4.21: Constructing a BIBD

$a = 5$ gasoline additives, $b = 5$ cars (blocks), $k = 4$ additives per car.

$$N = bk = 5 \cdot 4 = 20, \qquad r = \frac{N}{a} = \frac{20}{5} = 4, \qquad \lambda = \frac{r(k-1)}{a-1} = \frac{4 \cdot 3}{4} = 3.$$

Design: block j omits additive j:

$$\text{B1} : \{2,3,4,5\},\ \text{B2} : \{1,3,4,5\},\ \text{B3} : \{1,2,4,5\},\ \text{B4} : \{1,2,3,5\},\ \text{B5} : \{1,2,3,4\}.$$

Check: the pair $(1,2)$ appears in every block except B1 and B2, i.e. in $5 - 2 = 3 = \lambda$ blocks; the same holds for every pair.

7 Summary

Design	Nuisance factors	Error df
RCBD	1 (blocks)	$(a-1)(b-1)$
Latin square	2 (rows, columns)	$(p-1)(p-2)$
Graeco-Latin square	3	$(p-1)(p-3)$
BIBD	1 (incomplete blocks)	$N-a-b+1$

Chapter 5. Introduction to Factorial Designs

1 Introduction

In a **factorial design**, every complete replicate investigates all possible combinations of the levels of the factors.

- **Main effect**: change in response produced by a change in the level of a factor.
- **Interaction**: the difference in response between the levels of one factor is not the same at all levels of the other factor.

Example 5.1: Main effects and interaction in a 2×2 layout

Responses: $(A^-, B^-) = 20$, $(A^+, B^-) = 40$, $(A^-, B^+) = 30$, $(A^+, B^+) = 52$.

$$\begin{aligned} A &= \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{40+52}{2} - \frac{20+30}{2} = 46 - 25 = 21, \\ B &= \frac{30+52}{2} - \frac{20+40}{2} = 41 - 30 = 11, \\ AB &= \frac{1}{2}[(\text{A effect at } B^+) - (\text{A effect at } B^-)] \\ &= \frac{1}{2}[(52-30) - (40-20)] = \frac{1}{2}(22-20) = 1. \end{aligned}$$

If instead $(A^+, B^+) = 12$: A effect at B^- is 20, at B^+ is $12 - 30 = -18$; $AB = \frac{1}{2}(-18 - 20) = -19$. A large interaction makes the main effect $A = \frac{40+12}{2} - 25 = 1$ misleading.

Advantages of factorial experiments

1. More efficient than one-factor-at-a-time experiments.
2. All data are used in computing *both* effects (all observations are used in determining the average effect of A and the average effect of B).
3. Some information is provided on possible interaction between the factors; one-factor-at-a-time cannot detect interaction and may lead to wrong conclusions.

2 The Two-Factor Factorial Design

2.1 Battery design example

Factor A = plate material type ($a = 3$), factor B = temperature ($b = 3$: 15, 70, 125°F), $n = 4$ replicates, all $abn = 36$ runs in random order (a CRD on the ab cells). Response: battery life (hours).

Material	15°F	70°F	125°F	$y_{i\cdot\cdot}$
1	130, 155 74, 180	34, 40 80, 75	20, 70 82, 58	998
	$y_{11\cdot} = 539$	229	230	
2	150, 188 159, 126	136, 122 106, 115	25, 70 58, 45	1300
	623	479	198	
3	138, 110 168, 160	174, 120 150, 139	96, 104 82, 60	1501
	576	583	342	
$y_{\cdot j\cdot}$	1738	1291	770	$y_{\cdots} = 3799$

Questions:

(1) What effects do material type and temperature have on life? (2) Is there a material giving long life regardless of temperature (a *robust* product)?

2.2 Model and hypotheses

Definition 5.2: Two-factor factorial (fixed effects)

Effects model:

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}, \quad i = 1, \ldots, a;\ j = 1, \ldots, b;\ k = 1, \ldots, n,$$

$\sum_i \tau_i = 0$, $\sum_j \beta_j = 0$, $\sum_i (\tau\beta)_{ij} = \sum_j (\tau\beta)_{ij} = 0$, $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.
Means model: $y_{ijk} = \mu_{ij} + \varepsilon_{ijk}$, $\mu_{ij} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij}$.

Three hypotheses:

1. $H_0 : \tau_1 = \cdots = \tau_a = 0$ vs H_1: at least one $\tau_i \neq 0$;
2. $H_0 : \beta_1 = \cdots = \beta_b = 0$ vs H_1: at least one $\beta_j \neq 0$;
3. $H_0 : (\tau\beta)_{ij} = 0$ for all i, j vs H_1: at least one $(\tau\beta)_{ij} \neq 0$.

2.3 Least squares estimation

Derivation

$S(\theta) = \sum_i \sum_j \sum_k \left(y_{ijk} - \mu - \tau_i - \beta_j - (\tau\beta)_{ij}\right)^2$.

$$\left.\frac{\partial S}{\partial \mu}\right|_{\hat\theta} = -2 \sum_i \sum_j \sum_k \left(y_{ijk} - \hat\mu - \hat\tau_i - \hat\beta_j - \widehat{(\tau\beta)}_{ij}\right) = 0$$

$$\Rightarrow\ abn\hat\mu = y_{\cdots} \ \Rightarrow\ \hat\mu = \bar y_{\cdots},$$

$$\left.\frac{\partial S}{\partial \tau_i}\right|_{\hat\theta} = -2 \sum_j \sum_k \left(y_{ijk} - \hat\mu - \hat\tau_i - \hat\beta_j - \widehat{(\tau\beta)}_{ij}\right) = 0$$

$$\Rightarrow\ bn\hat\tau_i = y_{i\cdot\cdot} - bn\hat\mu \ \Rightarrow\ \hat\tau_i = \bar y_{i\cdot\cdot} - \bar y_{\cdots},$$

$$\left.\frac{\partial S}{\partial \beta_j}\right|_{\hat\theta} = 0 \ \Rightarrow\ \hat\beta_j = \bar y_{\cdot j\cdot} - \bar y_{\cdots},$$

$$\left.\frac{\partial S}{\partial (\tau\beta)_{ij}}\right|_{\hat\theta} = -2 \sum_k \left(y_{ijk} - \hat\mu - \hat\tau_i - \hat\beta_j - \widehat{(\tau\beta)}_{ij}\right) = 0$$

$$\Rightarrow\ \widehat{(\tau\beta)}_{ij} = \bar y_{ij\cdot} - \bar y_{\cdots} - (\bar y_{i\cdot\cdot} - \bar y_{\cdots}) - (\bar y_{\cdot j\cdot} - \bar y_{\cdots})$$

$$= \bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}.$$

(The zero-sum constraints on $\hat{\tau}, \hat{\beta}, \widehat{\tau\beta}$ kill the other sums.) Hence

$$\hat{\mu}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \widehat{(\tau\beta)}_{ij} = \bar{y}_{ij\cdot}, \qquad y_{ijk} = \hat{\mu}_{ij} + e_{ijk}, \quad e_{ijk} = y_{ijk} - \bar{y}_{ij\cdot}.$$

2.4 Partition of the total sum of squares

Derivation

$$\begin{aligned} y_{ijk} - \bar{y}_{\cdots} &= (\bar{y}_{ij\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot}) \\ &= \underbrace{(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})}_{\hat{\tau}_i} + \underbrace{(\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})}_{\hat{\beta}_j} + \underbrace{(\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})}_{\widehat{(\tau\beta)}_{ij}} + \underbrace{(y_{ijk} - \bar{y}_{ij\cdot})}_{e_{ijk}}. \end{aligned}$$

Theorem 5.3

$$\begin{aligned} \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{\cdots})^2 &= bn \sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 + an \sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2 \\ &\quad + n \sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2 + \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{ij\cdot})^2, \\ SS_T &= SS_A + SS_B + SS_{AB} + SS_E, \\ abn - 1 &= (a-1) + (b-1) + (a-1)(b-1) + ab(n-1). \end{aligned}$$

Proof [Supplement]. All six cross products vanish:

- any cross product with e_{ijk}: the other factor depends only on (i,j) and $\sum_k (y_{ijk} - \bar{y}_{ij\cdot}) = 0$;
- $\hat{\tau}_i \times \hat{\beta}_j$: $n \sum_i \hat{\tau}_i \sum_j \hat{\beta}_j = 0$;
- $\hat{\tau}_i \times \widehat{(\tau\beta)}_{ij}$: $n \sum_i \hat{\tau}_i \sum_j \widehat{(\tau\beta)}_{ij}$ and $\sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) = b\bar{y}_{i\cdot\cdot} - b\bar{y}_{i\cdot\cdot} - b\bar{y}_{\cdots} + b\bar{y}_{\cdots} = 0$;
- $\hat{\beta}_j \times \widehat{(\tau\beta)}_{ij}$: similarly $\sum_i \widehat{(\tau\beta)}_{ij} = 0$.

df of SS_{AB}: $SS_{AB} = n \sum\sum \widehat{(\tau\beta)}_{ij}^2$ involves ab terms subject to $\sum_i \widehat{(\tau\beta)}_{ij} = 0$ $(j = 1, \ldots, b)$ and $\sum_j \widehat{(\tau\beta)}_{ij} = 0$ $(i = 1, \ldots, a)$. Only $b + (a-1)$ of these $a + b$ conditions are independent (the sum of all row conditions equals the sum of all column conditions), so df $= ab - (a + b - 1) = (a-1)(b-1)$. ■

2.5 Expected mean squares

Theorem 5.4

$$\begin{aligned} \mathrm{E}(MS_A) &= \mathrm{E}\Big(\frac{SS_A}{a-1}\Big) = \sigma^2 + \frac{bn \sum_i \tau_i^2}{a-1}, & \mathrm{E}(MS_B) &= \sigma^2 + \frac{an \sum_j \beta_j^2}{b-1}, \\ \mathrm{E}(MS_{AB}) &= \sigma^2 + \frac{n \sum_i \sum_j (\tau\beta)_{ij}^2}{(a-1)(b-1)}, & \mathrm{E}(MS_E) &= \mathrm{E}\Big(\frac{SS_E}{ab(n-1)}\Big) = \sigma^2. \end{aligned}$$

Proof [Supplement]. MS_A**.** Averaging the model, $\bar{y}_{i\cdot\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot\cdot}$ (since $\sum_j \beta_j = \sum_j (\tau\beta)_{ij} = 0$) and $\bar{y}_{\cdots} = \mu + \bar{\varepsilon}_{\cdots}$. Thus

$$\begin{aligned}\mathrm{E}(SS_A) &= \mathrm{E}\Big[bn\sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2\Big] = \mathrm{E}\Big[bn\sum_i \big(\tau_i + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})\big)^2\Big] \\ &= \mathrm{E}\Big[bn\sum_i \tau_i^2 + bn\sum_i (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})^2 + 2bn\sum_i \tau_i(\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})\Big] \\ &= bn\sum_i \tau_i^2 + bn\,\mathrm{E}\Big[\sum_{i=1}^{a} (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})^2\Big] = bn\sum_i \tau_i^2 + bn(a-1)\frac{\sigma^2}{bn},\end{aligned}$$

since $\bar{\varepsilon}_{i\cdot\cdot}$ are iid with variance $\sigma^2/(bn)$. Dividing by $a-1$: $\mathrm{E}(MS_A) = \sigma^2 + \frac{bn}{a-1}\sum_i \tau_i^2$. MS_B is symmetric.

MS_{AB}**.** $\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots} = (\tau\beta)_{ij} + (\bar{\varepsilon}_{ij\cdot} - \bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdot j\cdot} + \bar{\varepsilon}_{\cdots})$. The $\bar{\varepsilon}_{ij\cdot}$ form an $a \times b$ table of iid $N(0, \sigma^2/n)$ variables, and the second bracket is exactly the RCBD residual of that table. By the RCBD result of Ch. 4, $\mathrm{E}\sum\sum(\cdot)^2 = (a-1)(b-1)\sigma^2/n$; the cross product has mean 0. Hence

$$\mathrm{E}(SS_{AB}) = n\sum\sum(\tau\beta)_{ij}^2 + n(a-1)(b-1)\frac{\sigma^2}{n} \;\Rightarrow\; \mathrm{E}(MS_{AB}) = \sigma^2 + \frac{n\sum\sum(\tau\beta)_{ij}^2}{(a-1)(b-1)}.$$

MS_E**.**

$$\mathrm{E}(MS_E) = \frac{1}{ab(n-1)}\mathrm{E}\Big[\sum\sum\sum(y_{ijk} - \bar{y}_{ij\cdot})^2\Big] = \frac{1}{ab}\sum_i\sum_j \mathrm{E}\Big[\frac{\sum_k (y_{ijk} - \bar{y}_{ij\cdot})^2}{n-1}\Big] = \frac{1}{ab}\cdot ab\,\sigma^2 = \sigma^2.$$

■

Under each H_0, the corresponding MS and MS_E both estimate σ^2; under H_1 the ratio exceeds 1.

2.6 Extra sum of squares for interaction

Derivation

Full: $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk} \Rightarrow SS_{E(F)} = \sum\sum\sum(y_{ijk} - \bar{y}_{ij\cdot})^2$.
Reduced: $y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon_{ijk} \Rightarrow SS_{E(R)} = \sum\sum\sum(y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2$.

$$SS_H = SS_{E(R)} - SS_{E(F)} = n\sum_i\sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2 = SS_{AB},$$

$$F_0 = \frac{SS_H/(a-1)(b-1)}{SS_{E(F)}/ab(n-1)} \overset{H_0}{\sim} F_{(a-1)(b-1),\,ab(n-1)}.$$

Proof [Supplement]. $y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots} = (y_{ijk} - \bar{y}_{ij\cdot}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})$; the cross product vanishes because $\sum_k (y_{ijk} - \bar{y}_{ij\cdot}) = 0$, so $SS_{E(R)} = SS_{E(F)} + SS_{AB}$. (For balanced data the additive-model LSEs are the same marginal means.) ■

2.7 ANOVA table

Source	SS	DF	MS	F_0
A	SS_A	$a-1$	MS_A	MS_A/MS_E
B	SS_B	$b-1$	MS_B	MS_B/MS_E
AB	SS_{AB}	$(a-1)(b-1)$	MS_{AB}	MS_{AB}/MS_E
Error	SS_E	$ab(n-1)$	MS_E	
Total	SS_T	$abn-1$		

Hand formulas:

$$SS_T = \sum\sum\sum y_{ijk}^2 - \frac{y_{...}^2}{abn}, \quad SS_A = \frac{\sum_i y_{i..}^2}{bn} - \frac{y_{...}^2}{abn}, \quad SS_B = \frac{\sum_j y_{.j.}^2}{an} - \frac{y_{...}^2}{abn},$$

$$SS_{\text{Subtotals}} = \frac{\sum_i \sum_j y_{ij.}^2}{n} - \frac{y_{...}^2}{abn}, \quad SS_{AB} = SS_{\text{Subtotals}} - SS_A - SS_B, \quad SS_E = SS_T - SS_{\text{Subtotals}}.$$

Example 5.5: Battery life ANOVA, line by line

$y_{...}^2/abn = 3799^2/36 = 400{,}900.03$, $\sum y^2 = 478{,}547$.

$$\begin{aligned}
SS_T &= 478547 - 400900.03 = 77646.97,\\
SS_{\text{Material}} &= \frac{998^2 + 1300^2 + 1501^2}{12} - 400900.03 = 411583.75 - 400900.03 = 10683.72,\\
SS_{\text{Temp}} &= \frac{1738^2 + 1291^2 + 770^2}{12} - 400900.03 = 440018.75 - 400900.03 = 39118.72,\\
SS_{\text{Subtotals}} &= \frac{539^2 + 229^2 + \cdots + 342^2}{4} - 400900.03\\
&= 460316.25 - 400900.03 = 59416.22,\\
SS_{AB} &= 59416.22 - 10683.72 - 39118.72 = 9613.78,\\
SS_E &= 77646.97 - 59416.22 = 18230.75.
\end{aligned}$$

Source	SS	DF	MS	F_0
Material type	10683.72	2	5341.86	7.91
Temperature	39118.72	2	19559.36	28.97
Interaction	9613.78	4	2403.44	3.56
Error	18230.75	27	675.21	
Total	77646.97	35		

$F_{0.05,4,27} = 2.73$, $F_{0.05,2,27} = 3.35$: interaction ($P = 0.019$) and both main effects are significant. Interaction plot: life decreases with temperature for all materials, but material 3 degrades least at high temperature — the significant interaction means the material effect depends on temperature.

2.8 Multiple comparisons when interaction is present

If interaction is significant, compare one factor's levels *at a fixed level of the other*. For a fixed j, test $H_0 : \mu_{ij} = \mu_{i'j}$ with Fisher's LSD

$$|\bar{y}_{ij.} - \bar{y}_{i'j.}| > LSD = t_{\alpha/2,ab(n-1)}\sqrt{\frac{2MS_E}{n}};$$

for a fixed i, compare $\bar{y}_{ij\cdot}$ and $\bar{y}_{ij'\cdot}$ similarly. (If interaction is not significant, compare main-effect means $\bar{y}_{i\cdot\cdot}$ with $\sqrt{2MS_E/(bn)}$.)

Example 5.6: Materials at 70°F

$LSD = t_{0.025,27}\sqrt{2(675.21)/4} = 2.052(18.37) = 37.70$. Cell means at 70°F: $\bar{y}_{12\cdot} = 57.25$, $\bar{y}_{22\cdot} = 119.75$, $\bar{y}_{32\cdot} = 145.75$.

$$|\bar{y}_{12\cdot} - \bar{y}_{22\cdot}| = 62.50 > 37.70, \quad |\bar{y}_{12\cdot} - \bar{y}_{32\cdot}| = 88.50 > 37.70, \quad |\bar{y}_{22\cdot} - \bar{y}_{32\cdot}| = 26.00 < 37.70.$$

At 70°F, types 2 and 3 have the same mean life, and type 1 is significantly lower.

A second example (problem 5.7, pressure × temperature, $n = 2$): F(pressure)= 8.47, F(temp)= 21.59, F(interaction)= 0.97 ($P = 0.47$): no interaction, so main effects are interpreted directly.

2.9 Residual analysis

Residuals $e_{ijk} = y_{ijk} - \bar{y}_{ij\cdot}$; normal Q–Q plot; residuals vs material, vs temperature, vs fitted values $\hat{y}_{ijk} = \bar{y}_{ij\cdot}$.

3 Independence of $\hat{\mu}$ and $\hat{\beta}_j$

Motivation. Independence of the estimated grand mean and effects, from orthogonality.

Theorem 5.7

In the two-factor model (M1), $\hat{\mu} = \bar{y}_{\cdots}$ and $\hat{\beta}_j = \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}$ are statistically independent.

Proof [Supplement]. Both are linear functions of the normal vector $\mathbf{y}$, so they are jointly normal and independence $\iff$ zero covariance. Since $\bar{y}_{\cdots} = \frac{1}{b}\sum_{j'} \bar{y}_{\cdot j'\cdot}$ and the $\bar{y}_{\cdot j'\cdot}$ are independent (disjoint observations) with variance $\sigma^2/(an)$,

$$\mathrm{Cov}(\bar{y}_{\cdots}, \bar{y}_{\cdot j\cdot}) = \frac{1}{b}\sum_{j'} \mathrm{Cov}(\bar{y}_{\cdot j'\cdot}, \bar{y}_{\cdot j\cdot}) = \frac{1}{b}\mathrm{Var}(\bar{y}_{\cdot j\cdot}) = \frac{\sigma^2}{abn} = \mathrm{Var}(\bar{y}_{\cdots}),$$

$$\mathrm{Cov}(\hat{\mu}, \hat{\beta}_j) = \mathrm{Cov}(\bar{y}_{\cdots}, \bar{y}_{\cdot j\cdot}) - \mathrm{Var}(\bar{y}_{\cdots}) = \frac{\sigma^2}{abn} - \frac{\sigma^2}{abn} = 0.$$

■

4 Comparing Three Models

Model	Error SS	Error df
M1: $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	$SS_E^1 = \sum\sum\sum(y_{ijk} - \bar{y}_{ij\cdot})^2$	$ab(n-1)$
M2: $y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon_{ijk}$ $(n > 1)$	$SS_E^2 = \sum\sum\sum(y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2$	$abn - a - b + 1$
M3: $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ $(n = 1)$	$SS_E^3 = \sum\sum(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$	$(a-1)(b-1)$

For M2: $\hat{\mu} = \bar{y}_{\cdots}$, $\hat{\tau}_i = \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}$, $\hat{\beta}_j = \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}$, $\hat{y}_{ijk} = \bar{y}_{i\cdot\cdot} + \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}$, $SS_A = bn\sum\hat{\tau}_i^2$ ($a-1$ df), $SS_B = an\sum\hat{\beta}_j^2$ ($b-1$ df).

Test $H_0 : \tau_i = 0$ when the within-cell variation is very small, i.e. $y_{ijk} \approx \bar{y}_{ij\cdot} \approx y_{ij1}$ (the single observation in M3):

$$F_0^{(1)} = \frac{bn\sum_i(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2/(a-1)}{SS_E^1/ab(n-1)}, \qquad SS_E^1 \approx 0 \;\Rightarrow\; F_0^{(1)} \text{ huge},$$

$$F_0^{(2)} = \frac{bn\sum_i(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2/(a-1)}{SS_E^2/(abn-a-b+1)}, \qquad SS_E^2 \approx n\sum_i\sum_j(y_{ij1} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 = n\,SS_E^3,$$

$$F_0^{(3)} = \frac{b\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2/(a-1)}{SS_E^3/(a-1)(b-1)}.$$

Since the numerators satisfy $bn\sum(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 \approx n \cdot b\sum(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$,

$$F_0^{(2)} \approx \frac{n \cdot b\sum(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2/(a-1)}{n\,SS_E^3/(abn-a-b+1)} = F_0^{(3)} \cdot \frac{abn-a-b+1}{(a-1)(b-1)} \;\geq\; F_0^{(3)}.$$

Conclusion: $F_0^{(1)} \gg F_0^{(2)} \geq F_0^{(3)}$. M1 separates pure error from interaction; M2 lumps interaction into error but gains df from replication; M3 has neither replication nor a separate interaction term, so it is the least powerful (and is biased if interaction exists).

5 The General Factorial: Three Factors

Derivation

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl},$$

$$SS_T = SS_A + SS_B + SS_C + SS_{AB} + SS_{AC} + SS_{BC} + SS_{ABC} + SS_E,$$

with estimates built hierarchically: $\widehat{(\tau\beta)}_{ij} = \hat{\mu}_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j$, …, $\widehat{(\tau\beta\gamma)}_{ijk} = \hat{\mu}_{ijk} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j - \hat{\gamma}_k - \widehat{(\tau\beta)}_{ij} - \widehat{(\tau\gamma)}_{ik} - \widehat{(\beta\gamma)}_{jk}$.

DF: $a-1$, $b-1$, $c-1$, $(a-1)(b-1)$, $(a-1)(c-1)$, $(b-1)(c-1)$, $(a-1)(b-1)(c-1)$, $abc(n-1)$; total $abcn-1$. For fixed effects every F_0 uses MS_E; e.g. $\mathrm{E}(MS_A) = \sigma^2 + \frac{bcn\sum\tau_i^2}{a-1}$, $\mathrm{E}(MS_{ABC}) = \sigma^2 + \frac{n\sum\sum\sum(\tau\beta\gamma)_{ijk}^2}{(a-1)(b-1)(c-1)}$. Note: $n \geq 2$ is required to estimate pure error.

6 Tukey's Test for Additivity (One Observation per Cell)

6.1 The problem

With $n = 1$ and model $y_{ij} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ij}$, interaction and error are inseparable (ab observations, ab parameters: $SS_E = 0$ on 0 df). Assuming no interaction, $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$, and $H_0 : \tau_i = 0$ is tested with $F_0 = MS_A/MS_E$ where MS_E is the residual ($= MS_{AB}$) mean square. Tukey's test checks that assumption using **one** degree of freedom.

6.2 Model

$$y_{ij} = \mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}, \qquad \mathrm{E}(y_{ij}) = \mu + \tau_i + \beta_j + \gamma\tau_i\beta_j, \quad \mathrm{Var}(y_{ij}) = \sigma^2,$$

$H_0 : \gamma = 0$ vs $H_1 : \gamma \neq 0$. The interaction is assumed to have the special multiplicative form $(\tau\beta)_{ij} = \gamma\tau_i\beta_j$.

6.3 Estimating μ, τ_i, β_j

Derivation

Since $\sum \tau_i = \sum \beta_j = 0$, the product term averages out:

$$\begin{aligned}
\bar{y}_{\cdot\cdot} &= \frac{1}{ab}\sum_i\sum_j(\mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}) = \mu + \bar{\varepsilon}_{\cdot\cdot} \\
&\Rightarrow\ \mathrm{E}(\bar{y}_{\cdot\cdot}) = \mu, \quad \hat{\mu} = \bar{y}_{\cdot\cdot} \text{ unbiased}, \\
\bar{y}_{i\cdot} &= \frac{1}{b}\sum_j(\mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}) = \mu + \tau_i + \bar{\varepsilon}_{i\cdot} \\
&\Rightarrow\ \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} \text{ unbiased}; \quad \text{similarly } \hat{\beta}_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot} \text{ unbiased}.
\end{aligned}$$

6.4 Least squares estimator of γ

Derivation

$S(\theta) = \sum_i\sum_j(y_{ij} - \mu - \tau_i - \beta_j - \gamma\tau_i\beta_j)^2 = \sum\sum\varepsilon_{ij}^2$. Write $S_{ij} := y_{ij} - \mu - \tau_i - \beta_j - \gamma\tau_i\beta_j$.

$$\frac{\partial S}{\partial \gamma}\Big|_{\hat{\theta}} = 0 \ \Rightarrow\ \sum_i\sum_j \tau_i\beta_j S_{ij} = 0 \ \Rightarrow$$

$$\sum\sum\tau_i\beta_j y_{ij} - \mu\underbrace{\sum\sum\tau_i\beta_j}_{0} - \underbrace{\sum\sum\tau_i^2\beta_j}_{0} - \underbrace{\sum\sum\tau_i\beta_j^2}_{0} - \gamma\sum\tau_i^2\sum\beta_j^2 = 0.$$

Hence (plugging in the estimated effects)

$$\hat{\gamma} = \frac{\sum_i\sum_j\hat{\tau}_i\hat{\beta}_j y_{ij}}{\sum_i\hat{\tau}_i^2\sum_j\hat{\beta}_j^2} = \frac{ab\sum\sum\hat{\tau}_i\hat{\beta}_j y_{ij}}{SS_A \cdot SS_B},$$

because $SS_A = b\sum_i\hat{\tau}_i^2$ and $SS_B = a\sum_j\hat{\beta}_j^2$.

6.5 Sum of squares for non-additivity

Theorem 5.8

$$SS_N = \sum_i\sum_j\hat{\gamma}^2\hat{\tau}_i^2\hat{\beta}_j^2 = \frac{\left(\sum\sum\hat{\tau}_i\hat{\beta}_j y_{ij}\right)^2}{\sum\hat{\tau}_i^2\sum\hat{\beta}_j^2} = \frac{\left[\sum_{i=1}^a\sum_{j=1}^b(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})y_{ij}\right]^2}{\sum_{i=1}^a(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2\sum_{j=1}^b(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2}$$

$$= \frac{ab\left(\sum\sum\hat{\tau}_i\hat{\beta}_j y_{ij}\right)^2}{SS_A\,SS_B}.$$

SS_N has 1 df; $SS_E = SS_{\text{Residual}} - SS_N$ with $(a-1)(b-1) - 1$ df, where $SS_{\text{Residual}} = \sum\sum(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$. $SS_N = SS_H = SS_{E(R)} - SS_{E(F)}$ is the extra SS for γ.

Proof [Supplement]. $SS_N = \hat{\gamma}^2\sum_i\hat{\tau}_i^2\sum_j\hat{\beta}_j^2 = \frac{(\sum\sum\hat{\tau}_i\hat{\beta}_j y_{ij})^2}{(\sum\hat{\tau}_i^2\sum\hat{\beta}_j^2)^2}\sum\hat{\tau}_i^2\sum\hat{\beta}_j^2$, which simplifies to the stated forms. ■

6.6 Distribution of SS_N under H_0

Unconditional (oracle) version. If τ_i, β_j were known, $\tilde{\gamma} = \frac{\sum\sum \tau_i\beta_j y_{ij}}{\sum\tau_i^2 \sum\beta_j^2}$ satisfies

$$\mathrm{E}(\tilde{\gamma}) = \frac{\sum\sum \tau_i\beta_j(\mu+\tau_i+\beta_j+\gamma\tau_i\beta_j)}{\sum\tau_i^2\sum\beta_j^2} = \gamma,$$

$$\mathrm{Var}(\tilde{\gamma}) = \Big(\frac{1}{\sum\tau_i^2\sum\beta_j^2}\Big)^2 \sum\sum \tau_i^2\beta_j^2\sigma^2 = \frac{\sigma^2}{\sum\tau_i^2\sum\beta_j^2},$$

so under H_0, $\tilde{\gamma} \sim N(0, \mathrm{Var}\tilde{\gamma})$ and $\tilde{\gamma}^2/\mathrm{Var}(\tilde{\gamma}) \sim \chi^2_1$.

With estimated effects. Condition on $\hat{\tau}, \hat{\beta}$. Under H_0,

$$\mathrm{E}(\hat{\gamma} \mid \hat{\tau}, \hat{\beta}, H_0) = \frac{ab\sum\sum \hat{\tau}_i\hat{\beta}_j(\mu+\tau_i+\beta_j)}{SS_A\, SS_B} = 0$$

(the weights $c_{ij} = \hat{\tau}_i\hat{\beta}_j$ have zero row and column sums, so they annihilate any additive table). Moreover $\sum\sum c_{ij}y_{ij} = \sum\sum c_{ij}e_{ij}$ depends only on the residuals, which are independent of the row and column means under normality; hence

$$\mathrm{Var}(\hat{\gamma} \mid \hat{\tau}, \hat{\beta}) = \Big(\frac{1}{\sum\hat{\tau}_i^2\sum\hat{\beta}_j^2}\Big)^2 \sum\sum \hat{\tau}_i^2\hat{\beta}_j^2\sigma^2 = \frac{\sigma^2}{\sum\hat{\tau}_i^2\sum\hat{\beta}_j^2} = \frac{ab\sigma^2}{SS_A\, SS_B} =: \widehat{\mathrm{Var}}(\hat{\gamma}).$$

Therefore

$$\frac{\hat{\gamma}^2}{\widehat{\mathrm{Var}}(\hat{\gamma})} = \frac{(ab)^2(\sum\sum \hat{\tau}_i\hat{\beta}_j y_{ij})^2}{SS_A^2\, SS_B^2} \cdot \frac{SS_A\, SS_B}{ab\,\sigma^2} = \frac{ab(\sum\sum \hat{\tau}_i\hat{\beta}_j y_{ij})^2}{SS_A\, SS_B\,\sigma^2} = \frac{SS_N}{\sigma^2} \overset{H_0}{\sim} \chi^2_1,$$

conditionally on $(\hat{\tau}, \hat{\beta})$, and since this conditional law does not depend on $(\hat{\tau}, \hat{\beta})$, also unconditionally.

Theorem 5.9: Cochran-type lemma

If Q, Q_1, Q_2 are quadratic forms with $Q = Q_1 + Q_2$, $Q \sim \chi^2_n$, $Q_1 \sim \chi^2_{n_1}$ and $Q_2 \geq 0$, then $Q_2 \sim \chi^2_{n-n_1}$ and $Q_1 \perp\!\!\!\perp Q_2$.

Apply it with $Q = SS_{\text{Residual}}/\sigma^2 \sim \chi^2_{(a-1)(b-1)}$ (under H_0, the additive model is true), $Q_1 = SS_N/\sigma^2 \sim \chi^2_1$, $Q_2 = SS_E/\sigma^2$:

$$\frac{SS_E}{\sigma^2} \sim \chi^2_{(a-1)(b-1)-1}, \quad SS_N \perp\!\!\!\perp SS_E$$

$$\Rightarrow \quad F_0 = \frac{MS_N}{MS_E} = \frac{SS_N/1}{SS_E/[(a-1)(b-1)-1]} \overset{H_0}{\sim} F_{1,(a-1)(b-1)-1}.$$

Reject $H_0 : \gamma = 0$ if $F_0 > F_{\alpha,1,(a-1)(b-1)-1}$.

Source	SS	DF	MS	F_0
A	SS_A	$a-1$	MS_A	MS_A/MS_E
B	SS_B	$b-1$	MS_B	MS_B/MS_E
Nonadditivity	SS_N	1	MS_N	MS_N/MS_E
Error	SS_E	$(a-1)(b-1)-1$	MS_E	
Total	SS_T	$ab-1$		

Example 5.10: Impurity data, temperature ($a = 3$) $\times$ pressure ($b = 5$), $n = 1$

Temp	25	30	35	40	45	$y_{i\cdot}$
100	5	4	6	3	5	23
125	3	1	4	2	3	13
150	1	1	3	1	2	8
$y_{\cdot j}$	9	6	13	6	10	44

$\bar{y}_{\cdot\cdot} = 44/15 = 2.933$; $\hat{\tau} = (4.6, 2.6, 1.6) - 2.933 = (1.667, -0.333, -1.333)$, $\sum \hat{\tau}_i^2 = 4.667$; $\hat{\beta} = (3, 2, 4.333, 2, 3.333) - 2.933 = (0.067, -0.933, 1.400, -0.933, 0.400)$, $\sum \hat{\beta}_j^2 = 3.867$.

$$
\begin{aligned}
SS_A &= 5(4.667) = 23.33, \quad SS_B = 3(3.867) = 11.60,\\
SS_{\text{Residual}} &= 166 - \tfrac{44^2}{15} - 23.33 - 11.60 = 2.00,\\
\textstyle\sum\sum \hat{\tau}_i \hat{\beta}_j y_{ij} &= 1.333,\\
SS_N &= \frac{1.333^2}{4.667 \times 3.867} = \frac{1.778}{18.04} = 0.0985,\\
SS_E &= 2.00 - 0.0985 = 1.9015 \text{ (7 df)}, \quad MS_E = 0.2716,\\
F_0 &= \frac{0.0985}{0.2716} = 0.36 < F_{0.05,1,7} = 5.59.
\end{aligned}
$$

No evidence of interaction; the additive analysis ($F_{\text{temp}} = 11.667/0.25 = 46.67$, $F_{\text{press}} = 2.9/0.25 = 11.6$) is justified.

Chapter 6. The 2^k Factorial Design

1 The 2^2 Factorial Design

Motivation. Every run informs every effect; with two levels the effects are sign-table contrasts.

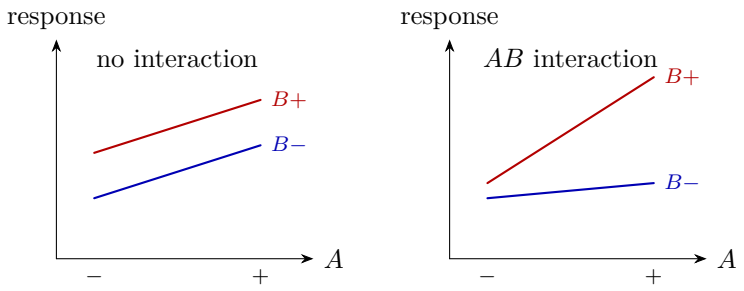


Figure. *Interaction plots for a 2^2 design: parallel lines, no interaction; non-parallel lines, an AB interaction, and the main effect of A alone no longer describes the data.*

- The 2^k design is a special case of the general factorial: k factors, each at **two levels** ("low" $-$ and "high" $+$; quantitative or qualitative). Very widely used in industry (screening).
- Simplest case: 2^2. Effects model

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk}, \qquad i = 1, 2;\ j = 1, 2;\ k = 1, \ldots, n.$$

1.1 Notation for treatment combinations

Lowercase letters name the factors at their high level: (1) both low, a, b, ab both high; the symbol also denotes the **total** of the n observations there.

$$\begin{array}{ccc} b\ (-,+) & — & ab\ (+,+) \\ | & & | \\ (1)\ (-,-) & — & a\ (+,-) \end{array}$$

Example 6.1: Chemical process data ($n = 3$)

A = reactant concentration, B = amount of catalyst, response = yield.

A	B	I	II	III	Total
$-$	$-$	28	25	27	$(1) = 80$
$+$	$-$	36	32	32	$a = 100$
$-$	$+$	18	19	23	$b = 60$
$+$	$+$	31	30	29	$ab = 90$

Analysis procedure:

(1) estimate factor effects (main effects A, B, interaction AB);

(2) statistical testing (ANOVA: SS for A, B, AB, error);

(3) residual analysis (normality, constant variance).

1.2 Estimating effects

Derivation

The effect of A at the low level of B is $\dfrac{a-(1)}{n}$; at the high level of B it is $\dfrac{ab-b}{n}$. The main effect of A is their average:

$$A = \frac{1}{2}\Big(\frac{a-(1)}{n} + \frac{ab-b}{n}\Big) = \frac{ab+a-b-(1)}{2n}.$$

Equivalently $A = \bar{y}_{A^+} - \bar{y}_{A^-} = \dfrac{ab+a}{2n} - \dfrac{b+(1)}{2n}$.

$$A = \frac{1}{2n}[ab+a-b-(1)], \qquad B = \frac{1}{2n}[ab+b-a-(1)], \qquad AB = \frac{1}{2n}[ab+(1)-a-b].$$

AB is half the difference between the A effect at high B and at low B: $\frac{1}{2}\big[\frac{ab-b}{n} - \frac{a-(1)}{n}\big]$.

Table of algebraic signs

Treatment combination	I	A	B	AB
(1)	+	−	−	+
a	+	+	−	−
b	+	−	+	−
ab	+	+	+	+

The AB column is the elementwise product of the A and B columns. Every pair of columns is orthogonal.

1.3 Sums of squares via contrasts

Theorem 6.2: SS of a contrast

For a contrast $C = \sum_\ell c_\ell S_\ell$ in the treatment totals S_ℓ (each a sum of n observations), $\sum_\ell c_\ell = 0$,

$$SS_C = \frac{(\text{contrast})^2}{n\sum_\ell c_\ell^2}.$$

In the 2^2 design: $\text{Contrast}_A = ab+a-b-(1)$, $\text{Contrast}_B = ab+b-a-(1)$, $\text{Contrast}_{AB} = ab+(1)-a-b$, $\sum c_\ell^2 = 4$, so

$$SS_A = \frac{[ab+a-b-(1)]^2}{4n}, \quad SS_B = \frac{[ab+b-a-(1)]^2}{4n}, \quad SS_{AB} = \frac{[ab+(1)-a-b]^2}{4n}.$$

Direct proof for SS_A [Supplement]. In the two-factor notation $SS_A = \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{n}(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 = 2n\sum_{i=1}^{2}(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 = 2n[(\bar{y}_{A^+} - \bar{y}_{\cdots})^2 + (\bar{y}_{A^-} - \bar{y}_{\cdots})^2]$, with $\bar{y}_{A^+} = \frac{ab+a}{2n}$, $\bar{y}_{A^-} = \frac{b+(1)}{2n}$, $\bar{y}_{\cdots} = \frac{ab+a+b+(1)}{4n}$:

$$\bar{y}_{A^+} - \bar{y}_{\cdots} = \frac{2(ab+a) - (ab+a+b+(1))}{4n} = \frac{ab+a-b-(1)}{4n}, \qquad \bar{y}_{A^-} - \bar{y}_{\cdots} = -\frac{ab+a-b-(1)}{4n},$$

$$SS_A = 2n\cdot 2\cdot\frac{[ab+a-b-(1)]^2}{16n^2} = \frac{[ab+a-b-(1)]^2}{4n}.$$

■

General: $t_0^2 = F_0$ **[Supplement].** Let $S_\ell = \sum_{k=1}^{n} y_{\ell k} = n\bar{y}_\ell$ (ℓ indexes the cells), and $C = \sum_\ell c_\ell S_\ell = n\sum_\ell c_\ell \bar{y}_\ell$. Then

$$\mathrm{Var}(C) = \mathrm{Var}\Big(n\sum_\ell c_\ell \bar{y}_\ell\Big) = n^2 \sum_\ell c_\ell^2 \mathrm{Var}(\bar{y}_\ell) = n^2 \sum_\ell c_\ell^2 \frac{\sigma^2}{n} = n\sigma^2 \sum_\ell c_\ell^2.$$

Let $t_0 = \dfrac{C - \mathrm{E}(C)}{\sqrt{\widehat{\mathrm{Var}}(C)}} = \dfrac{\sum c_\ell S_\ell - \mathrm{E}(C)}{\sqrt{n\, MS_E \sum c_\ell^2}}$. Assuming $\mathrm{E}(C) = 0$ (true under H_0, see below),

$$t_0^2 = \frac{(\sum_\ell c_\ell S_\ell)^2/(n\sum c_\ell^2)}{MS_E} = \frac{SS_C/1}{MS_E} \sim F_{1,\,2^2(n-1)}.$$

So each effect's F-test in the ANOVA is the square of a t-test of the contrast. ■

Derivation: $\mathrm{E}(\mathrm{Contrast}_A) = 0$ **under** H_0

With effects A_1, A_2 (low/high) etc.,

$$ab = \textstyle\sum_k y_{22k} = n(\mu + A_2 + B_2 + (AB)_{22}) + \varepsilon_{22\cdot}, \qquad a = n(\mu + A_2 + B_1 + (AB)_{21}) + \varepsilon_{21\cdot},$$
$$b = n(\mu + A_1 + B_2 + (AB)_{12}) + \varepsilon_{12\cdot}, \qquad (1) = n(\mu + A_1 + B_1 + (AB)_{11}) + \varepsilon_{11\cdot}.$$

Hence

$$\mathrm{E}(\mathrm{Contrast}_A) = n(2A_2 - 2A_1 + \underbrace{(AB)_{22} + (AB)_{21} - (AB)_{12} - (AB)_{11}}_{=0 \text{ by } \sum_i (AB)_{ij} = 0}) = n(2A_2 - 2A_1) \overset{H_0}{=} 0.$$

Since $A_1 + A_2 = 0$, $\mathrm{E}(\mathrm{Contrast}_A) = 4nA_2$ and $\mathrm{E}(A \text{ effect}) = \frac{4nA_2}{2n} = 2A_2 = A_2 - A_1$.

1.4 ANOVA

DF: $2^2 n - 1 = 1 + 1 + 1 + 2^2(n-1)$.

Source	SS	DF	MS	F_0
A	SS_A	1	MS_A	MS_A/MS_E
B	SS_B	1	MS_B	MS_B/MS_E
AB	SS_{AB}	1	MS_{AB}	MS_{AB}/MS_E
Error	SS_E	$2^2(n-1)$	MS_E	
Total	SS_T	$2^2 n - 1$		

Example 6.3: Chemical process, line by line

$$A = \frac{90 + 100 - 60 - 80}{2(3)} = \frac{50}{6} = 8.33,$$
$$B = \frac{90 + 60 - 100 - 80}{6} = \frac{-30}{6} = -5.00,$$
$$AB = \frac{90 + 80 - 100 - 60}{6} = \frac{10}{6} = 1.67,$$
$$SS_A = \frac{50^2}{4(3)} = 208.33, \quad SS_B = \frac{(-30)^2}{12} = 75.00, \quad SS_{AB} = \frac{10^2}{12} = 8.33,$$

$$SS_T = \sum y^2 - \frac{y_{\cdots}^2}{12} = 9398 - \frac{330^2}{12} = 9398 - 9075 = 323.00,$$
$$SS_E = 323.00 - 208.33 - 75.00 - 8.33 = 31.34.$$

Source	SS	DF	MS	F_0
A	208.33	1	208.33	53.15
B	75.00	1	75.00	19.13
AB	8.33	1	8.33	2.13
Error	31.34	8	3.92	
Total	323.00	11		

$F_{0.05,1,8} = 5.32$: A and B significant, AB not. Increasing concentration raises yield; increasing catalyst lowers it.

2 The 2^3 Design

Motivation. 2^k runs estimate all main effects and interactions as orthogonal contrasts of equal variance.

Factors A, B, C at two levels: 8 runs $(1), a, b, ab, c, ac, bc, abc$.

Trt	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

Properties:

(1) except for I, every column has equally many + and − (a contrast);

(2) the product of any two columns is a column of the table ($A \cdot B = AB$, $AB \cdot BC = AB^2C = AC$, with squares $= I$);

(3) I times any column leaves it unchanged;

(4) all columns are mutually orthogonal.

$$A = \frac{1}{4n}\left[(abc + ac + ab + a) - (bc + c + b + (1))\right], \qquad SS_A = \frac{(\text{Contrast}_A)^2}{8n}.$$

Theorem 6.4: General 2^k

$$\text{Effect} = \frac{\text{Contrast}}{2^{k-1}n}, \qquad SS = \frac{(\text{Contrast})^2}{2^k n}, \qquad \text{Var(Effect)} = \frac{2^k n\sigma^2}{(2^{k-1}n)^2} = \frac{\sigma^2}{2^{k-2}n},$$

$\text{df(Total)} = n2^k - 1$, $\text{df(Error)} = 2^k(n-1)$, each of the $2^k - 1$ effects has 1 df.

Proof [Supplement]. Each contrast uses all 2^k totals with $c_\ell = \pm 1$, so $\sum c_\ell^2 = 2^k$ and $SS = \text{Contrast}^2/(n2^k)$. The effect is (sum of 2^{k-1} totals at +)/$(2^{k-1}n)$ minus (sum at −)/$(2^{k-1}n)$, i.e. $\text{Contrast}/(2^{k-1}n)$. Variance: $\text{Var(Contrast)} = n\sigma^2 \sum c_\ell^2 = 2^k n\sigma^2$. ■

Source	SS	DF
A, B, C	SS_A, SS_B, SS_C	1 each
AB, AC, BC	$SS_{AB}, SS_{AC}, SS_{BC}$	1 each
ABC	SS_{ABC}	1
Error	SS_E	$2^3(n-1)$
Total	SS_T	$n2^3 - 1$

Example 6.5: A 2^3 with $n = 2$ (soft-drink fill height)

Totals: $(1) = -4$, $a = 1$, $b = -1$, $ab = 5$, $c = -1$, $ac = 3$, $bc = 2$, $abc = 11$ (sum $= 16$).

$$\begin{aligned}
\text{Contrast}_A &= a + ab + ac + abc - (1) - b - c - bc = 1 + 5 + 3 + 11 + 4 + 1 + 1 - 2 = 24,\\
\text{Contrast}_B &= b + ab + bc + abc - (1) - a - c - ac = -1 + 5 + 2 + 11 + 4 - 1 + 1 - 3 = 18,\\
\text{Contrast}_C &= c + ac + bc + abc - (1) - a - b - ab = -1 + 3 + 2 + 11 + 4 - 1 + 1 - 5 = 14,\\
\text{Contrast}_{AB} &= (1) + ab + c + abc - a - b - ac - bc = -4 + 5 - 1 + 11 - 1 + 1 - 3 - 2 = 6,\\
\text{Contrast}_{AC} &= (1) + b + ac + abc - a - ab - c - bc = -4 - 1 + 3 + 11 - 1 - 5 + 1 - 2 = 2,\\
\text{Contrast}_{BC} &= (1) + a + bc + abc - b - ab - c - ac = -4 + 1 + 2 + 11 + 1 - 5 + 1 - 3 = 4,\\
\text{Contrast}_{ABC} &= a + b + c + abc - (1) - ab - ac - bc = 1 - 1 - 1 + 11 + 4 - 5 - 3 - 2 = 4.
\end{aligned}$$

Effects $=$ Contrast$/(4 \cdot 2)$: $A = 3.00$, $B = 2.25$, $C = 1.75$, $AB = 0.75$, $AC = 0.25$, $BC = 0.50$, $ABC = 0.50$. SS $=$ Contrast$^2/16$: 36.00, 20.25, 12.25, 2.25, 0.25, 1.00, 1.00. With $SS_T = 78.00$, $SS_E = 78 - 73 = 5.00$ on 8 df, $MS_E = 0.625$; $F_A = 57.6$, $F_B = 32.4$, $F_C = 19.6$, $F_{AB} = 3.6$ ($F_{0.05,1,8} = 5.32$): only the main effects are significant.

3 Regression Approach to the 2^k Design

Derivation

2^2: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{12} x_{1i} x_{2i} + \varepsilon_i$ with coded $x = \pm 1$. Using the totals as responses,

$$\mathbf{X} = \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} (1) \\ a \\ b \\ ab \end{pmatrix}$$

$$\Rightarrow \ \mathbf{X}^T\mathbf{y} = \begin{pmatrix} (1) + a + b + ab \\ -(1) + a - b + ab \\ -(1) - a + b + ab \\ (1) - a - b + ab \end{pmatrix} = \begin{pmatrix} \text{Contrast}_{\text{total}} \\ \text{Contrast}_A \\ \text{Contrast}_B \\ \text{Contrast}_{AB} \end{pmatrix}.$$

Since $\mathbf{X}^T\mathbf{X} = 4\mathbf{I}$, $\hat{\boldsymbol{\beta}} = \frac{1}{4}\mathbf{X}^T\mathbf{y}$, e.g.

$$\hat{\beta}_1 = \tfrac{1}{4}\text{Contrast}_A = \tfrac{1}{4}(2n \cdot \text{main effect } A) = \tfrac{n}{2} \cdot \text{main effect } A.$$

2^3: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_3 x_3 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{123} x_1 x_2 x_3 + \varepsilon$, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \frac{1}{2^3}(\text{Contrast}_{\text{total}}, \text{Contrast}_A, \dots)^T$ and since $A = \frac{1}{2^{k-1}n}\text{Contrast}_A$, $\hat{\beta}_1 = \frac{n}{2}$ main effect$_A$.

Remark 6.6: Using cell averages instead

If the responses are the cell *averages* $\bar{y}_\ell = S_\ell/n$, then $\hat{\beta}_1 = \frac{1}{4}\sum_\ell c_\ell \bar{y}_\ell = \frac{1}{4n}\text{Contrast}_A = \frac{A}{2}$: a regression coefficient is **half the effect**, because x moves from -1 to $+1$ (two units). Chemical data: $\mathbf{y} = (80/3, 100/3, 60/3, 90/3)^T$ gives $\hat{\beta}_0 = 27.5$, $\hat{\beta}_1 = 4.1667 = 8.33/2$, $\hat{\beta}_2 = -2.5 = -5/2$, $\hat{\beta}_{12} = 0.8333 = 1.67/2$ (matching the R output).

4 Residual Analysis

Fitted values from the regression model (only significant terms): chemical process $\hat{y} = 27.5 + \frac{8.33}{2}x_1 - \frac{5.00}{2}x_2$. Check normality (Q–Q plot of residuals) and constant variance (residuals vs fitted).

5 Summary of Designs So Far

Design	Model	Analysis
CRD	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	one-way ANOVA / regression
RCBD, BIBD	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	two-way ANOVA / regression
Latin square	$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}$	three-way ANOVA
Graeco-Latin	$y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}$	four-way ANOVA
Factorial	$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	two-way ANOVA with interaction
2^k	contrasts of $(1), a, b, ab, \ldots$	ANOVA via contrasts / regression

6 Selected exercises

Motivation. Sign-table properties, normal plots, center points, and precision of a first-order fit.

Exercise 6.1 (Sign table and the unreplicated 2^k)

Code each factor as $x_j = \pm 1$ and let the column of the effect with letter set $J \subseteq \{1, \ldots, k\}$ be $c_J(\mathbf{x}) = \prod_{j\in J} x_j$ over the 2^k runs ($c_\emptyset = I$).
(a) Prove properties (1)–(4) of the sign table: $\sum_{\mathbf{x}} c_J(\mathbf{x}) = 0$ for $J \neq \emptyset$, $c_J c_K = c_{J\triangle K}$, and $\sum_{\mathbf{x}} c_J(\mathbf{x})c_K(\mathbf{x}) = 2^k\,\mathbf{1}\{J = K\}$.
(b) In a single replicate ($n = 1$) with $y(\mathbf{x}) = \mu(\mathbf{x}) + \varepsilon(\mathbf{x})$, $\varepsilon \overset{\text{iid}}{\sim} N(0, \sigma^2)$, show that the $2^k - 1$ effect estimates are independent $N(\text{effect}, \sigma^2/2^{k-2})$, and that their sums of squares add up to SS_T.

Solution [Supplement].

(a)

Step 1. For $J \neq \emptyset$, summing over the 2^k points factor by factor,

$$\sum_{\mathbf{x}\in\{\pm1\}^k} \prod_{j\in J} x_j = \prod_{j\in J}\Big(\sum_{x_j=\pm1} x_j\Big)\cdot 2^{k-|J|} = 0.$$

Step 2. $c_J(\mathbf{x})c_K(\mathbf{x}) = \prod_{j\in J\cap K} x_j^2 \prod_{j\in J\triangle K} x_j = c_{J\triangle K}(\mathbf{x})$ since $x_j^2 = 1$ (e.g. $AB \cdot BC = AC$), and $c_\emptyset c_J = c_J$.

Step 3. Hence $\sum_{\mathbf{x}} c_J c_K = \sum_{\mathbf{x}} c_{J\triangle K}$, which is 2^k if $J = K$ and 0 otherwise by Step 1.

(b)

Step 1. The estimate of effect J is $\hat{E}_J = \mathbf{c}_J^T \mathbf{y}/2^{k-1}$, where $\mathbf{c}_J$ is the column. Being linear in $\mathbf{y} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, the estimates are jointly normal with $\mathrm{Cov}(\hat{E}_J, \hat{E}_K) = \sigma^2 \mathbf{c}_J^T \mathbf{c}_K / 2^{2k-2}$, which is 0 for $J \neq K$ and $\sigma^2/2^{k-2}$ for $J = K$ by (a). Uncorrelated jointly normal variables are independent.

Step 2. By (a) the 2^k columns $\mathbf{c}_J/2^{k/2}$ form an orthonormal basis of $\mathbb{R}^{2^k}$, so $\mathbf{y}^T\mathbf{y} = \sum_J (\mathbf{c}_J^T \mathbf{y})^2/2^k$. The $J = \emptyset$ term is $(\sum y)^2/2^k$, and for $J \neq \emptyset$ the term is $SS_J = \mathrm{Contrast}_J^2/2^k$. Hence $\sum_{J\neq\emptyset} SS_J = \mathbf{y}^T\mathbf{y} - (\sum y)^2/2^k = SS_T$.

All $2^k - 1$ df go to effects: no error estimate. By Step 1 null effects are a sample from $N(0, \sigma^2/2^{k-2})$, on a line in a normal plot, while real effects stand out; or negligible high-order interactions are pooled into error. ■

Exercise 6.2 (Center points and curvature)

A 2^k design ($n_F = 2^k$ runs) is augmented with n_C runs at the center $\mathbf{x} = \mathbf{0}$. Suppose the true mean is the second-order model $\mu(\mathbf{x}) = \beta_0 + \sum_j \beta_j x_j + \sum_{i<j} \beta_{ij} x_i x_j + \sum_j \beta_{jj} x_j^2$ and errors are iid $N(0, \sigma^2)$.
(a) Show $\mathrm{E}(\bar{y}_F - \bar{y}_C) = \sum_j \beta_{jj}$.
(b) Show that under $H_0 : \sum_j \beta_{jj} = 0$,

$$F_0 = \frac{SS_{\text{pure quad}}}{s_C^2} \sim F_{1,\, n_C - 1}, \qquad SS_{\text{pure quad}} = \frac{n_F n_C (\bar{y}_F - \bar{y}_C)^2}{n_F + n_C},$$

where s_C^2 is the sample variance of the center runs.
(c) Show that adding center runs changes none of the factorial effect estimates.

Solution [Supplement].

(a) Over the factorial points $x_j^2 = 1$, while $\sum x_j = 0$ and $\sum x_i x_j = 0$ $(i \neq j)$ by Exercise 6.1(a). Hence $\mathrm{E}\bar{y}_F = \beta_0 + \sum_j \beta_{jj}$. At the center $\mu(\mathbf{0}) = \beta_0$, so $\mathrm{E}\bar{y}_C = \beta_0$.

(b)

Step 1. $\bar{y}_F$ and $\bar{y}_C$ are independent with variances σ^2/n_F and σ^2/n_C, so $\mathrm{Var}(\bar{y}_F - \bar{y}_C) = \sigma^2(n_F + n_C)/(n_F n_C)$ and, under H_0, $SS_{\text{pure quad}}/\sigma^2 = (\bar{y}_F - \bar{y}_C)^2/\mathrm{Var}(\bar{y}_F - \bar{y}_C) \sim \chi_1^2$.

Step 2. The center runs are an iid normal sample, so $(n_C - 1)s_C^2/\sigma^2 \sim \chi_{n_C-1}^2$ independently of $\bar{y}_C$; it is also independent of the factorial runs. Hence s_C^2 is independent of $\bar{y}_F - \bar{y}_C$ and $F_0 \sim F_{1,n_C-1}$.

Step 3. A significant F_0 signals curvature but cannot say which β_{jj} is nonzero; that needs a response surface design with more than two levels per factor.

(c) Each effect estimate is a contrast over the factorial runs only (center runs have $x_j = 0$ and contribute 0 to every column c_J, $J \neq \emptyset$). So the centers buy a pure-error estimate and a curvature check without disturbing the factorial analysis. ■

Exercise 6.3 (Precision of the fitted first-order model)

The first-order model $y = \beta_0 + \sum_{j=1}^k \beta_j x_j + \varepsilon$ is fitted to n replicates of a 2^k design ($N = n2^k$ runs). Show that the variance of the fitted value at a point $\mathbf{x}$ of the design region is

$$\mathrm{Var}\{\hat{y}(\mathbf{x})\} = \frac{\sigma^2}{N}(1 + \|\mathbf{x}\|^2),$$

so it depends only on the distance from the center.

Solution [Supplement].

Step 1. The columns of $\mathbf{X}$ are $I, x_1, \ldots, x_k$, each repeated over the n replicates. By Exercise 6.1(a) they are orthogonal, and every entry is ± 1, so $\mathbf{X}^T\mathbf{X} = N\mathbf{I}_{k+1}$.

Step 2. $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} = (\sigma^2/N)\mathbf{I}_{k+1}$: the coefficient estimates are uncorrelated with equal variance.

Step 3. $\hat{y}(\mathbf{x}) = (1, \mathbf{x}^T)\hat{\boldsymbol{\beta}}$, so $\mathrm{Var}\{\hat{y}(\mathbf{x})\} = (\sigma^2/N)(1, \mathbf{x}^T)(1, \mathbf{x}^T)^T = (\sigma^2/N)(1 + \|\mathbf{x}\|^2)$.

The variance is smallest at the center (σ^2/N) and largest at the corners ($\sigma^2(1+k)/N$); a design with this property (equal precision on spheres) is called rotatable. ■

Chapter 7. Blocking and Confounding in the 2^k Factorial Design

1 Blocking a Replicated Design

- If there are n replicates of the 2^k design, each replicate can be run as a **block** (e.g. one batch of raw material, one day).
- Replicates are run in block order; runs *within* a block are randomized.

Example 7.1: Chemical process in three blocks

Block 1	Block 2	Block 3
$(1) = 28$	$(1) = 25$	$(1) = 27$
$a = 36$	$a = 32$	$a = 32$
$b = 18$	$b = 19$	$b = 23$
$ab = 31$	$ab = 30$	$ab = 29$
$B_1 = 113$	$B_2 = 106$	$B_3 = 111$

Without blocks (Ch. 6): $SS_A = \frac{1}{2^2 \cdot 3}(ab+a-b-(1))^2 = 208.33$, $A = \frac{1}{2\cdot 3}(ab+a-b-(1)) = 8.33$; error df $2^2(3-1) = 8$.
Replicate = block, an RCBD (Ch. 4) $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ with the 4 combinations as treatments. In Ch. 4, $SS_{\text{Block}} = \sum_i \sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 = \frac{1}{a}\sum_j y_{\cdot j}^2 - \frac{1}{N} y_{\cdot\cdot}^2$. Here each block has $2^2 = 4$ runs:

$$SS_{\text{Blocks}} = \frac{1}{4}\sum_{j=1}^{3} B_j^2 - \frac{1}{12} y_{\cdot\cdot}^2 = \frac{113^2 + 106^2 + 111^2}{4} - \frac{330^2}{12}$$
$$= \frac{36326}{4} - 9075 = 9081.5 - 9075 = 6.50, \qquad \text{df} = b - 1 = 2,$$
$$SS_E = 31.34 - 6.50 = 24.84, \qquad \text{df} = 8 - 2 = 6.$$

Source	SS	DF	MS	F_0
Blocks	6.50	2	3.25	
A	208.33	1	208.33	50.32
B	75.00	1	75.00	18.12
AB	8.33	1	8.33	2.01
Error	24.84	6	4.14	
Total	323.00	11		

Conclusions unchanged (small block effect); estimates identical, as each block is a full replicate.

2 Confounding in Blocks: the Unreplicated Case

If block size $< 2^k$, some effect becomes **confounded** (indistinguishable) with blocks, usually a high-order interaction.

2.1 2^2 in two blocks

Block 1	Block 2
(1)	a
ab	b

Trt	I	A	B	AB	Block
(1)	+	−	−	+	1
a	+	+	−	−	2
b	+	−	+	−	2
ab	+	+	+	+	1

$A = \frac{1}{2}(ab + a - (1) - b)$ and $B = \frac{1}{2}(ab + b - (1) - a)$ are **unaffected by blocking** (one + and one − run per block), but $AB = \frac{1}{2}(ab + (1) - a - b) = \frac{1}{2}[(\text{block 1}) - (\text{block 2})]$ is **confounded with blocks**.

Formal argument [Supplement]. Add block effects δ_1, δ_2 to every observation in blocks 1 and 2. Then

$$\begin{aligned} \mathrm{E}(\text{Contrast}_A) &= \mathrm{E}[ab] + \mathrm{E}[a] - \mathrm{E}[(1)] - \mathrm{E}[b] \ \ni\ \delta_1 + \delta_2 - \delta_1 - \delta_2 = 0, \\ \mathrm{E}(\text{Contrast}_{AB}) &= \mathrm{E}[ab] + \mathrm{E}[(1)] - \mathrm{E}[a] - \mathrm{E}[b] \ \ni\ \delta_1 + \delta_1 - \delta_2 - \delta_2 = 2(\delta_1 - \delta_2) \neq 0. \end{aligned}$$

So A, B are block-free, while the AB contrast estimates interaction + block difference, inseparably. ■

2.2 2^3 in two blocks with ABC confounded

Trt	I	A	B	AB	C	AC	BC	ABC	Block
(1)	+	−	−	+	−	+	+	−	1
a	+	+	−	−	−	−	+	+	2
b	+	−	+	−	−	+	−	+	2
ab	+	+	+	+	−	−	−	−	1
c	+	−	−	+	+	−	−	+	2
ac	+	+	−	−	+	+	−	−	1
bc	+	−	+	−	+	−	+	−	1
abc	+	+	+	+	+	+	+	+	2

Block 1 = $\{(1), ab, ac, bc\}$ (the − signs of ABC), Block 2 = $\{a, b, c, abc\}$ (the + signs). Every other column has two + and two − per block, so all effects but ABC are block-free.

3 Constructing the Blocks: the Defining Contrast

Definition 7.2: Defining contrast

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_k x_k,$$

$x_i \in \{0, 1\}$ is the level of the ith factor in a treatment combination, and $\alpha_i \in \{0, 1\}$ is the exponent of the ith factor in the effect to be confounded. Treatment combinations with the same $L \pmod 2$ go to the same block.

For ABC: $\alpha_1 = \alpha_2 = \alpha_3 = 1$, $L = x_1 + x_2 + x_3$.

Trt	L	mod 2
(1)	$0+0+0=0$	0
a	$1+0+0=1$	1
b	$0+1+0=1$	1
ab	$1+1+0=2$	0
c	$0+0+1=1$	1
ac	$1+0+1=2$	0
bc	$0+1+1=2$	0
abc	$1+1+1=3$	1

Block 1 ($L=0$): $(1), ab, ac, bc$; Block 2 ($L=1$): a, b, c, abc.

Remark 7.3: Why L works

The sign of a treatment combination in the column of effect $A^{\alpha_1}B^{\alpha_2}C^{\alpha_3}$ is $\prod_i (2x_i-1)^{\alpha_i} = (-1)^{\sum_i \alpha_i(1-x_i)}$, which depends only on $\sum \alpha_i x_i \pmod 2$ (for fixed α). So L mod 2 splits the runs exactly by the sign of the confounded column.

3.1 The principal block and group structure

- The **principal block** is the block containing (1).
- It is a *group* under multiplication with exponents mod 2: each element but (1) is a product of two others. For ABC:

$$ab \cdot bc = ab^2c = ac, \qquad ab \cdot ac = a^2bc = bc, \qquad bc \cdot ac = abc^2 = ab.$$

- The other block is a coset: one of its elements times the principal block. With b:

$$b \cdot (1) = b, \quad b \cdot ab = ab^2 = a, \quad b \cdot ac = abc, \quad b \cdot bc = b^2c = c.$$

Closure [Supplement]. If $L(u) \equiv 0$ and $L(v) \equiv 0$, then for the product uv (exponents added mod 2) $L(uv) \equiv L(u) + L(v) \equiv 0$, because L is linear mod 2. Similarly if $L(w) \equiv 1$ then $L(wu) \equiv 1$ for every u in the principal block. ■

3.2 More blocks (four blocks)

For 2^p blocks confound p independent effects; their **generalized interactions** (products mod 2) are confounded too. A 2^3 in 4 blocks confounding AB, AC also loses $AB \cdot AC = A^2BC = BC$: $3 = 2^2 - 1$ effects, one per block df. Blocks by $(L_1, L_2) = (x_1+x_2,\ x_1+x_3)$ mod 2: $(0,0) : \{(1), abc\}$, $(1,1) : \{a, bc\}$, $(1,0) : \{b, ac\}$, $(0,1) : \{c, ab\}$.

4 Block Confounding with Replication

4.1 Complete confounding

The same effect is confounded in every replicate. 2^3 in two blocks, three replicates, ABC confounded in all:

Rep 1		Rep 2		Rep 3	
Bl 1	Bl 2	Bl 1	Bl 2	Bl 1	Bl 2
(1)	a	ac	a	c	(1)
ab	b	(1)	c	abc	ac
ac	c	ab	abc	b	bc
bc	abc	bc	b	a	ab

(Labels are arbitrary; each replicate splits $\{(1), ab, ac, bc\}$ vs $\{a, b, c, abc\}$, randomized within blocks.)

Source	DF
Replications	2
Blocks (or ABC)	1
Replication × blocks	2
A, B, AB, C, AC, BC	1 each (6)
Error (replication × all other effects)	12
Total	23

The 6 block df ($= 2 \cdot 3$ blocks $-1 = 5$) split as 2 (replicates) +1 (ABC/blocks) +2 (rep × blocks); error df $= 3 \cdot 6 - 6 = 12$ (replicate × the six estimable effects). ABC cannot be tested.

4.2 Partial confounding

A *different* interaction per replicate; each is estimated from the replicates where it is unconfounded. 2^3, four replicates, confounding ABC, AB, AC, BC in replicates 1–4:

Rep 1 (ABC)		Rep 2 (AB)		Rep 3 (AC)		Rep 4 (BC)	
Bl 1	Bl 2	Bl 1	Bl 2	Bl 1	Bl 2	Bl 1	Bl 2
(1)	a	(1)	a	(1)	a	(1)	b
ab	b	c	b	b	c	a	c
ac	c	ab	ac	ac	ab	bc	ab
bc	abc	abc	bc	abc	bc	abc	ac

(Check rep 2 with $L = x_1 + x_2$: $(1), c, ab, abc$ have $L \equiv 0$.)

Source	DF
Replications	3
Blocks within replications	4
ABC (rep 1), AB (rep 2), AC (rep 3), BC (rep 4)	1 each
A, B, C	1 each
AB (from reps 1,3,4), AC (reps 1,2,4), BC (reps 1,2,3), ABC (reps 2,3,4)	1 each
Error	17
Total	31

DF: $31 = 3 + 4 + 7 + 17$. A partially confounded interaction uses its 3 unconfounded replicates: SS $(\text{Contrast}^{(3\text{ reps})})^2/(3 \cdot 2^3)$ and it has relative information 3/4.

Example 7.4: *ABC* in rep 1, *AB* in rep 2, *BC* in rep 3

(a) *Blocks.* Rep 1 ($L = x_1 + x_2 + x_3$): $\{(1), ab, ac, bc\} \,|\, \{a, b, c, abc\}$. Rep 2 ($L = x_1 + x_2$): $\{(1), c, ab, abc\} \,|\, \{a, b, ac, bc\}$. Rep 3 ($L = x_2 + x_3$): $\{(1), a, bc, abc\} \,|\, \{b, c, ab, ac\}$.
(b) *ANOVA skeleton* ($N = 24$):

Source	DF
Replications	2
Blocks within replications (*ABC* rep 1, *AB* rep 2, *BC* rep 3)	3
A, *B*, *C*, *AC* (never confounded)	1 each (4)
AB (reps 1, 3), *BC* (reps 1, 2), *ABC* (reps 2, 3)	1 each (3)
Error	11
Total	23

Error df $= 23 - 2 - 3 - 7 = 11$.

5 Selected exercises

Motivation. Confounded effect = block contrast, the 2^p-block counting rule, and where block noise goes.

Exercise 7.1 (Confounded effect = blocks; Montgomery 7.22)

(a) For the 2^2 design in two blocks with AB confounded, prove algebraically that $SS_{AB} = SS_{\text{Blocks}}$.
(b) More generally, for an unreplicated 2^k in two blocks confounding effect E, show $SS_E = SS_{\text{Blocks}}$.

Solution.

(b) contains (a) ($k = 2$, $E = AB$, blocks $\{(1), ab\}$ and $\{a, b\}$).

Step 1. Let B_+ and B_- be the totals of the two blocks, the runs with sign $+$ and $-$ in column E, each of 2^{k-1} runs. Then $\text{Contrast}_E = B_+ - B_-$ and $SS_E = (B_+ - B_-)^2/2^k$.

Step 2. As for blocks in Ch. 4, with $y_{\cdot\cdot} = B_+ + B_-$,

$$SS_{\text{Blocks}} = \frac{B_+^2 + B_-^2}{2^{k-1}} - \frac{(B_+ + B_-)^2}{2^k} = \frac{2B_+^2 + 2B_-^2 - (B_+ + B_-)^2}{2^k} = \frac{(B_+ - B_-)^2}{2^k}.$$

For $k = 2$: $SS_{AB} = \frac{1}{4}\{ab + (1) - a - b\}^2 = SS_{\text{Blocks}}$. One and the same degree of freedom: the ANOVA lists only one. ■

Exercise 7.2 (The algebra of 2^p blocks)

Identify an effect $A^{\alpha_1} \cdots K^{\alpha_k}$ with $\alpha \in \{0,1\}^k \setminus \{\mathbf{0}\}$ and a run with $\mathbf{x} \in \{0,1\}^k$. Blocks are formed by p linearly independent (over arithmetic mod 2) defining contrasts $\alpha^{(1)}, \ldots, \alpha^{(p)}$: runs with equal $(L_1(\mathbf{x}), \ldots, L_p(\mathbf{x}))$, $L_r(\mathbf{x}) = \alpha^{(r)} \cdot \mathbf{x} \bmod 2$, share a block.
(a) Show that there are 2^p blocks of 2^{k-p} runs each.
(b) Show that an effect α is constant in sign within every block iff α is a nonzero sum (mod 2) of some of the $\alpha^{(r)}$ (a generalized interaction), so exactly $2^p - 1$ effects are confounded.
(c) Show that every other effect has equally many $+$ and $-$ runs in every block, so its estimate is free of additive block effects.

Solution [Supplement]. Work in $\mathbb{F}_2^k$. By "Why L works", the sign of run $\mathbf{x}$ in column α is $\pm(-1)^{\alpha\cdot\mathbf{x}}$ ($\pm$ depending only on α), determined by $\alpha\cdot\mathbf{x} \bmod 2$.

(a) $T(\mathbf{x}) = (L_1(\mathbf{x}), \dots, L_p(\mathbf{x}))$ is linear $\mathbb{F}_2^k \to \mathbb{F}_2^p$ of rank p, hence onto; the blocks are its 2^p fibres, cosets of $\ker T$ with 2^{k-p} elements.

(b)

Step 1 ($\Leftarrow$). If $\alpha = \sum_{r\in R}\alpha^{(r)}$, then $\alpha\cdot\mathbf{x} = \sum_{r\in R} L_r(\mathbf{x})$, which is constant on each block.

Step 2 ($\Rightarrow$). If $\alpha\cdot\mathbf{x}$ is constant on blocks, it is 0 on $\ker T$ (its value at $\mathbf{0}$), so $\alpha \in (\ker T)^\perp$, of dimension p. The row space of T lies in $(\ker T)^\perp$ and has dimension p, so they coincide: α is a combination of the $\alpha^{(r)}$.

Step 3 (count). The row space has 2^p elements; removing $\mathbf{0}$ leaves $2^p - 1$ confounded effects, one per block degree of freedom.

(c) Otherwise, by Step 2, $\mathbf{x} \mapsto \alpha\cdot\mathbf{x}$ is a nonzero functional on $\ker T$, taking 0 and 1 equally often (2^{k-p-1} times; its kernel has index 2). On $\mathbf{x}_0 + \ker T$, $\alpha\cdot\mathbf{x} = \alpha\cdot\mathbf{x}_0 + \alpha\cdot\mathbf{z}$ keeps the balance, so block effects cancel in Contrast_α. ■

Exercise 7.3 (Where does the block noise go?)

An unreplicated 2^k ($N = 2^k$) must be run on two batches of $N/2$ runs, with random batch effects δ_1, δ_2 iid with variance σ_δ^2, independent of the errors (variance σ^2). Each effect is estimated by $\hat{E} = \text{Contrast}/2^{k-1}$.
(a) If the highest-order interaction is confounded with batches, show that every other effect has variance $\sigma^2/2^{k-2}$, while the confounded contrast picks up an extra $2\sigma_\delta^2$.
(b) If instead the runs are split into the two batches completely at random, show that every effect has average variance $\sigma^2/2^{k-2} + 2\sigma_\delta^2/(N-1)$.
(c) Compare the total extra variance summed over the $N-1$ effects in (a) and (b).

Solution [Supplement].

Step 1 (general form). With column $\mathbf{c}$ ($N/2$ entries $+1$, $N/2$ entries -1) and $s = \sum_{\mathbf{x}\in\text{batch }1} c(\mathbf{x})$, the batch part of the contrast is $\delta_1 s + \delta_2(0-s) = (\delta_1 - \delta_2)s$. The errors contribute $\sigma^2\mathbf{c}^T\mathbf{c} = N\sigma^2$ to Var(Contrast) and are independent of the batches, so given the assignment

$$\text{Var}(\hat{E}) = \frac{N\sigma^2 + 2\sigma_\delta^2 s^2}{(N/2)^2} = \frac{\sigma^2}{2^{k-2}} + \frac{8\sigma_\delta^2 s^2}{N^2}.$$

(a) By Exercise 7.2(c) ($p = 1$), $s = 0$ for every unconfounded effect. For the confounded one $s = \pm N/2$, giving the extra $8\sigma_\delta^2(N/2)^2/N^2 = 2\sigma_\delta^2$.

(b) Batch 1 is a random half of the runs, so the number H of $+1$ entries in it is hypergeometric (population N, $N/2$ successes, $N/2$ draws), with $\text{Var}H = \frac{N}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{N/2}{N-1} = \frac{N^2}{16(N-1)}$ and $\text{E}H = N/4$. Since $s = 2H - N/2$, $\text{E}s^2 = 4\text{Var}H = N^2/\{4(N-1)\}$, and the average extra variance is $8\sigma_\delta^2\text{E}s^2/N^2 = 2\sigma_\delta^2/(N-1)$.

(c) Both totals are $2\sigma_\delta^2$: spread over all $N-1$ effects in (b), concentrated in the sacrificed effect in (a), which leaves the others as precise as in a homogeneous experiment. ■

Chapter 8. Fitting Regression Models

1 Introduction

- Regression analysis studies relationships between a response y (dependent variable) and regressors $x_1, \dots, x_k$ (independent variables); we build a model $y = f(x_1, \dots, x_k) + \varepsilon$.
- Simple linear regression ($k = 1$): $y = \beta_0 + \beta_1 x + \varepsilon$.
- Polynomial regression ($k = 1$): $y = \beta_0 + \beta_1 x + \cdots + \beta_p x^p + \varepsilon$.
- Multiple linear regression ($k \geq 2$): $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \varepsilon$.

Steps for a simple linear regression analysis

1. Plot a scatter diagram (y vs x). If linear go to step 2; if not, consider a nonlinear $f(x)$ or transform to linearize.
2. Estimate β_0, β_1.
3. Perform residual analysis to assess model adequacy.
4. Go back to step 1 if the model does not pass the residual analysis.

2 Linear Regression Models

2.1 Simple linear regression

$y = \beta_0 + \beta_1 x + \varepsilon$, errors iid $(0, \sigma^2)$. **Important**: x is *not* random; y is. Hence

$$\mathrm{E}(y \mid x) = \beta_0 + \beta_1 x, \qquad \mathrm{Var}(y \mid x) = \sigma^2;$$

the true regression line is the line of mean responses; β_0 (intercept) and β_1 (slope) are the regression coefficients. Example: viscosity (y) of a polymer vs temperature (x).

Theorem 8.1: Scalar least squares

Minimizing $L = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$ gives

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar x)(y_i - \bar y)}{\sum (x_i - \bar x)^2}, \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x.$$

Proof [Supplement]. $\frac{\partial L}{\partial \beta_0} = -2\sum (y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0 \Rightarrow n\hat\beta_0 + \hat\beta_1 \sum x_i = \sum y_i \Rightarrow \hat\beta_0 = \bar y - \hat\beta_1 \bar x$. $\frac{\partial L}{\partial \beta_1} = -2\sum x_i (y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0$. Substitute $\hat\beta_0$: $\sum x_i [(y_i - \bar y) - \hat\beta_1 (x_i - \bar x)] = 0$. Since $\sum \bar x (y_i - \bar y) = 0$ and $\sum \bar x (x_i - \bar x) = 0$, this is $\sum (x_i - \bar x)(y_i - \bar y) = \hat\beta_1 \sum (x_i - \bar x)^2$. ∎

2.2 Multiple linear regression in matrix form

Data $(y_i, x_{i1}, \dots, x_{ik})$, $i = 1, \dots, n$: $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i$, i.e.

$$\mathbf{y} = \mathbf{X}\boldsymbol\beta + \boldsymbol\varepsilon, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \ \mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}, \ \boldsymbol\beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_k \end{pmatrix}, \ \boldsymbol\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

$\mathbf{y}$: $n \times 1$, $\mathbf{X}$: $n \times p$ ($p = k + 1$), $\boldsymbol\beta$: $p \times 1$.

3 Least Squares Estimation

Motivation. Every design is a linear model; least squares unifies the ANOVA computations.

Theorem 8.2: Normal equations

$$L = \sum_{i=1}^{n} \varepsilon_i^2 = \boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon} = (\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^T(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}) \quad \Longrightarrow \quad \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}, \quad \hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

Proof [Supplement].

$$L = \mathbf{y}^T\mathbf{y} - \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta},$$

$$\frac{\partial L}{\partial \boldsymbol{\beta}}\Big|_{\hat{\boldsymbol{\beta}}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}.$$

(Using $\partial(\mathbf{a}^T\boldsymbol{\beta})/\partial\boldsymbol{\beta} = \mathbf{a}$ and $\partial(\boldsymbol{\beta}^T\mathbf{A}\boldsymbol{\beta})/\partial\boldsymbol{\beta} = 2\mathbf{A}\boldsymbol{\beta}$ for symmetric $\mathbf{A}$.) The Hessian $2\mathbf{X}^T\mathbf{X}$ is positive definite when $\mathbf{X}$ has full column rank, so this is a minimum. ■

Fitted values $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$ (hat matrix $\mathbf{H}$, $n \times n$). Residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}-\mathbf{H})\mathbf{y}$.

Theorem 8.3: Properties of $\hat{\beta}$

If $\mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$,

$$\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

Proof [Supplement]. $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{X}\boldsymbol{\beta}+\boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}$. Hence $\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\,\sigma^2\mathbf{I}\,\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

■

3.1 Estimating σ^2

Theorem 8.4

$$SS_E = \sum_{i=1}^{n} e_i^2 = \mathbf{e}^T\mathbf{e} = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}, \qquad \hat{\sigma}^2 = MS_E = \frac{SS_E}{n-k-1}, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement].

$$\begin{aligned} SS_E &= (\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}})^T(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{y}^T\mathbf{y} - 2\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} + \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} \\ &= \mathbf{y}^T\mathbf{y} - 2\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} + \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} \qquad (\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}) \\ &= \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}. \end{aligned}$$

Also $SS_E = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}$, so by $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \mathrm{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$,

$$\mathrm{E}(SS_E) = \sigma^2\mathrm{tr}(\mathbf{I}-\mathbf{H}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta} = \sigma^2(n-p) + 0 = \sigma^2(n-k-1).$$

■

Example 8.5: A two-regressor fit

With $n = 16$ observations and two regressors,

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} 16 & 1458 & 164 \\ 1458 & 133560 & 14946 \\ 164 & 14946 & 1726 \end{pmatrix}, \qquad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} 37577 \\ 3429550 \\ 385562 \end{pmatrix},$$

solving $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$ gives $\hat{\boldsymbol{\beta}} = (1566.078,\ 7.621,\ 8.585)^T$, i.e. $\hat{y} = 1566.08 + 7.62x_1 + 8.58x_2$ (same as `lm(y~x1+x2)`).

4 Regression Approach to the CRD with a Single Factor

Motivation. One-way ANOVA as an indicator regression: the same model.

4.1 Effects coding (constraint $\sum \tau_i = 0$)

$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, $\sum_{i=1}^{a} \tau_i = 0$. For $a = 3$, $n = 2$:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{y} = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \\ y_{31} \\ y_{32} \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \mu \\ \tau_1 \\ \tau_2 \end{pmatrix},$$

i.e. $y = \mu + \tau_1 x_1 + \tau_2 x_2 + \varepsilon$ with $x_1 = 1$ (level 1), -1 (level 3), 0 otherwise; $x_2 = 1$ (level 2), -1 (level 3), 0 otherwise.

Theorem 8.6: Effects coding reproduces the ANOVA estimates

For general a (balanced, n per level) with $x_i = 1$ at level i, -1 at level a, 0 otherwise ($i = 1, \ldots, a-1$):

$$\hat{\mu} = \bar{y}_{\cdot\cdot}, \qquad \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} \quad (i = 1, \ldots, a-1), \qquad \hat{\tau}_a = -\sum_{i<a} \hat{\tau}_i = \bar{y}_{a\cdot} - \bar{y}_{\cdot\cdot}.$$

Proof [Supplement]. Entries of $\mathbf{X}^T\mathbf{X}$: $(\mathbf{1}, \mathbf{1}) = N$; $(\mathbf{1}, x_i) = n - n = 0$; $(x_i, x_i) = n + n = 2n$; $(x_i, x_l) = n$ for $i \neq l$ (both equal -1 on the n runs at level a). So

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} N & \mathbf{0}^T \\ \mathbf{0} & n(\mathbf{I}_{a-1} + \mathbf{J}_{a-1}) \end{pmatrix}, \qquad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} y_{\cdot\cdot} \\ y_{1\cdot} - y_{a\cdot} \\ \vdots \\ y_{a-1,\cdot} - y_{a\cdot} \end{pmatrix}.$$

Since $(\mathbf{I} + \mathbf{J})(\mathbf{I} - \frac{1}{a}\mathbf{J}) = \mathbf{I} + \mathbf{J} - \frac{1}{a}\mathbf{J} - \frac{a-1}{a}\mathbf{J} = \mathbf{I}$ (as $\mathbf{J}\mathbf{J} = (a-1)\mathbf{J}$), $[n(\mathbf{I} + \mathbf{J})]^{-1} = \frac{1}{n}(\mathbf{I} - \frac{1}{a}\mathbf{J})$. Thus $\hat{\mu} = y_{\cdot\cdot}/N$ and

$$\hat{\tau}_i = \frac{1}{n}\Big[(y_{i\cdot} - y_{a\cdot}) - \frac{1}{a}\sum_{l=1}^{a-1}(y_{l\cdot} - y_{a\cdot})\Big] = \frac{1}{n}\Big[y_{i\cdot} - y_{a\cdot} - \frac{y_{\cdot\cdot} - a\,y_{a\cdot}}{a}\Big] = \frac{y_{i\cdot}}{n} - \frac{y_{\cdot\cdot}}{an} = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}.$$

■

Example 8.7: Four treatments, $n = 4$

Trt	Observations				$\bar{y}_{i\cdot}$
1	3129	3000	2865	2890	2971.00
2	3200	3300	2975	3150	3156.25
3	2800	2900	2985	3050	2933.75
4	2600	2700	2600	2765	2666.25

$y = \mu + \tau_1 x_1 + \tau_2 x_2 + \tau_3 x_3 + \varepsilon$ with effects coding ($x_i = -1$ for level 4). The 16×4 design matrix has rows $(1,1,0,0) \times 4$, $(1,0,1,0) \times 4$, $(1,0,0,1) \times 4$, $(1,-1,-1,-1) \times 4$ and

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} 16 & 0 & 0 & 0 \\ 0 & 8 & 4 & 4 \\ 0 & 4 & 8 & 4 \\ 0 & 4 & 4 & 8 \end{pmatrix}, \qquad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} 46909 \\ 1219 \\ 1960 \\ 1070 \end{pmatrix}$$

($1219 = y_{1\cdot} - y_{4\cdot} = 11884 - 10665$, etc.). Solving: $\hat{\mu} = 2931.8125$, $\hat{\tau}_1 = 39.1875$, $\hat{\tau}_2 = 224.4375$, $\hat{\tau}_3 = 1.9375$, and $\hat{\tau}_4 = -(39.1875 + 224.4375 + 1.9375) = -265.5625$ — exactly $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$.

4.2 Reference-cell (dummy) coding (constraint $\tau_1 = 0$)

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \mu \\ \tau_2 \\ \tau_3 \end{pmatrix}, \qquad x_1 = \mathbb{1}(\text{level 2}),\ x_2 = \mathbb{1}(\text{level 3}).$$

Now $\hat{\mu} = \bar{y}_{1\cdot}$ and $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{1\cdot}$: μ is the mean of the reference level. The fitted values $\hat{y}_{ij} = \bar{y}_{i\cdot}$, the residuals and all SS are identical under both codings (same column space), only the meaning of the parameters differs.

Remark 8.8: fitted values do not depend on the coding

CRD with $a = 4$, $n = 2$, $\mathbf{y} = (5,5,0,0,0,0,-5,-5)^T$ and effects coding. Since $\hat{y}_{ij} = \bar{y}_{i\cdot}$ for any full-rank coding of the one-way model: $\bar{y}_{1\cdot} = 5$, $\bar{y}_{2\cdot} = 0$, $\bar{y}_{3\cdot} = 0$, $\bar{y}_{4\cdot} = -5$, so $\hat{\mathbf{y}} = (5,5,0,0,0,0,-5,-5)^T$ and $\mathbf{e} = \mathbf{0}$. (Directly: $\hat{\mu} = 0$, $\hat{\tau} = (5,0,0)$, $\hat{\tau}_4 = -5$.)

5 Regression Approach to the 2^2 Design

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk} \iff y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon,$$

with the chemical-process cell averages:

$$\mathbf{y} = \begin{pmatrix} 80/3 \\ 100/3 \\ 60/3 \\ 90/3 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{X}^T\mathbf{X} = 4\mathbf{I}_4,$$

$\hat{\boldsymbol{\beta}} = \frac{1}{4}\mathbf{X}^T\mathbf{y} = (27.5,\ 4.1667,\ -2.5,\ 0.8333)^T$: the coefficients are half the effects $A = 8.33$, $B = -5.00$, $AB = 1.67$ (Ch. 6).

6 Test for Significance of Regression

Motivation. Does any regressor matter? The analogue of equal treatment means.

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$ vs $H_1 : \beta_j \neq 0$ for at least one j. Rejecting H_0 means at least one regressor contributes significantly.

Theorem 8.9: ANOVA decomposition

$$\underbrace{\mathbf{y}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}}_{SS_T} = \underbrace{\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}}_{SS_R} + \underbrace{\mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}}_{SS_E}, \qquad (n-1) = k + (n-k-1).$$

Proof [Supplement]. Add and subtract $\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$. In matrix form $SS_T = \mathbf{y}^T(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{y}$, $SS_R = \mathbf{y}^T(\mathbf{H} - \frac{1}{n}\mathbf{J})\mathbf{y}$, $SS_E = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$ (because $\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} = \mathbf{y}^T\mathbf{H}\mathbf{y}$ and $\mathbf{1} \in \Omega(\mathbf{X})$). $\mathbf{H} - \frac{1}{n}\mathbf{J}$ and $\mathbf{I} - \mathbf{H}$ are idempotent with ranks k and $n-k-1$ and $(\mathbf{I}-\mathbf{H})(\mathbf{H} - \frac{1}{n}\mathbf{J}) = \mathbf{0}$, so (Theorems 3.11 and 3.12) under H_0 (where $(\mathbf{H} - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta} = \beta_0\mathbf{1} - \beta_0\mathbf{1} = \mathbf{0}$), $SS_R/\sigma^2 \sim \chi^2_k$, $SS_E/\sigma^2 \sim \chi^2_{n-k-1}$, independent. ■

$$MS_R = \frac{SS_R}{k}, \qquad MS_E = \frac{SS_E}{n-k-1}, \qquad F_0 = \frac{MS_R}{MS_E} \overset{H_0}{\sim} F_{k,n-k-1}; \quad \text{reject if } F_0 > F_{\alpha,k,n-k-1}.$$

Source	SS	DF	MS	F_0
Regression	$\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}$	k	MS_R	MS_R/MS_E
Error	$\mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$	$n-k-1$	MS_E	
Total	$\mathbf{y}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}$	$n-1$		

6.1 Tests on individual coefficients

$H_0 : \beta_j = 0$ vs $H_1 : \beta_j \neq 0$:

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}, \qquad C_{jj} = \left[(\mathbf{X}^T\mathbf{X})^{-1}\right]_{jj};$$

reject if $|t_0| > t_{\alpha/2,n-k-1}$.

Why t [Supplement]. $\hat{\beta}_j \sim N(\beta_j, \sigma^2 C_{jj})$; $(n-p)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-p}$; and $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp SS_E$ because

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\sigma^2\mathbf{I}(\mathbf{I}-\mathbf{H}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^T - \mathbf{X}^T\mathbf{H}) = \mathbf{0}$$

($\mathbf{X}^T\mathbf{H} = (\mathbf{H}\mathbf{X})^T = \mathbf{X}^T$). Hence $\dfrac{(\hat{\beta}_j - \beta_j)/\sqrt{\sigma^2 C_{jj}}}{\sqrt{\hat{\sigma}^2/\sigma^2}} = \dfrac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t_{n-p}$. ■

7 Confidence Intervals in Regression

On coefficients. $\dfrac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t_{n-(k+1)}$, $j = 0, \ldots, k$, so

$$\hat{\beta}_j - t_{\alpha/2,n-(k+1)}\,se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{\alpha/2,n-(k+1)}\,se(\hat{\beta}_j).$$

On the mean response at $\mathbf{x}_0 = (1, x_{01}, \ldots, x_{0k})^T$:

1. $\hat{y}_0 = \mathbf{x}_0^T \hat{\boldsymbol{\beta}}$;
2. $\mathrm{E}(\hat{y}_0) = \mathbf{x}_0^T \boldsymbol{\beta} = \mathrm{E}(y \mid \mathbf{x}_0)$, $\mathrm{Var}(\hat{y}_0) = \mathbf{x}_0^T \mathrm{Var}(\hat{\boldsymbol{\beta}}) \mathbf{x}_0 = \sigma^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0$;
3. $\dfrac{\hat{y}_0 - \mathrm{E}(y \mid \mathbf{x}_0)}{\sqrt{\hat{\sigma}^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}} \sim t_{n-(k+1)}$;
4. $100(1-\alpha)\%$ CI:

$$\hat{y}_0 - t_{\alpha/2, n-(k+1)} \sqrt{\hat{\sigma}^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0} \le \mathrm{E}(y \mid \mathbf{x}_0) \le \hat{y}_0 + t_{\alpha/2, n-(k+1)} \sqrt{\hat{\sigma}^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}.$$

8 The Extra Sum of Squares Principle (General)

Partition $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ (full model) and test $H_0 : \boldsymbol{\beta}_1 = \mathbf{0}$ via the reduced model $\mathbf{y} = \mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$:

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})] / [df(\mathrm{R}) - df(\mathrm{F})]}{SS_E(\mathrm{F}) / df(\mathrm{F})} \overset{H_0}{\sim} F_{df(\mathrm{R}) - df(\mathrm{F}),\ df(\mathrm{F})}.$$

Proof: $SS_E(\mathrm{R}) - SS_E(\mathrm{F}) = \mathbf{y}^T (\mathbf{H}_F - \mathbf{H}_R) \mathbf{y}$, with $\mathbf{H}_F - \mathbf{H}_R$ idempotent, orthogonal to $\mathbf{I} - \mathbf{H}_F$ and $(\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ under H_0 (exactly as for the RCBD in Ch. 4). This principle underlies the RCBD, BIBD (adjusted SS), Tukey's 1-df test and ANCOVA.

9 Selected exercises

Motivation. Placement of x's, unequal variances, and leverage identities.

Exercise 8.1 (Where to put the points; Montgomery 10.15)

A straight line $y = \beta_0 + \beta_1 x + \varepsilon$ is fitted with n (even) runs at levels $x_1, \ldots, x_n$ in the coded region $[-1, 1]$. Show that $\mathrm{Var}(\hat{\beta}_1)$ is minimized by putting $n/2$ runs at each end, and discuss the price of this design.

Solution.

Step 1. Since $\sum_i (x_i - \bar{x})\bar{y} = 0$, the scalar least squares estimate is $\hat{\beta}_1 = \sum_i (x_i - \bar{x}) y_i / S_{xx}$, a linear combination of independent y_i, so $\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \sum_i (x_i - \bar{x})^2 / S_{xx}^2 = \sigma^2 / S_{xx}$ (with $S_{xx} > 0$, i.e. not all x_i equal). We therefore maximize S_{xx}.

Step 2. $S_{xx} = \sum_i x_i^2 - n\bar{x}^2 \le \sum_i x_i^2 \le n$, since $|x_i| \le 1$. Equality in both requires $x_i^2 = 1$ for all i and $\bar{x} = 0$, i.e. half the runs at -1 and half at $+1$; this design attains $S_{xx} = n$ and $\mathrm{Var}(\hat{\beta}_1) = \sigma^2 / n$.

Step 3 (price). With two levels, $\beta_{11} x^2$ is indistinguishable from an intercept shift: no lack-of-fit check. Center runs (Ch. 6) buy a curvature check for a little precision; use the end-point design only when linearity is certain. ■

Exercise 8.2 (Weighted least squares; Montgomery 10.16)

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of full rank p and $\mathrm{Var}(\varepsilon_i) = \sigma^2 / w_i$ (independent; known $w_i > 0$), and $\mathbf{W} = \mathrm{diag}(w_1, \ldots, w_n)$.

(a) Show that minimizing $\sum_i w_i(y_i - \mathbf{x}_i^T\boldsymbol{\beta})^2$ gives $\hat{\boldsymbol{\beta}}_W = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\mathbf{y}$. For a straight line, write the two normal equations.
(b) Show $\mathrm{E}\hat{\boldsymbol{\beta}}_W = \boldsymbol{\beta}$ and $\mathrm{Var}(\hat{\boldsymbol{\beta}}_W) = \sigma^2(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}$.

Solution.

(a)

Step 1. With $\mathbf{W}^{1/2} = \mathrm{diag}(\sqrt{w_i})$, $\mathbf{y}^* = \mathbf{W}^{1/2}\mathbf{y}$ and $\mathbf{X}^* = \mathbf{W}^{1/2}\mathbf{X}$, the criterion is $\|\mathbf{y}^* - \mathbf{X}^*\boldsymbol{\beta}\|^2$, an ordinary least squares problem; $\mathbf{X}^*$ has full rank because $\mathbf{W}^{1/2}$ is invertible.

Step 2. By the normal-equations theorem, $\mathbf{X}^{*T}\mathbf{X}^*\hat{\boldsymbol{\beta}} = \mathbf{X}^{*T}\mathbf{y}^*$, i.e. $\mathbf{X}^T\mathbf{W}\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{W}\mathbf{y}$, with the stated unique solution. For $y = \beta_0 + \beta_1 x$,

$$\hat{\beta}_0\sum_i w_i + \hat{\beta}_1\sum_i w_i x_i = \sum_i w_i y_i, \qquad \hat{\beta}_0\sum_i w_i x_i + \hat{\beta}_1\sum_i w_i x_i^2 = \sum_i w_i x_i y_i.$$

(b) $\hat{\boldsymbol{\beta}}_W = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\boldsymbol{\varepsilon}$, so $\mathrm{E}\hat{\boldsymbol{\beta}}_W = \boldsymbol{\beta}$, and with $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{W}^{-1}$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}_W) = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}(\sigma^2\mathbf{W}^{-1})\mathbf{W}\mathbf{X}(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}.$$

The transformed errors $\mathbf{W}^{1/2}\boldsymbol{\varepsilon}$ have covariance $\sigma^2\mathbf{I}$, so all the ordinary least squares results (tests, intervals) apply to the starred model; observations with small variance get large weight. ■

Exercise 8.3 (Leverage and deleted residuals)

In the model with an intercept, let h_{ii} be the diagonal entries of $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ and e_i the residuals.
(a) Show $\sum_i h_{ii} = p$, $\frac{1}{n} \le h_{ii} \le 1$ and $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$.
(b) Let $\hat{y}_{(i)}$ be the prediction of y_i from the fit without observation i (assume $h_{ii} < 1$). Show $y_i - \hat{y}_{(i)} = e_i/(1 - h_{ii})$, so PRESS $= \sum_i\{e_i/(1 - h_{ii})\}^2$ needs only one fit.

Solution [Supplement].

(a)

Step 1. $\mathbf{H}$ is symmetric and idempotent, and $\mathrm{tr}\mathbf{H} = \mathrm{tr}\{(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\} = \mathrm{tr}\mathbf{I}_p = p$.

Step 2. $h_{ii} = (\mathbf{H}^2)_{ii} = \sum_j h_{ij}^2 \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. Because $\mathbf{1} \in$ the column space of $\mathbf{X}$, $\mathbf{H} - \frac{1}{n}\mathbf{J}$ is also symmetric idempotent ($\mathbf{H}\mathbf{J} = \mathbf{J}\mathbf{H} = \mathbf{J}$), so its diagonal entries $h_{ii} - \frac{1}{n}$ are ≥ 0 by the same argument.

Step 3. $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$ and $\mathrm{Var}(\mathbf{e}) = \sigma^2(\mathbf{I} - \mathbf{H})(\mathbf{I} - \mathbf{H})^T = \sigma^2(\mathbf{I} - \mathbf{H})$, whose diagonal is $\sigma^2(1 - h_{ii})$. High-leverage points have small residual variance, which is why residuals are standardized by $\sqrt{MS_E(1 - h_{ii})}$.

(b)

Step 1. Let $\mathbf{A} = \mathbf{X}^T\mathbf{X}$ and $\mathbf{x}_i^T$ the ith row. Without observation i, $\mathbf{X}_{(i)}^T\mathbf{X}_{(i)} = \mathbf{A} - \mathbf{x}_i\mathbf{x}_i^T$ and $\mathbf{X}_{(i)}^T\mathbf{y}_{(i)} = \mathbf{X}^T\mathbf{y} - \mathbf{x}_i y_i$. Multiplying out shows (Sherman–Morrison)

$$(\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^T)^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1}\mathbf{x}_i\mathbf{x}_i^T\mathbf{A}^{-1}}{1 - h_{ii}}, \qquad h_{ii} = \mathbf{x}_i^T\mathbf{A}^{-1}\mathbf{x}_i.$$

Step 2. Hence, with $\hat{y}_i = \mathbf{x}_i^T\mathbf{A}^{-1}\mathbf{X}^T\mathbf{y}$,

$$\hat{y}_{(i)} = \mathbf{x}_i^T(\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^T)^{-1}(\mathbf{X}^T\mathbf{y} - \mathbf{x}_i y_i) = \Big(1 + \frac{h_{ii}}{1 - h_{ii}}\Big)\mathbf{x}_i^T\mathbf{A}^{-1}(\mathbf{X}^T\mathbf{y} - \mathbf{x}_i y_i)$$

$$= \frac{\hat{y}_i - h_{ii} y_i}{1 - h_{ii}}.$$

Step 3. $y_i - \hat{y}_{(i)} = \dfrac{(1 - h_{ii}) y_i - \hat{y}_i + h_{ii} y_i}{1 - h_{ii}} = \dfrac{y_i - \hat{y}_i}{1 - h_{ii}} = \dfrac{e_i}{1 - h_{ii}}.$ ■

Chapter 9. Experiments with Random Factors

1 Introduction: Fixed vs Random Factors

- **Fixed factor**: a specific set of levels is chosen; inference is confined to those levels. Quantitative factors are usually fixed (levels chosen to span a region for a response curve).
- **Random factor**: levels are chosen at random from a larger population of potential levels; inference is about the *entire population* of levels. Industrial applications include measurement-system (gauge R&R) studies.

2 The One-Way Random Effects Model

2.1 Example: looms

Fabric is woven on hundreds of looms; do looms differ in strength? Four looms chosen *at random*, four determinations each ($a = 4$, $n = 4$): a random factor.

Loom	Observations				$y_{i\cdot}$
1	98	97	99	96	390
2	91	90	93	92	366
3	96	95	97	95	383
4	95	96	99	98	388
					1527

2.2 Model and covariance structure

Definition 9.1

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \qquad i = 1, \dots, a;\ j = 1, \dots, n,$$

$\tau_i \overset{\text{iid}}{\sim} N(0, \sigma_\tau^2)$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, $\tau_i \perp\!\!\!\perp \varepsilon_{ij}$. σ_τ^2 and σ^2 are **variance components**.

Derivation

$$\begin{aligned}
\mathrm{E}(y_{ij}) &= \mu, \qquad \mathrm{Var}(y_{ij}) = \mathrm{Var}(\tau_i) + \mathrm{Var}(\varepsilon_{ij}) = \sigma_\tau^2 + \sigma^2, \\
\mathrm{Cov}(y_{ij}, y_{ij'}) &= \mathrm{Cov}(\tau_i + \varepsilon_{ij}, \tau_i + \varepsilon_{ij'}) = \mathrm{E}\big[(\tau_i + \varepsilon_{ij})(\tau_i + \varepsilon_{ij'})\big] = \mathrm{E}(\tau_i^2) = \sigma_\tau^2 \quad (j \neq j'), \\
\mathrm{Cov}(y_{ij}, y_{i'j'}) &= \mathrm{E}\big[(\tau_i + \varepsilon_{ij})(\tau_{i'} + \varepsilon_{i'j'})\big] = 0 \quad (i \neq i').
\end{aligned}$$

So with $\mathbf{y} = (y_{11}, \dots, y_{1n}, \dots, y_{a1}, \dots, y_{an})^T$,

$$\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{y}) = \begin{pmatrix} \sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n & & \\ & \ddots & \\ & & \sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n \end{pmatrix} = \mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n),$$

a block-diagonal matrix: observations from the same loom are correlated (intraclass correlation $\rho = \sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$), different looms are independent. Under $H_0 : \sigma_\tau^2 = 0$, $\mathrm{Cov}(\mathbf{y}) = \sigma^2\mathbf{I}$.

2.3 Hypotheses

The individual treatments are not of specific interest; we care about the population:

$H_0 : \sigma_\tau^2 = 0$ (all treatments identical) vs $H_1 : \sigma_\tau^2 > 0$ (variability exists between treatments).

2.4 ANOVA and expected mean squares

The partition $SS_T = SS_{\text{Treatments}} + SS_E$ with df $an - 1 = (a-1) + a(n-1)$ is purely algebraic and still holds.

Theorem 9.2

$$\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + n\sigma_\tau^2, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$ and $\bar{y}_{\cdot\cdot} = \mu + \bar{\tau} + \bar{\varepsilon}_{\cdot\cdot}$ (now $\bar{\tau} = \frac{1}{a}\sum \tau_i$ is random, not 0). Thus

$$\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = (\tau_i - \bar{\tau}) + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}),$$

and the two brackets are independent with mean zero, so the cross term has expectation zero:

$$\mathrm{E}(MS_{\text{Treatments}}) = \mathrm{E}\Big[\frac{n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}{a-1}\Big] = n\underbrace{\mathrm{E}\Big[\frac{\sum_i(\tau_i - \bar{\tau})^2}{a-1}\Big]}_{=\mathrm{Var}(\tau_i)=\sigma_\tau^2} + n\underbrace{\mathrm{E}\Big[\frac{\sum_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2}{a-1}\Big]}_{=\mathrm{Var}(\bar{\varepsilon}_{i\cdot})=\sigma^2/n} = n\sigma_\tau^2 + \sigma^2.$$

(A sample variance of iid variables is unbiased for their variance.) For the error, $y_{ij} - \bar{y}_{i\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}$ (τ_i cancels), so

$$\mathrm{E}(MS_E) = \mathrm{E}\Big[\frac{\sum\sum(y_{ij} - \bar{y}_{i\cdot})^2}{a(n-1)}\Big] = \frac{1}{a}\sum_{i=1}^{a}\mathrm{E}\Big[\frac{\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2}{n-1}\Big] = \sigma^2.$$

■

Test: $F_0 = MS_{\text{Treatments}}/MS_E \overset{H_0}{\sim} F_{a-1,N-a}$; reject $H_0 : \sigma_\tau^2 = 0$ if $F_0 > F_{\alpha,a-1,N-a}$. (Same computation as the fixed case; only the hypothesis and interpretation differ.)

2.5 Quadratic-form treatment

Theorem 9.3

Let $\mathbf{A} = \frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n - \frac{1}{N}\mathbf{J}_N$. Then (a) $SS_{\text{Treatments}} = \mathbf{y}^T\mathbf{A}\mathbf{y}$;
(b) $\mathbf{A}\boldsymbol{\Sigma} = (\sigma^2 + n\sigma_\tau^2)\mathbf{A}$, so $\mathbf{A}\boldsymbol{\Sigma}/(\sigma^2 + n\sigma_\tau^2) = \mathbf{A}$ is idempotent and $SS_{\text{Treatments}}/(\sigma^2 + n\sigma_\tau^2) \sim \chi^2_{a-1}$;
(c) $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = (a-1)(\sigma^2 + n\sigma_\tau^2)$.

Proof [Supplement]. (a) $\frac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n)\mathbf{y}$ replaces each y_{ij} by $\bar{y}_{i\cdot}$, and $\frac{1}{N}\mathbf{J}_N\mathbf{y} = \bar{y}_{\cdot\cdot}\mathbf{1}$. So $\mathbf{A}\mathbf{y}$ is the vector of $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$. $\mathbf{A}$ is symmetric and idempotent (it is $\mathbf{H} - \frac{1}{N}\mathbf{J}_N$ of Ch. 3), hence $\mathbf{y}^T\mathbf{A}\mathbf{y} = (\mathbf{A}\mathbf{y})^T(\mathbf{A}\mathbf{y}) = \sum_i\sum_j(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

(b) Using $(\mathbf{P}\otimes\mathbf{Q})(\mathbf{R}\otimes\mathbf{S}) = \mathbf{PR}\otimes\mathbf{QS}$ and $\mathbf{J}_n\mathbf{J}_n = n\mathbf{J}_n$:

$$\tfrac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n)(\mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n)) = \mathbf{I}_a \otimes \tfrac{1}{n}(\sigma^2\mathbf{J}_n + n\sigma_\tau^2\mathbf{J}_n) = (\sigma^2 + n\sigma_\tau^2)\,\tfrac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n,$$
$$\tfrac{1}{N}(\mathbf{J}_a \otimes \mathbf{J}_n)(\mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n)) = \tfrac{1}{N}\mathbf{J}_a \otimes (\sigma^2 + n\sigma_\tau^2)\mathbf{J}_n = (\sigma^2 + n\sigma_\tau^2)\tfrac{1}{N}\mathbf{J}_N.$$

Subtracting, $\mathbf{A\Sigma} = (\sigma^2 + n\sigma_\tau^2)\mathbf{A}$. For $\mathbf{y} \sim N(\mu\mathbf{1}, \mathbf{\Sigma})$, $\mathbf{y}^T\mathbf{B}\mathbf{y} \sim \chi^2_{\text{rank}\mathbf{B}}(\cdot)$ when $\mathbf{B\Sigma}$ is idempotent; with $\mathbf{B} = \mathbf{A}/(\sigma^2 + n\sigma_\tau^2)$, $\mathbf{B\Sigma} = \mathbf{A}$ is idempotent of rank $\text{tr}\mathbf{A} = a - 1$, and the noncentrality $\mu^2\mathbf{1}^T\mathbf{B}\mathbf{1} = 0$ since $\mathbf{A}\mathbf{1} = \mathbf{0}$.

(c) $\text{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \text{tr}(\mathbf{A\Sigma}) + \mu^2\mathbf{1}^T\mathbf{A}\mathbf{1} = (\sigma^2 + n\sigma_\tau^2)\text{tr}(\mathbf{A}) + 0 = (a-1)(\sigma^2 + n\sigma_\tau^2)$. ■

2.6 Estimating variance components

ANOVA (moment) estimators, from equating MS to EMS:

$$\hat\sigma^2 = MS_E, \qquad \hat\sigma_\tau^2 = \frac{MS_{\text{Treatments}} - MS_E}{n}, \qquad \widehat{\text{Var}}(y_{ij}) = \hat\sigma_y^2 = \hat\sigma_\tau^2 + \hat\sigma^2.$$

Unbiased: $\text{E}(\hat\sigma_\tau^2) = \frac{(\sigma^2 + n\sigma_\tau^2) - \sigma^2}{n} = \sigma_\tau^2$. Also $\hat\sigma_\tau^2 = \widehat{\text{Cov}}(y_{ij}, y_{ij'})$ and $\widehat{\text{Cov}}(\mathbf{y})$ has $\hat\sigma^2 + \hat\sigma_\tau^2$ on the diagonal blocks and $\hat\sigma_\tau^2$ off-diagonal within blocks.

Potential problems:

(1) **negative estimates** when $MS_{\text{Treatments}} < MS_E$;

(2) they are moment estimators and do not have the best statistical properties (REML/ML via `lme` are alternatives).

2.7 Confidence intervals

For σ^2. Under normality $\frac{(N-a)MS_E}{\sigma^2} \sim \chi^2_{N-a}$, so

$$P\Big(\chi^2_{1-\alpha/2,N-a} \le \frac{(N-a)MS_E}{\sigma^2} \le \chi^2_{\alpha/2,N-a}\Big) = 1-\alpha \;\Rightarrow\; \frac{(N-a)MS_E}{\chi^2_{\alpha/2,N-a}} \le \sigma^2 \le \frac{(N-a)MS_E}{\chi^2_{1-\alpha/2,N-a}}.$$

For σ_τ^2: no exact interval exists.

For the intraclass correlation $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$. Since $\frac{(a-1)MS_{\text{Treatments}}}{\sigma^2 + n\sigma_\tau^2} \sim \chi^2_{a-1}$ and is independent of $\frac{(N-a)MS_E}{\sigma^2} \sim \chi^2_{N-a}$,

$$\mathcal{A} = \frac{MS_{\text{Treatments}}/(n\sigma_\tau^2 + \sigma^2)}{MS_E/\sigma^2} \sim F_{a-1,N-a} \quad\Rightarrow\quad P(F_{1-\alpha/2,a-1,N-a} \le \mathcal{A} \le F_{\alpha/2,a-1,N-a}) = 1-\alpha.$$

Inverting the pivot [Supplement]. Let $\theta = \sigma_\tau^2/\sigma^2$ and $F^* = MS_{\text{Treatments}}/MS_E$. Then $\mathcal{A} = F^*/(1+n\theta)$ and

$$F_{1-\alpha/2} \le \frac{F^*}{1+n\theta} \le F_{\alpha/2} \iff \frac{1}{n}\Big(\frac{F^*}{F_{\alpha/2}} - 1\Big) \le \theta \le \frac{1}{n}\Big(\frac{F^*}{F_{1-\alpha/2}} - 1\Big).$$

Call these bounds L and U. Since $\frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} = \frac{\theta}{1+\theta}$ is increasing in θ,

$$\frac{L}{1+L} \le \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} \le \frac{U}{1+U},$$

$$L = \frac{1}{n}\Big(\frac{MS_{\text{Treatments}}}{MS_E}\frac{1}{F_{\alpha/2,a-1,N-a}} - 1\Big), \quad U = \frac{1}{n}\Big(\frac{MS_{\text{Treatments}}}{MS_E}\frac{1}{F_{1-\alpha/2,a-1,N-a}} - 1\Big).$$

■

Example 9.4: Looms, line by line

$\sum y^2 = 145845$, $y_{\cdot\cdot}^2/N = 1527^2/16 = 145733.06$.

$$SS_T = 145845 - 145733.06 = 111.94,$$
$$SS_{\text{Looms}} = \frac{390^2 + 366^2 + 383^2 + 388^2}{4} - 145733.06 = \frac{583289}{4} - 145733.06 = 89.19,$$
$$SS_E = 111.94 - 89.19 = 22.75.$$

Source	SS	DF	MS	F_0
Looms	89.19	3	29.73	15.68
Error	22.75	12	1.90	
Total	111.94	15		

$F_0 = 15.68 > F_{0.05,3,12} = 3.49$ ($P = 0.00019$): reject $H_0 : \sigma_\tau^2 = 0$.

$$\hat\sigma^2 = 1.90, \qquad \hat\sigma_\tau^2 = \frac{29.73 - 1.90}{4} = 6.96, \qquad \hat\sigma_y^2 = 6.96 + 1.90 = 8.86$$

(`lme`: 6.958 and 1.896). Most of the variability ($6.96/8.86 = 78.6\%$) is due to looms.
95% CI for σ^2: $\chi^2_{0.025,12} = 23.34$, $\chi^2_{0.975,12} = 4.40$: $[\frac{12(1.896)}{23.34}, \frac{12(1.896)}{4.40}] = [0.975,\ 5.17]$.
95% CI for $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$: $F_{0.025,3,12} = 4.47$, $F_{0.975,3,12} = 0.0698$, $F^* = 15.68$:

$$L = \tfrac{1}{4}(\tfrac{15.68}{4.47} - 1) = 0.626, \quad U = \tfrac{1}{4}(\tfrac{15.68}{0.0698} - 1) = 55.95, \quad \left[\tfrac{0.626}{1.626},\ \tfrac{55.95}{56.95}\right] = [0.385,\ 0.982].$$

3 Two-Factor Factorial with Random Factors

Motivation. Randomly sampled levels contribute a variance; expected mean squares decide the F-ratio.

Definition 9.5

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk},$$

$\tau_i \sim N(0, \sigma_\tau^2)$, $\beta_j \sim N(0, \sigma_\beta^2)$, $(\tau\beta)_{ij} \sim N(0, \sigma_{\tau\beta}^2)$, $\varepsilon_{ijk} \sim N(0, \sigma^2)$, all independent.
$\text{Var}(y_{ijk}) = \sigma_\tau^2 + \sigma_\beta^2 + \sigma_{\tau\beta}^2 + \sigma^2$.

Hypotheses: $H_0 : \sigma_\tau^2 = 0$, $H_0 : \sigma_\beta^2 = 0$, $H_0 : \sigma_{\tau\beta}^2 = 0$. The ANOVA computations are unchanged; the tests change.

Theorem 9.6

$$\text{E}(MS_A) = \sigma^2 + n\sigma_{\tau\beta}^2 + bn\sigma_\tau^2, \quad \text{E}(MS_B) = \sigma^2 + n\sigma_{\tau\beta}^2 + an\sigma_\beta^2,$$
$$\text{E}(MS_{AB}) = \sigma^2 + n\sigma_{\tau\beta}^2, \quad \text{E}(MS_E) = \sigma^2.$$

$\text{E}(MS_A)$ **[Supplement].**

$$\bar y_{i\cdot\cdot} = \mu + \tau_i + \bar\beta + \overline{(\tau\beta)}_{i\cdot} + \bar\varepsilon_{i\cdot\cdot}, \qquad \bar y_{\cdots} = \mu + \bar\tau + \bar\beta + \overline{(\tau\beta)}_{\cdot\cdot} + \bar\varepsilon_{\cdots},$$
$$SS_A = bn \sum_i \left[(\tau_i - \bar\tau) + (\overline{(\tau\beta)}_{i\cdot} - \overline{(\tau\beta)}_{\cdot\cdot}) + (\bar\varepsilon_{i\cdot\cdot} - \bar\varepsilon_{\cdots})\right]^2.$$

$\bar\beta$ cancels. The three brackets are independent with mean 0, so cross terms vanish in expectation, and each is a sample-variance-type sum of iid terms:

$$\mathrm{E}(MS_A) = bn\big[\mathrm{Var}(\tau_i) + \mathrm{Var}(\overline{(\tau\beta)}_{i\cdot}) + \mathrm{Var}(\bar\varepsilon_{i\cdot\cdot})\big] = bn\Big(\sigma_\tau^2 + \frac{\sigma_{\tau\beta}^2}{b} + \frac{\sigma^2}{bn}\Big) = bn\sigma_\tau^2 + n\sigma_{\tau\beta}^2 + \sigma^2.$$

By symmetry $\mathrm{E}(MS_B) = an\sigma_\beta^2 + n\sigma_{\tau\beta}^2 + \sigma^2$. For MS_{AB}: $\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot}$ removes τ_i, β_j and leaves the RCBD-type residual of the table $\{(\tau\beta)_{ij} + \bar\varepsilon_{ij\cdot}\}$ of iid variables with variance $\sigma_{\tau\beta}^2 + \sigma^2/n$; its expected sum of squares is $(a-1)(b-1)(\sigma_{\tau\beta}^2 + \sigma^2/n)$, hence $\mathrm{E}(MS_{AB}) = n\sigma_{\tau\beta}^2 + \sigma^2$. ■

Tests (choose the denominator whose EMS equals the numerator's under H_0):

$$H_0 : \sigma_\tau^2 = 0 : \ F_0 = \frac{MS_A}{MS_{AB}}, \qquad H_0 : \sigma_\beta^2 = 0 : \ F_0 = \frac{MS_B}{MS_{AB}}, \qquad H_0 : \sigma_{\tau\beta}^2 = 0 : \ F_0 = \frac{MS_{AB}}{MS_E}.$$

Estimators:

$$\hat\sigma_\tau^2 = \frac{MS_A - MS_{AB}}{bn}, \quad \hat\sigma_\beta^2 = \frac{MS_B - MS_{AB}}{an}, \quad \hat\sigma_{\tau\beta}^2 = \frac{MS_{AB} - MS_E}{n}, \quad \hat\sigma^2 = MS_E.$$

Example 9.7: Gauge capability study

20 parts (random) × 3 operators (random), $n = 2$ measurements each.

Source	SS	DF	MS	F_0	denominator
Parts	1185.43	19	62.39	$62.39/0.71 = 87.9$	MS_{AB}
Operators	2.62	2	1.31	$1.31/0.71 = 1.84$	MS_{AB}
P×O	27.05	38	0.71	$0.71/0.99 = 0.72$	MS_E
Error	59.50	60	0.99		

$$\hat\sigma_\tau^2 = \frac{62.39 - 0.71}{3\cdot 2} = 10.28, \quad \hat\sigma_\beta^2 = \frac{1.31 - 0.71}{20\cdot 2} = 0.015,$$
$$\hat\sigma_{\tau\beta}^2 = \frac{0.71 - 0.99}{2} = -0.14, \quad \hat\sigma^2 = 0.99.$$

The negative $\hat\sigma_{\tau\beta}^2$ is consistent with a zero interaction component; one may refit the reduced model $y = \mu + \tau_i + \beta_j + \varepsilon$. Gauge variability $\hat\sigma_{\text{gauge}}^2 = \hat\sigma^2 + \hat\sigma_\beta^2$ (repeatability + reproducibility) is small relative to part variability.

4 Two-Factor Mixed Models

Motivation. In a mixed model the fixed factor is tested against the interaction mean square.

4.1 Restricted mixed model

Definition 9.8: A fixed, B random (restricted model)

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk},$$

$\sum_i \tau_i = 0$; $\beta_j \overset{\text{iid}}{\sim} N(0, \sigma_\beta^2)$; $(\tau\beta)_{ij} \sim N\big(0, \frac{a-1}{a}\sigma_{\tau\beta}^2\big)$ with the restriction $\sum_{i=1}^{a}(\tau\beta)_{ij} = (\tau\beta)_{\cdot j} = 0$ for each j; $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

Interaction terms in the same column are therefore *not* independent:

Theorem 9.9

$\operatorname{Cov}[(\tau\beta)_{ij},(\tau\beta)_{i'j}] = -\frac{1}{a}\sigma^2_{\tau\beta}$ $(i \neq i')$, and $\operatorname{Cov}[(\tau\beta)_{ij},(\tau\beta)_{ij'}] = 0$ $(j \neq j')$.

Proof [Supplement]. Let $x = \operatorname{Cov}[(\tau\beta)_{ij},(\tau\beta)_{i'j}]$ (same for all $i \neq i'$ by symmetry). Since $\sum_i (\tau\beta)_{ij} = 0$,

$$0 = \operatorname{Var}\Big(\sum_{i=1}^{a}(\tau\beta)_{ij}\Big) = \sum_{i=1}^{a}\operatorname{Var}((\tau\beta)_{ij}) + 2\binom{a}{2}x = a\cdot\frac{a-1}{a}\sigma^2_{\tau\beta} + a(a-1)x \;\Rightarrow\; x = -\frac{\sigma^2_{\tau\beta}}{a}.$$

Different columns j correspond to independently drawn random levels, so their interactions are uncorrelated. ■

Theorem 9.10: EMS, restricted model

$$\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\tau\beta} + \frac{bn\sum_i \tau_i^2}{a-1}, \quad \mathrm{E}(MS_B) = \sigma^2 + an\sigma^2_{\beta},$$
$$\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma^2_{\tau\beta}, \quad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. MS_A. $\bar{y}_{i\cdot\cdot} = \mu + \tau_i + \bar{\beta} + \overline{(\tau\beta)}_{i\cdot} + \bar{\varepsilon}_{i\cdot\cdot}$; $\bar{y}_{\cdots} = \mu + \bar{\tau} + \bar{\beta} + \overline{(\tau\beta)}_{\cdot\cdot} + \bar{\varepsilon}_{\cdots}$ with $\bar{\tau} = 0$ and $\overline{(\tau\beta)}_{\cdot\cdot} = 0$ (restriction). So $\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots} = \tau_i + \overline{(\tau\beta)}_{i\cdot} + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})$ and

$$\mathrm{E}(MS_A) = bn\frac{\sum\tau_i^2}{a-1} + bn\underbrace{\mathrm{E}\Big(\frac{\sum_i \overline{(\tau\beta)}_{i\cdot}^2}{a-1}\Big)}_{(*)} + bn\underbrace{\mathrm{E}\Big(\frac{\sum_i(\bar{\varepsilon}_{i\cdot\cdot}-\bar{\varepsilon}_{\cdots})^2}{a-1}\Big)}_{=\sigma^2/(bn)}.$$

$(*)$: $\overline{(\tau\beta)}_{i\cdot} = \frac{1}{b}\sum_j(\tau\beta)_{ij}$ is an average of b independent terms (different j), so

$$\mathrm{E}\Big(\sum_{i=1}^{a}\overline{(\tau\beta)}_{i\cdot}^2\Big) = \sum_i \operatorname{Var}(\overline{(\tau\beta)}_{i\cdot}) = \sum_{i=1}^{a}\frac{\operatorname{Var}((\tau\beta)_{ij})}{b} = \frac{a}{b}\cdot\frac{a-1}{a}\sigma^2_{\tau\beta} = \frac{a-1}{b}\sigma^2_{\tau\beta},$$

so $bn(*) = bn\cdot\frac{\sigma^2_{\tau\beta}}{b} = n\sigma^2_{\tau\beta}$. Hence $\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\tau\beta} + \frac{bn\sum\tau_i^2}{a-1}$.

MS_B. $\bar{y}_{\cdot j\cdot} = \mu + \bar{\tau} + \beta_j + \overline{(\tau\beta)}_{\cdot j} + \bar{\varepsilon}_{\cdot j\cdot} = \mu + \beta_j + \bar{\varepsilon}_{\cdot j\cdot}$ because $\bar{\tau} = 0$ and $\overline{(\tau\beta)}_{\cdot j} = 0$ by the restriction. Thus $\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots} = (\beta_j - \bar{\beta}) + (\bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}_{\cdots})$ and

$$\mathrm{E}(MS_B) = an\,\mathrm{E}\Big(\frac{\sum_j(\beta_j-\bar{\beta})^2}{b-1}\Big) + an\,\mathrm{E}\Big(\frac{\sum_j(\bar{\varepsilon}_{\cdot j\cdot}-\bar{\varepsilon}_{\cdots})^2}{b-1}\Big) = an\sigma^2_{\beta} + an\frac{\sigma^2}{an} = \sigma^2 + an\sigma^2_{\beta}.$$

The interaction does *not* appear in $\mathrm{E}(MS_B)$ in the restricted model.

MS_{AB}. $\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots} = ((\tau\beta)_{ij} - \overline{(\tau\beta)}_{i\cdot}) + (\varepsilon\text{-part})$. For fixed i, $\{(\tau\beta)_{ij}\}_{j=1}^{b}$ are iid with variance $\frac{a-1}{a}\sigma^2_{\tau\beta}$, so $\mathrm{E}\sum_j((\tau\beta)_{ij} - \overline{(\tau\beta)}_{i\cdot})^2 = (b-1)\frac{a-1}{a}\sigma^2_{\tau\beta}$; summing over i: $(a-1)(b-1)\sigma^2_{\tau\beta}$. With the error part contributing $(a-1)(b-1)\sigma^2/n$, $\mathrm{E}(MS_{AB}) = n\sigma^2_{\tau\beta} + \sigma^2$. ■

Tests: $H_0 : \tau_i = 0$: $F_0 = MS_A/MS_{AB}$; $H_0 : \sigma^2_{\beta} = 0$: $F_0 = MS_B/MS_E$; $H_0 : \sigma^2_{\tau\beta} = 0$: $F_0 = MS_{AB}/MS_E$.
Estimators: $\hat{\sigma}^2_{\beta} = \frac{MS_B - MS_E}{an}$, $\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB}-MS_E}{n}$, $\hat{\sigma}^2 = MS_E$, $\hat{\tau}_i = \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}$.

4.2 Unrestricted mixed model

Definition 9.11

$$y_{ijk} = \mu + \alpha_i + \gamma_j + (\alpha\gamma)_{ij} + \varepsilon_{ijk},$$
$$\sum_i \alpha_i = 0, \quad \gamma_j \sim N(0, \sigma^2_\gamma), \quad (\alpha\gamma)_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2_{\alpha\gamma}), \quad \varepsilon_{ijk} \sim N(0, \sigma^2),$$

with *no* sum-to-zero restriction on the interactions (all mutually independent).

Theorem 9.12: EMS, unrestricted model

$$\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\alpha\gamma} + \frac{bn\sum \alpha_i^2}{a-1}, \quad \mathrm{E}(MS_B) = \sigma^2 + n\sigma^2_{\alpha\gamma} + an\sigma^2_\gamma,$$
$$\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma^2_{\alpha\gamma}, \quad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. $\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots} = \alpha_i + [\overline{(\alpha\gamma)}_{i\cdot} - \overline{(\alpha\gamma)}_{\cdot\cdot}] + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})$ ($\bar{\gamma}$ cancels, $\bar{\alpha} = 0$). The $\overline{(\alpha\gamma)}_{i\cdot}$ are iid with variance $\sigma^2_{\alpha\gamma}/b$, so

$$\mathrm{E}(MS_A) = \frac{bn\sum \alpha_i^2}{a-1} + bn \cdot \frac{\sigma^2_{\alpha\gamma}}{b} + bn \cdot \frac{\sigma^2}{bn} = \frac{bn\sum \alpha_i^2}{a-1} + n\sigma^2_{\alpha\gamma} + \sigma^2.$$

$\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots} = (\gamma_j - \bar{\gamma}) + [\overline{(\alpha\gamma)}_{\cdot j} - \overline{(\alpha\gamma)}_{\cdot\cdot}] + (\bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}_{\cdots})$ with $\mathrm{Var}(\overline{(\alpha\gamma)}_{\cdot j}) = \sigma^2_{\alpha\gamma}/a$, so

$$\mathrm{E}(MS_B) = an\sigma^2_\gamma + an \cdot \frac{\sigma^2_{\alpha\gamma}}{a} + an \cdot \frac{\sigma^2}{an} = an\sigma^2_\gamma + n\sigma^2_{\alpha\gamma} + \sigma^2.$$

■

Consequently, in the unrestricted model $H_0 : \sigma^2_\gamma = 0$ is tested by $F_0 = MS_B/MS_{AB}$ and $\hat{\sigma}^2_\gamma = \frac{MS_B - MS_{AB}}{an}$. (Restricted vs unrestricted differ only in the test/estimator for the random main effect.)

5 Rules for Expected Mean Squares

1. The error term $\varepsilon_{ijk\ldots m}$ is written $\varepsilon_{(ijk\ldots)m}$, where m is the replication subscript.
2. For each term, classify subscripts: **live** (present, not in parentheses), **dead** (present, in parentheses), **absent** (in the model but not in that term). E.g. $(\tau\beta)_{ij}$: i, j live, k absent; $\varepsilon_{(ij)k}$: k live, i, j dead.
3. DF of a term $= \prod$(#levels of dead subscripts) $\times \prod$(#levels -1 of live subscripts). E.g. $(\tau\beta)_{ij}$: $(a-1)(b-1)$; $\varepsilon_{(ij)k}$: $ab(n-1)$.
4. If an interaction contains at least one random effect, the entire interaction is random.
5. Variance components: $\sigma^2_\tau, \sigma^2_\beta, \ldots$; fixed factors: $\sum \tau_i^2/(a-1)$, $\sum\sum(\tau\beta)_{ij}^2/((a-1)(b-1))$, …
6. Replicates are always random.

Steps (restricted model):

S1. Write the terms of the model as row labels; column headings are the subscripts, with F/R and the number of levels above them.

S2. In each row, write 1 in a column if the column's subscript is *dead* in that row's term.

S3. In each row, if the column subscript is *live* in the term, write 0 if the column factor is fixed (F) and 1 if random (R).

S4. Fill the remaining cells (absent subscripts) with the number of levels shown above the column.

S5. For the EMS of a term, cover the columns headed by the term's live subscripts.

S6. Consider every row that contains at least all the subscripts of the term; multiply the remaining numbers in that row and attach the row's fixed factor or variance component. Sum these.

Example 9.13: Two-factor mixed model via the algorithm (A fixed, B random)

	F	R	R	
	a	b	n	
Term	i	j	k	EMS
τ_i	0	b	n	$\sigma^2 + n\sigma^2_{\tau\beta} + bn\sum \tau_i^2/(a-1)$
β_j	a	1	n	$\sigma^2 + an\sigma^2_{\beta}$
$(\tau\beta)_{ij}$	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta}$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

E.g. for β_j: cover column j; rows containing j: β_j (remaining $a\cdot n$), $(\tau\beta)_{ij}$ (remaining $0\cdot n = 0$), ε $(1\cdot 1)$. So $\mathrm{E}(MS_B) = an\sigma^2_\beta + 0 + \sigma^2$ — the zero in the fixed column is exactly the restriction $\sum_i (\tau\beta)_{ij} = 0$. For τ_i: cover i; rows containing i: τ_i (bn), $(\tau\beta)_{ij}$ $(1\cdot n)$, ε (1).
With A random and B fixed the table is $\tau_i : [1, b, n]$, $\beta_j : [a, 0, n]$, $(\tau\beta)_{ij} : [1, 0, n]$, $\varepsilon : [1, 1, 1]$, df $a-1$, $b-1$, $(a-1)(b-1)$, $ab(n-1)$.

Example 9.14: Three factors, A and C random, B fixed

$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl}$.

	R	F	R	R	
	a	b	c	n	
Term	i	j	k	l	EMS
τ_i	1	b	c	n	$\sigma^2 + bn\sigma^2_{\tau\gamma} + bcn\sigma^2_{\tau}$
β_j	a	0	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma} + acn\frac{\sum \beta_j^2}{b-1}$
γ_k	a	b	1	n	$\sigma^2 + bn\sigma^2_{\tau\gamma} + abn\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	1	0	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta}$
$(\tau\gamma)_{ik}$	1	b	1	n	$\sigma^2 + bn\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	a	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + an\sigma^2_{\beta\gamma}$
$(\tau\beta\gamma)_{ijk}$	1	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

Derivation for β_j: cover column j; the rows containing j are β_j $(a\cdot c\cdot n)$, $(\tau\beta)_{ij}$ $(1\cdot c\cdot n)$, $(\beta\gamma)_{jk}$ $(a\cdot 1\cdot n)$, $(\tau\beta\gamma)_{ijk}$ $(1\cdot 1\cdot n)$, ε (1). For τ_i: rows with i: τ_i (bcn), $(\tau\beta)_{ij}$ $(0\cdot cn = 0)$, $(\tau\gamma)_{ik}$ $(b\cdot 1\cdot n)$, $(\tau\beta\gamma)$ (0), ε.
Test for $H_0 : \beta_j = 0$. No single mean square has expectation $\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma}$. But

$$\mathrm{E}(MS_{AB}) + \mathrm{E}(MS_{BC}) - \mathrm{E}(MS_{ABC}) = \sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma},$$

so use the approximate (Satterthwaite) test

$$F_0 = \frac{MS_B}{MS_{AB} + MS_{BC} - MS_{ABC}} \quad \text{or} \quad F_0' = \frac{MS_B + MS_{ABC}}{MS_{AB} + MS_{BC}},$$

with denominator df $\nu = \dfrac{(MS_{AB} + MS_{BC} - MS_{ABC})^2}{\frac{MS_{AB}^2}{f_{AB}} + \frac{MS_{BC}^2}{f_{BC}} + \frac{MS_{ABC}^2}{f_{ABC}}}$ (and for F_0' both numerator and denominator df are of this form).

Chapter 10. Nested and Split-Plot Designs

1 Introduction

- Multifactor experiments with important industrial applications.
- They frequently involve one or more random factors, so the methodology of Ch. 9 (expected mean squares, variance components) is essential.
- Many variations exist; we consider the basic situations.

2 The Two-Stage Nested Design

2.1 Crossed vs nested

In a **nested** (hierarchical) design the levels of factor B are similar but *not identical* at different levels of factor A: batch 1 of supplier 1 has nothing to do with batch 1 of supplier 2.

Derivation: layout

	1	2	3	4	5	6	7	8	9	10	11	12
Supplier 1	○	○	○	○								
Supplier 2					○	○	○	○				
Supplier 3									○	○	○	○

Each supplier has its own 4 batches: 12 distinct batches, labeled 1–4 within each supplier.

Example 10.1: Purity of raw material

A company buys raw material from 3 suppliers; the material comes in batches. Is purity uniform? Select 4 batches at random from each supplier and make 3 purity determinations on each batch (coded data):

Supplier	1				2				3			
Batch	1	2	3	4	1	2	3	4	1	2	3	4
	1	−2	−2	1	1	0	−1	0	2	−2	1	3
	−1	−3	0	4	−2	4	0	3	4	0	−1	2
	0	−4	1	0	−3	2	−2	2	0	2	2	1
$y_{ij\cdot}$	0	−9	−1	5	−4	6	−3	5	6	0	2	6
$y_{i\cdot\cdot}$		−5				4				14		

2.2 Model and decomposition

Definition 10.2

$$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}, \qquad i = 1, \ldots, a;\ j = 1, \ldots, b;\ k = 1, \ldots, n,$$

$j(i)$: the jth level of B nested under the ith level of A; each replicate is nested within the (i, j) combination, hence $\varepsilon_{(ij)k}$. There is **no** $A \times B$ interaction (every level of B appears with only one level of A).

Theorem 10.3

$$\sum_i\sum_j\sum_k (y_{ijk}-\bar y_{\cdots})^2 = bn\sum_i(\bar y_{i\cdot\cdot}-\bar y_{\cdots})^2 + n\sum_i\sum_j(\bar y_{ij\cdot}-\bar y_{i\cdot\cdot})^2 + \sum_i\sum_j\sum_k(y_{ijk}-\bar y_{ij\cdot})^2,$$

$$SS_T = SS_A + SS_{B(A)} + SS_E, \quad abn-1 = (a-1)+a(b-1)+ab(n-1).$$

Proof [Supplement]. $y_{ijk}-\bar y_{\cdots} = (\bar y_{i\cdot\cdot}-\bar y_{\cdots}) + (\bar y_{ij\cdot}-\bar y_{i\cdot\cdot}) + (y_{ijk}-\bar y_{ij\cdot})$. Cross products:

$$\sum_i\sum_j\sum_k (\bar y_{i\cdot\cdot}-\bar y_{\cdots})(\bar y_{ij\cdot}-\bar y_{i\cdot\cdot}) = n\sum_i(\bar y_{i\cdot\cdot}-\bar y_{\cdots})\underbrace{\sum_j(\bar y_{ij\cdot}-\bar y_{i\cdot\cdot})}_{=0} = 0,$$

and any product with $(y_{ijk}-\bar y_{ij\cdot})$ vanishes after summing over k. DF of $B(A)$: within each of the a levels of A, $b-1$ df, total $a(b-1)$. Note $SS_{B(A)} = SS_B + SS_{AB}$ of a (wrongly) crossed analysis: $a(b-1) = (b-1)+(a-1)(b-1)$. ■

Source	SS	DF	MS
A	SS_A	$a-1$	MS_A
B within A	$SS_{B(A)}$	$a(b-1)$	$MS_{B(A)}$
Error	SS_E	$ab(n-1)$	MS_E
Total	SS_T	$abn-1$	

2.3 Expected mean squares: three cases

Case 1: A fixed, B fixed ($\sum_i \tau_i = 0$, $\sum_j \beta_{j(i)} = 0$ for each i).

Term	F a i	F b j	R n k	EMS
τ_i	0	b	n	$\sigma^2 + bn\sum\tau_i^2/(a-1)$
$\beta_{j(i)}$	1	0	n	$\sigma^2 + n\sum\sum\beta_{j(i)}^2/[a(b-1)]$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

$H_0:\tau_i=0$: $F_0 = MS_A/MS_E$; $H_0:\beta_{j(i)}=0$: $F_0 = MS_{B(A)}/MS_E$. Estimates $\hat\tau_i = \bar y_{i\cdot\cdot}-\bar y_{\cdots}$, $\hat\beta_{j(i)} = \bar y_{ij\cdot}-\bar y_{i\cdot\cdot}$, $\hat\sigma^2 = MS_E$.

Case 2: A random, B random ($\tau_i \sim N(0,\sigma_\tau^2)$, $\beta_{j(i)} \sim N(0,\sigma_\beta^2)$).

Term	R i	R j	R k	EMS
τ_i	1	b	n	$\sigma^2 + n\sigma_\beta^2 + bn\sigma_\tau^2$
$\beta_{j(i)}$	1	1	n	$\sigma^2 + n\sigma_\beta^2$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

$H_0:\sigma_\tau^2=0$: $F_0 = MS_A/MS_{B(A)}$; $H_0:\sigma_\beta^2=0$: $F_0 = MS_{B(A)}/MS_E$. $\hat\sigma^2 = MS_E$, $\hat\sigma_\beta^2 = \frac{MS_{B(A)}-MS_E}{n}$, $\hat\sigma_\tau^2 = \frac{MS_A-MS_{B(A)}}{bn}$.

Case 3: A fixed, B random (mixed).

Term	F i	R j	R k	EMS
τ_i	0	b	n	$\sigma^2 + n\sigma^2_\beta + bn\sum \tau_i^2/(a-1)$
$\beta_{j(i)}$	1	1	n	$\sigma^2 + n\sigma^2_\beta$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

$H_0 : \tau_i = 0$: $F_0 = MS_A/MS_{B(A)}$; $H_0 : \sigma^2_\beta = 0$: $F_0 = MS_{B(A)}/MS_E$. $\hat\tau_i = \bar y_{i\cdot\cdot} - \bar y_{\cdots}$, $\hat\sigma^2 = MS_E$, $\hat\sigma^2_\beta = \frac{MS_{B(A)} - MS_E}{n}$.

Direct derivation of $\mathrm{E}(MS_A)$ in Case 2 [Supplement]. $\bar y_{i\cdot\cdot} = \mu + \tau_i + \bar\beta_{\cdot(i)} + \bar\varepsilon_{i\cdot\cdot}$ with $\bar\beta_{\cdot(i)} = \frac{1}{b}\sum_j \beta_{j(i)}$; the $\bar\beta_{\cdot(i)}$ are iid with variance σ^2_β/b (each supplier has its own batches). So

$$\mathrm{E}(MS_A) = bn\Big[\mathrm{Var}(\tau_i) + \mathrm{Var}(\bar\beta_{\cdot(i)}) + \mathrm{Var}(\bar\varepsilon_{i\cdot\cdot})\Big] = bn\Big[\sigma^2_\tau + \frac{\sigma^2_\beta}{b} + \frac{\sigma^2}{bn}\Big] = bn\sigma^2_\tau + n\sigma^2_\beta + \sigma^2.$$

Similarly $\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} = (\beta_{j(i)} - \bar\beta_{\cdot(i)}) + (\bar\varepsilon_{ij\cdot} - \bar\varepsilon_{i\cdot\cdot})$ gives $\mathrm{E}(MS_{B(A)}) = n(\sigma^2_\beta + \sigma^2/n) = n\sigma^2_\beta + \sigma^2$. ■

Example 10.4: Purity data, line by line

$\sum y^2 = 153$, $y_{\cdots} = 13$, $y^2_{\cdots}/36 = 4.694$.

$$\begin{aligned}
SS_T &= 153 - 4.694 = 148.31,\\
SS_A &= \frac{(-5)^2 + 4^2 + 14^2}{12} - 4.694 = \frac{237}{12} - 4.694 = 19.750 - 4.694 = 15.06,\\
SS_{B(A)} &= \frac{\sum_i \sum_j y^2_{ij\cdot}}{3} - \frac{\sum_i y^2_{i\cdot\cdot}}{12}\\
&= \frac{0^2 + (-9)^2 + (-1)^2 + 5^2 + (-4)^2 + 6^2 + (-3)^2 + 5^2 + 6^2 + 0^2 + 2^2 + 6^2}{3} - \frac{237}{12}\\
&= \frac{269}{3} - 19.75 = 69.92,\\
SS_E &= 148.31 - 15.06 - 69.92 = 63.33.
\end{aligned}$$

Source	SS	DF	MS	F_0 A,B fixed	F_0 A fixed, B random	F_0 A,B random
Suppliers	15.06	2	7.53	$\frac{7.53}{2.64} = 2.85$	$\frac{7.53}{7.77} = 0.97$	0.97
Batches(Suppliers)	69.92	9	7.77	$\frac{7.77}{2.64} = 2.94$	2.94	2.94
Error	63.33	24	2.64			
Total	148.31	35				

With suppliers fixed and batches random (the natural model): $F_0 = 0.97 < F_{0.05,2,9} = 4.26$ — no supplier difference; $F_0 = 2.94 > F_{0.05,9,24} = 2.30$ — significant batch-to-batch variability within suppliers. $\hat\sigma^2_\beta = \frac{7.77 - 2.64}{3} = 1.71$, $\hat\sigma^2 = 2.64$: the batch variance is a sizeable part of total variability — the suppliers should reduce batch-to-batch variation.
Warning: using a factorial (crossed) analysis here would wrongly split $SS_{B(A)}$ into SS_B and SS_{AB}.

Example 10.5: Plants and leaves

Three plants (random), two leaves per plant (random, nested), n determinations per leaf: $y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$, $\tau_i \sim N(0, \sigma^2_\tau)$, $\beta_{j(i)} \sim N(0, \sigma^2_\beta)$, $\varepsilon \sim N(0, \sigma^2)$; leaves are *nested* in plants (leaf 1 of plant 1 $\neq$ leaf 1 of plant 2), both random.

Source	DF	MS	EMS	F_0
Plants	2	MS_A	$\sigma^2 + n\sigma^2_\beta + 2n\sigma^2_\tau$	$MS_A/MS_{B(A)}$
Leaves(Plants)	$3(2-1)=3$	$MS_{B(A)}$	$\sigma^2 + n\sigma^2_\beta$	$MS_{B(A)}/MS_E$
Error	$6(n-1)$	MS_E	σ^2	
Total	$6n-1$			

3 The Three-Stage Nested Design

Derivation

$$y_{ijkl} = \mu + \tau_i + \beta_{j(i)} + \gamma_{k(ij)} + \varepsilon_{(ijk)l},$$

$$y_{ijkl} - \bar{y}_{....} = (\bar{y}_{i...} - \bar{y}_{....}) + (\bar{y}_{ij..} - \bar{y}_{i...}) + (\bar{y}_{ijk.} - \bar{y}_{ij..}) + (y_{ijkl} - \bar{y}_{ijk.}),$$

$SS_T = SS_A + SS_{B(A)} + SS_{C(B)} + SS_E$ with df $(a-1)$, $a(b-1)$, $ab(c-1)$, $abc(n-1)$.

EMS with A, B fixed and C random:

Term	F a i	F b j	R c k	R n l	EMS
τ_i	0	b	c	n	$\sigma^2 + n\sigma^2_\gamma + bcn\sum \tau_i^2/(a-1)$
$\beta_{j(i)}$	1	0	c	n	$\sigma^2 + n\sigma^2_\gamma + cn\sum\sum \beta^2_{j(i)}/[a(b-1)]$
$\gamma_{k(ij)}$	1	1	1	n	$\sigma^2 + n\sigma^2_\gamma$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

Tests: $H_0: \tau_i = 0$: $F_0 = MS_A/MS_{C(B)}$; $H_0: \beta_{j(i)} = 0$: $F_0 = MS_{B(A)}/MS_{C(B)}$; $H_0: \sigma^2_\gamma = 0$: $F_0 = MS_{C(B)}/MS_E$.

4 Designs with Both Nested and Factorial Factors

Example 10.6: Hand insertion of electronic components

Factors: 3 assembly fixtures (A, fixed), 2 workplace layouts (B, fixed), 4 operators per layout (C, random). Layouts are in different plant areas, so **operators are nested within layouts**; fixtures are crossed with both. $n = 2$.

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_{k(j)} + (\tau\beta)_{ij} + (\tau\gamma)_{ik(j)} + \varepsilon_{(ijk)l},$$

τ_i: fixture, β_j: layout, $\gamma_{k(j)}$: operator within layout. ($\sum \tau_i = \sum \beta_j = 0$, $\sum_i (\tau\beta)_{ij} = \sum_j (\tau\beta)_{ij} = 0$, $\gamma_{k(j)} \sim N(0, \sigma^2_\gamma)$, $(\tau\gamma)_{ik(j)} \sim N(0, \frac{a-1}{a}\sigma^2_{\tau\gamma})$ with $\sum_i (\tau\gamma)_{ik(j)} = 0$, $\varepsilon \sim N(0, \sigma^2)$.)

Term	F 3 i	F 2 j	R 4 k	R 2 l	DF	EMS
τ_i	0	2	4	2	$a-1=2$	$\sigma^2+2\sigma^2_{\tau\gamma}+16\sum\tau_i^2/2$
β_j	3	0	4	2	$b-1=1$	$\sigma^2+6\sigma^2_{\gamma}+24\sum\beta_j^2$
$\gamma_{k(j)}$	3	1	1	2	$b(c-1)=6$	$\sigma^2+6\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	0	0	4	2	$(a-1)(b-1)=2$	$\sigma^2+2\sigma^2_{\tau\gamma}+8\sum\sum(\tau\beta)_{ij}^2/2$
$(\tau\gamma)_{ik(j)}$	0	1	1	2	$(a-1)b(c-1)=12$	$\sigma^2+2\sigma^2_{\tau\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	$abc(n-1)=24$	σ^2

General form: $\mathrm{E}(MS_A)=\sigma^2+n\sigma^2_{\tau\gamma}+bcn\frac{\sum\tau_i^2}{a-1}$, $\mathrm{E}(MS_B)=\sigma^2+an\sigma^2_{\gamma}+acn\frac{\sum\beta_j^2}{b-1}$, $\mathrm{E}(MS_{C(B)})=\sigma^2+an\sigma^2_{\gamma}$, $\mathrm{E}(MS_{AB})=\sigma^2+n\sigma^2_{\tau\gamma}+cn\frac{\sum\sum(\tau\beta)^2}{(a-1)(b-1)}$, $\mathrm{E}(MS_{AC(B)})=\sigma^2+n\sigma^2_{\tau\gamma}$.

Source	SS	DF	MS	F_0
Fixtures	82.80	2	41.40	$41.40/5.49=7.54$
Layouts	4.08	1	4.08	$4.08/11.99=0.34$
Operators(layout)	71.91	6	11.99	$11.99/2.33=5.14$
Fixture×Layout	19.04	2	9.52	$9.52/5.49=1.73$
Fixture×Operator(layout)	65.84	12	5.49	$5.49/2.33=2.35$
Error	56.00	24	2.33	
Total	299.67	47		

Fixtures significant ($F_{0.05,2,12}=3.89$), layouts not ($F_{0.05,1,6}=5.99$), operators within layouts significant ($F_{0.05,6,24}=2.51$), fixture×operator significant ($F_{0.05,12,24}=2.18$). Variance components: $\hat\sigma^2_\gamma=\frac{11.99-2.33}{6}=1.61$, $\hat\sigma^2_{\tau\gamma}=\frac{5.49-2.33}{2}=1.58$, $\hat\sigma^2=2.33$.
Tests summary: $H_0:\tau_i=0$: $\frac{MS_A}{MS_{AC(B)}}$, $\hat\tau_i=\bar y_{i\cdots}-\bar y$; $H_0:\beta_j=0$: $\frac{MS_B}{MS_{C(B)}}$; $H_0:\sigma^2_\gamma=0$: $\frac{MS_{C(B)}}{MS_E}$, $\hat\sigma^2_\gamma=\frac{MS_{C(B)}-MS_E}{an}$; $H_0:\sigma^2_{\tau\gamma}=0$: $\frac{MS_{AC(B)}}{MS_E}$, $\hat\sigma^2_{\tau\gamma}=\frac{MS_{AC(B)}-MS_E}{n}$; $H_0:(\tau\beta)_{ij}=0$: $\frac{MS_{AB}}{MS_{AC(B)}}$, $\widehat{(\tau\beta)}_{ij}=\bar y_{ij\cdot\cdot}-\bar y_{i\cdots}-\bar y_{\cdot j\cdot\cdot}+\bar y$.

5 The Split-Plot Design

5.1 Motivation

- *Electrical components life test*: baking time (5, 10, 15 min) × oven temperature (580, 600, 620, 640). A true CRD (random temperature, one component, random time, repeat) is impractical; instead the oven is heated once and all 9 components go in (3 per time): a **restriction on randomization**.
- The **split-plot** design is a multifactor experiment in which it is not possible to completely randomize the order of the runs.
- *Paper manufacturing*: 3 pulp preparation methods × 4 cooking temperatures, 3 replicates. A CRD needs $3\times12=36$ pulp batches; pulp preparation is **hard to change**. Split-plot: per replicate and method, one batch split into 4 samples cooked at the 4 temperatures: $3\times3=9$ batches.
- Each replicate (block) is divided into three **whole plots**; pulp method is the **whole-plot treatment**.

- Each whole plot is divided into four **subplots** (split plots); temperature is the **subplot treatment**.
- Generally the hard-to-change factor is assigned to the whole plots.

Example 10.7: Paper tensile strength data

	Rep 1			Rep 2			Rep 3		
Temp	P1	P2	P3	P1	P2	P3	P1	P2	P3
200	30	34	29	28	31	31	31	35	32
225	35	41	26	32	36	30	37	40	34
250	37	38	33	40	42	32	41	39	39
275	36	42	36	41	40	40	40	44	45

5.2 Split-plot model I (full model)

Derivation

$$y_{ijk} = \mu + \underbrace{\tau_i + \beta_j + (\tau\beta)_{ij}}_{\text{whole-plot part}} + \underbrace{\gamma_k + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk}}_{\text{subplot part}} + \varepsilon_{(ijk)1},$$

$i = 1, \ldots, r$ (blocks/replicates, random), $j = 1, \ldots, a$ (factor A, whole plot, fixed), $k = 1, \ldots, b$ (factor B, subplot, fixed), one observation per cell.

- Whole-plot part: τ_i block effect ($\sim N(0, \sigma^2_\tau)$); β_j effect of A ($\sum \beta_j = 0$); $(\tau\beta)_{ij}$ whole-plot error ($\sim N(0, \sigma^2_{\tau\beta})$).
- Subplot part: γ_k effect of B ($\sum \gamma_k = 0$); $(\tau\gamma)_{ik}$ block×B ($\sim N(0, \sigma^2_{\tau\gamma})$); $(\beta\gamma)_{jk}$ A×B ($\sum_j = \sum_k = 0$); $(\tau\beta\gamma)_{ijk}$ subplot error ($\sim N(0, \sigma^2_{\tau\beta\gamma})$).

Decomposition:

$$\begin{aligned} y_{ijk} - \bar{y}_{\cdots} = {} & (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) \\ & + (\bar{y}_{i\cdot k} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot\cdot k} + \bar{y}_{\cdots}) + (\bar{y}_{\cdot jk} - \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot k} + \bar{y}_{\cdots}) \\ & + (y_{ijk} - \bar{y}_{ij\cdot} - \bar{y}_{i\cdot k} - \bar{y}_{\cdot jk} + \bar{y}_{i\cdot\cdot} + \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots}), \end{aligned}$$

$$SS_T = SS_{\text{Block}} + SS_A + SS_B + SS_{E(w)} + SS_{\text{Block}\times B} + SS_{AB} + SS_{E(s)},$$

$$rab - 1 = (r-1) + (a-1) + (b-1) + (r-1)(a-1) + (r-1)(b-1) + (a-1)(b-1) + (r-1)(a-1)(b-1).$$

	R	F	F	R	
	r	a	b	1	
Term	i	j	k	l	EMS
τ_i	1	a	b	1	$\sigma^2 + ab\sigma^2_\tau$
β_j	r	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta} + rb\sum \beta_j^2/(a-1)$
$(\tau\beta)_{ij}$	1	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta}$
γ_k	r	a	0	1	$\sigma^2 + a\sigma^2_{\tau\gamma} + ra\sum \gamma_k^2/(b-1)$
$(\tau\gamma)_{ik}$	1	a	0	1	$\sigma^2 + a\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	r	0	0	1	$\sigma^2 + \sigma^2_{\tau\beta\gamma} + r\sum\sum(\beta\gamma)^2_{jk}/[(a-1)(b-1)]$
$(\tau\beta\gamma)_{ijk}$	1	0	0	1	$\sigma^2 + \sigma^2_{\tau\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2 (not estimable)

Since there is one observation per cell, ε has 0 df and is inseparable from $(\tau\beta\gamma)_{ijk}$ (i.e. σ^2 is confounded with $\sigma^2_{\tau\beta\gamma}$). Tests:

- A: $MS_A/MS_{(\tau\beta)}$ (whole-plot error);
- B: $MS_B/MS_{(\tau\gamma)}$;
- AB: $MS_{AB}/MS_{(\tau\beta\gamma)}$;
- blocks: no exact test.

5.3 Split-plot model II (alternate model)

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_k + (\beta\gamma)_{jk} + \varepsilon_{ijk},$$

assuming block×B and block×A×B interactions are negligible, so they are pooled into the subplot error ε_{ijk} ($\tau_i \sim N(0,\sigma^2_\tau)$, $\sum\beta_j = 0$, $(\tau\beta)_{ij} \sim N(0,\sigma^2_{\tau\beta})$, $\sum\gamma_k = 0$, $\sum_j(\beta\gamma)_{jk} = \sum_k(\beta\gamma)_{jk} = 0$, $\varepsilon_{ijk} \sim N(0,\sigma^2)$).

$$\begin{aligned} y_{ijk} - \bar y_{\cdots} &= (\bar y_{i\cdot\cdot} - \bar y_{\cdots}) + (\bar y_{\cdot j\cdot} - \bar y_{\cdots}) + (\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdots}) \\ &\quad + (\bar y_{\cdot\cdot k} - \bar y_{\cdots}) + (\bar y_{\cdot jk} - \bar y_{\cdot j\cdot} - \bar y_{\cdot\cdot k} + \bar y_{\cdots}) + (y_{ijk} - \bar y_{ij\cdot} - \bar y_{\cdot jk} + \bar y_{\cdot j\cdot}), \end{aligned}$$

$$SS_T = \underbrace{SS_{\text{Block}} + SS_A + SS_{E(w)}}_{\text{whole plot}} + \underbrace{SS_B + SS_{AB} + SS_{E(s)}}_{\text{subplot}},$$

$$rab - 1 = (r-1) + (a-1) + (r-1)(a-1) + (b-1) + (a-1)(b-1) + a(r-1)(b-1).$$

(Check: $(r-1)+(a-1)+(r-1)(a-1) = ra-1$ and $(b-1)[1+(a-1)+a(r-1)] = (b-1)ar$; sum $= rab-1$.)

Term	R i	F j	F k	R l	EMS
τ_i	1	a	b	1	$\sigma^2 + ab\sigma^2_\tau$
β_j	r	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta} + rb\sum\beta_j^2/(a-1)$
$(\tau\beta)_{ij}$	1	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta}$
γ_k	r	a	0	1	$\sigma^2 + ra\sum\gamma_k^2/(b-1)$
$(\beta\gamma)_{jk}$	r	0	0	1	$\sigma^2 + r\sum\sum(\beta\gamma)_{jk}^2/[(a-1)(b-1)]$
$\varepsilon_{(ijk)}$	1	1	1	1	σ^2

Tests: A: $MS_A/MS_{E(w)}$; B: $MS_B/MS_{E(s)}$; AB: $MS_{AB}/MS_{E(s)}$.

Remark 10.8

The whole-plot factor is tested against the (usually larger) whole-plot error with few df, so it is less precise; subplot factor and interaction are more precise. Put the factor of most interest in the subplots.

Example 10.9: Paper data, model II, line by line

$y_{\cdots} = 1297$, $\sum y^2 = 47551$, $y_{\cdots}^2/36 = 46728.03$, $SS_T = 822.97$. Block totals $417, 423, 457$; pulp totals $428, 462, 407$; temperature totals $281, 311, 341, 364$; block×pulp totals $\begin{smallmatrix}138 & 155 & 124\\ 141 & 149 & 133\\ 149 & 158 & 150\end{smallmatrix}$.

$$SS_{\text{Block}} = \frac{417^2 + 423^2 + 457^2}{12} - 46728.03 = 77.56,$$

$$SS_A = \frac{428^2 + 462^2 + 407^2}{12} - 46728.03 = 128.39,$$
$$SS_{E(w)} = \frac{\sum\sum y_{ij\cdot}^2}{4} - 46728.03 - 77.56 - 128.39 = 36.28,$$
$$SS_B = \frac{281^2 + 311^2 + 341^2 + 364^2}{9} - 46728.03 = 434.08,$$
$$SS_{AB} = \frac{\sum_j \sum_k y_{\cdot jk}^2}{3} - 46728.03 - 128.39 - 434.08 = 75.17,$$
$$SS_{E(s)} = 822.97 - 77.56 - 128.39 - 36.28 - 434.08 - 75.17 = 71.50.$$

Stratum	Source	SS	DF	MS	F_0
Whole plot	Blocks	77.56	2	38.78	
	Pulp (A)	128.39	2	64.19	64.19/9.07 = 7.08
	Whole-plot error	36.28	4	9.07	
Subplot	Temperature (B)	434.08	3	144.69	144.69/3.97 = 36.43
	A×B	75.17	6	12.53	12.53/3.97 = 3.15
	Subplot error	71.50	18	3.97	
	Total	822.97	35		

Pulp: $P = 0.049$; temperature: $P < 10^{-7}$; interaction: $P = 0.027$ (same as `aov(y~pulp*temp+Error(block/pulp))`). Under model I the 18 subplot-error df would split into block×temp ($SS = 20.67$, 6 df) and block×pulp×temp (12 df).

Example 10.10: Paint dispersion

Four pigment mixes (A, fixed), three application methods (B, fixed: brushing, spraying, rolling), three days (blocks, random). A mix is prepared once per day and applied by all three methods: mix is the whole-plot factor, method is the subplot factor. $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_k + (\beta\gamma)_{jk} + \varepsilon_{ijk}$ (i: day, j: mix, k: method).

Source	DF	F_0
Days (blocks)	2	–
Mix (A)	3	$MS_A/MS_{E(w)}$
Whole-plot error (days×mix)	6	
Method (B)	2	$MS_B/MS_{E(s)}$
Mix×Method	6	$MS_{AB}/MS_{E(s)}$
Subplot error	$a(r-1)(b-1) = 4 \cdot 2 \cdot 2 = 16$	
Total	35	

Example 10.11: R degrees of freedom

Two replicates (blocks), A with 4 levels (whole plot), B with 3 levels (subplot): `summary(aov(obs~A*B+Error(block/A)))`. Error: block — residual df $r - 1 = 1$; Error: block:A — A df 3, residual df $(r-1)(a-1) = 3$; Error: Within — B df 2, A:B df 6, residual df $a(r-1)(b-1) = 4 \cdot 1 \cdot 2 = 8$. Total $= 1 + 3 + 3 + 2 + 6 + 8 = 23 = 24 - 1$.

Other examples of split-plot structure:

(i) TV-tube screen color difference: on each of 5 days (whole plots, with 3 replicates), 4 operators measure in random order — day is the whole-plot factor, operator the subplot factor;

(ii) the paint dispersion study above.

Chapter 11. The Analysis of Covariance (ANACOVA)

1 Introduction

- ANACOVA combines the features of ANOVA and regression.
- It reduces a large error variance in the ANOVA model by adding one or more quantitative variables (covariates) linearly related to the response.
- Setting: besides the response y there is a covariate x, linearly related to y, that *cannot be controlled* by the experimenter but can be *observed* along with y (recall Ch. 4: known but uncontrollable nuisance factor).

Example 11.1: Cracker promotion

A company studies three promotion types on cracker sales. 15 stores; each store is randomly assigned to one promotion (5 stores per type). y = number of cases sold during the promotional period; x = sales in the preceding period (covariate).

	Store 1		2		3		4		5	
Trt	y	x	y	x	y	x	y	x	y	x
1	38	21	39	26	36	22	45	28	33	19
2	43	34	38	26	38	29	27	18	34	25
3	24	23	32	29	31	30	21	16	28	29

The scatter of y around the treatment averages is large, but the scatter around the *treatment regression lines* (on x) is much smaller: part of the "error" is explained by x.

2 Model

Definition 11.2

$$y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}, \qquad i = 1, \ldots, a;\ j = 1, \ldots, n,$$

x_{ij} given (fixed), $\sum_i \tau_i = 0$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, β = common regression coefficient (slope). Hence

$$\mathrm{E}(y_{ij}) = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}), \qquad \mathrm{Var}(y_{ij}) = \sigma^2.$$

Each treatment has its own regression line, all **parallel** with slope β; τ_i is the vertical shift.

2.1 Notation: corrected sums of squares and cross products

$$S_{xx} = \sum_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot})^2, \quad S_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot})(y_{ij} - \bar{y}_{\cdot\cdot}), \quad S_{yy} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2,$$

$$T_{xx} = n \sum_i (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2, \quad T_{xy} = n \sum_i (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}), \quad T_{yy} = n \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2,$$

$$E_{xx} = \sum_i \sum_j (x_{ij} - \bar{x}_{i\cdot})^2, \quad E_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{i\cdot})(y_{ij} - \bar{y}_{i\cdot}), \quad E_{yy} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2,$$

with $S = T + E$ in each case (same proof as the one-way ANOVA identity, applied to products).

3 Least Squares Estimation

Motivation. Compare treatments at a common covariate value: parallel lines, covariate variation removed from the error.

Derivation

$S = \sum_i \sum_j \left(y_{ij} - \mu - \tau_i - \beta(x_{ij} - \bar{x}_{\cdot\cdot})\right)^2$.

$$\frac{\partial S}{\partial \mu} = 0 \Rightarrow \sum_i \sum_j \left(y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}(x_{ij} - \bar{x}_{\cdot\cdot})\right) = 0 \Rightarrow \hat{\mu} = \bar{y}_{\cdot\cdot},$$

$$\frac{\partial S}{\partial \tau_i} = 0 \Rightarrow n\hat{\mu} + n\hat{\tau}_i + \hat{\beta}\sum_{j=1}^{n}(x_{ij} - \bar{x}_{\cdot\cdot}) = y_{i\cdot} \Rightarrow \hat{\tau}_i = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) - \hat{\beta}(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}),$$

$$\frac{\partial S}{\partial \beta} = 0 \Rightarrow \sum_i \sum_j \left(y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}(x_{ij} - \bar{x}_{\cdot\cdot})\right)(x_{ij} - \bar{x}_{\cdot\cdot}) = 0.$$

Theorem 11.3

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}}, \qquad \hat{\mu} = \bar{y}_{\cdot\cdot}, \qquad \hat{\tau}_i = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) - \hat{\beta}(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}).$$

Proof [Supplement]. (In $\partial S/\partial\mu$, $\sum \hat{\tau}_i = 0$ and $\sum\sum(x_{ij} - \bar{x}_{\cdot\cdot}) = 0$ give $N\hat{\mu} = y_{\cdot\cdot}$.) Expand the β-equation:

$$\sum\sum y_{ij}(x_{ij} - \bar{x}_{\cdot\cdot}) - \hat{\mu}\underbrace{\sum\sum(x_{ij} - \bar{x}_{\cdot\cdot})}_{=0} - \sum_i \hat{\tau}_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot}) - \hat{\beta}\sum\sum(x_{ij} - \bar{x}_{\cdot\cdot})^2 = 0.$$

Since $\sum\sum y_{ij}(x_{ij} - \bar{x}_{\cdot\cdot}) = S_{xy}$ and $\sum_j (x_{ij} - \bar{x}_{\cdot\cdot}) = n(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})$,

$$\sum_i \hat{\tau}_i \, n(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}) + \hat{\beta} S_{xx} = S_{xy}.$$

Substitute $\hat{\tau}_i$:

$$\underbrace{n\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})}_{=T_{xy}} - \hat{\beta}\underbrace{n\sum_i (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2}_{=T_{xx}} + \hat{\beta} S_{xx} = S_{xy}$$

$$\Rightarrow \hat{\beta}(S_{xx} - T_{xx}) = S_{xy} - T_{xy} \;\Rightarrow\; \hat{\beta} = \frac{E_{xy}}{E_{xx}}.$$

So the slope is estimated *within* treatments (pooled within-group regression). ■

Theorem 11.4: Error sum of squares

$$SS_E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}, \qquad \text{df} = a(n-1) - 1, \qquad \text{Var}(\hat{\beta}) = \frac{\sigma^2}{E_{xx}}.$$

Proof [Supplement]. Fitted value $\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}(x_{ij} - \bar{x}_{\cdot\cdot}) = \bar{y}_{i\cdot} + \hat{\beta}(x_{ij} - \bar{x}_{i\cdot})$, so $e_{ij} = (y_{ij} - \bar{y}_{i\cdot}) - \hat{\beta}(x_{ij} - \bar{x}_{i\cdot})$ and

$$SS_E = \sum\sum e_{ij}^2 = E_{yy} - 2\hat{\beta}E_{xy} + \hat{\beta}^2 E_{xx} = E_{yy} - 2\frac{E_{xy}^2}{E_{xx}} + \frac{E_{xy}^2}{E_{xx}} = E_{yy} - \frac{E_{xy}^2}{E_{xx}}.$$

Parameters: μ, $a-1$ free τ's, $\beta \Rightarrow a+1$; df $= an-a-1 = a(n-1)-1$. Variance: $\hat\beta = \sum\sum(x_{ij}-\bar x_{i\cdot})y_{ij}/E_{xx}$ is linear in independent y's, so $\mathrm{Var}(\hat\beta) = \sigma^2\sum\sum(x_{ij}-\bar x_{i\cdot})^2/E_{xx}^2 = \sigma^2/E_{xx}$. ■

Adjusted treatment means: $\hat\mu + \hat\tau_i = \bar y_{i\cdot} - \hat\beta(\bar x_{i\cdot} - \bar x_{\cdot\cdot})$ — the treatment mean adjusted to the common covariate value $\bar x_{\cdot\cdot}$.

4 Regression (Matrix) Formulation

Derivation

With $a=3$, $n=5$ and $\tau_3 = -\tau_1-\tau_2$, index the 15 observations by ℓ:

$$y_\ell = \mu + \tau_1 z_{1\ell} + \tau_2 z_{2\ell} + \beta(x_\ell - \bar x) + \varepsilon_\ell, \qquad z_1 = \begin{cases} 1 & \text{trt 1} \\ -1 & \text{trt 3} \\ 0 & \text{otherwise} \end{cases}, \; z_2 = \begin{cases} 1 & \text{trt 2} \\ -1 & \text{trt 3} \\ 0 & \text{otherwise} \end{cases},$$

$$\mathbf{X} = \left[\begin{array}{ccc|c} 1 & 1 & 0 & x_{11}-\bar x_{\cdot\cdot} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & x_{15}-\bar x_{\cdot\cdot} \\ \hline 1 & 0 & 1 & x_{21}-\bar x_{\cdot\cdot} \\ \vdots & & & \vdots \\ 1 & 0 & 1 & x_{25}-\bar x_{\cdot\cdot} \\ \hline 1 & -1 & -1 & x_{31}-\bar x_{\cdot\cdot} \\ \vdots & & & \vdots \\ 1 & -1 & -1 & x_{35}-\bar x_{\cdot\cdot} \end{array}\right], \qquad \hat{\boldsymbol\beta} = \begin{pmatrix} \hat\mu \\ \hat\tau_1 \\ \hat\tau_2 \\ \hat\beta \end{pmatrix} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y},$$

$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol\beta} = \mathbf{H}\mathbf{y}$, $\mathbf{e} = (\mathbf{I}-\mathbf{H})\mathbf{y}$, $SS_E = \mathbf{e}^T\mathbf{e} = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}$.

Source	DF	
Model $(\tau_1,\ldots,\tau_{a-1},\beta)$	a	$F_0 = MS_{\text{Model}}/MS_E$
Error	$an-a-1$	
Total	$an-1$	

Cracker example (`lm`): $\hat\mu = 33.800$, $\hat\tau_1 = 6.017$, $\hat\tau_2 = 0.942$, $\hat\tau_3 = -6.959$, $\hat\beta = 0.899$.

5 Hypothesis Testing via Extra Sums of Squares

Testing of interest:

(i) $H_0 : \tau_1 = \cdots = \tau_a = 0$;

(ii) $H_0 : \beta = 0$.

Derivation

Full: $\mathbf{y} = \mathbf{X}\boldsymbol\beta + \boldsymbol\varepsilon = \mathbf{X}_1\boldsymbol\beta_1 + \mathbf{X}_2\boldsymbol\beta_2 + \boldsymbol\varepsilon$; Reduced: $\mathbf{y} = \mathbf{X}_2\boldsymbol\beta_2 + \boldsymbol\varepsilon$; $H_0 : \boldsymbol\beta_1 = \mathbf{0}$:

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})]/[df_R - df_F]}{SS_E(\mathrm{F})/df_F} \overset{H_0}{\sim} F_{df_R-df_F,\, df_F}.$$

(i) Treatments. Full $y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$ ($df_F = an - a - 1$); reduced $y_{ij} = \mu + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$ (simple regression, $df_R = an - 2$), whose error SS is $S_{yy} - S_{xy}^2/S_{xx}$:

$$F_0 = \frac{\left[(S_{yy} - \frac{S_{xy}^2}{S_{xx}}) - (E_{yy} - \frac{E_{xy}^2}{E_{xx}})\right]/(a-1)}{SS_E(\mathrm{F})/[a(n-1)-1]} \overset{H_0}{\sim} F_{a-1,\,a(n-1)-1}.$$

(ii) Slope. Reduced $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ (one-way ANOVA, error SS E_{yy}, df $a(n-1)$):

$$F_0 = \frac{[E_{yy} - SS_E(\mathrm{F})]/1}{SS_E(\mathrm{F})/[a(n-1)-1]} = \frac{E_{xy}^2/E_{xx}}{MS_E} \overset{H_0}{\sim} F_{1,\,a(n-1)-1} \qquad \Big(= t_0^2,\ t_0 = \frac{\hat\beta}{\sqrt{MS_E/E_{xx}}}\Big).$$

Example 11.5: Cracker promotion, line by line

Totals: $x_{i\cdot} = 116, 132, 127$ ($x_{\cdot\cdot} = 375$, $\bar{x}_{\cdot\cdot} = 25$); $y_{i\cdot} = 191, 180, 136$ ($y_{\cdot\cdot} = 507$, $\bar{y}_{\cdot\cdot} = 33.8$); $\sum x^2 = 9735$, $\sum y^2 = 17783$, $\sum xy = 12937$.

$$\begin{aligned}
S_{xx} &= 9735 - \frac{375^2}{15} = 9735 - 9375 = 360.0,\\
S_{yy} &= 17783 - \frac{507^2}{15} = 17783 - 17136.6 = 646.4,\\
S_{xy} &= 12937 - \frac{375 \cdot 507}{15} = 12937 - 12675 = 262.0,\\
T_{xx} &= \frac{116^2 + 132^2 + 127^2}{5} - 9375 = 26.8,\\
T_{yy} &= \frac{191^2 + 180^2 + 136^2}{5} - 17136.6 = 338.8,\\
T_{xy} &= \frac{116(191) + 132(180) + 127(136)}{5} - 12675 = -37.4,\\
E_{xx} &= 360.0 - 26.8 = 333.2, \quad E_{yy} = 646.4 - 338.8 = 307.6, \quad E_{xy} = 262.0 + 37.4 = 299.4.
\end{aligned}$$

$$\begin{aligned}
\hat\beta &= \frac{E_{xy}}{E_{xx}} = \frac{299.4}{333.2} = 0.8986,\\
SS_E(\mathrm{F}) &= 307.6 - \frac{299.4^2}{333.2} = 307.6 - 269.03 = 38.57 \quad (11\text{ df}), \qquad MS_E = 3.506,\\
SS_E(\mathrm{R}_\tau) &= 646.4 - \frac{262^2}{360} = 646.4 - 190.68 = 455.72 \quad (13\text{ df}).
\end{aligned}$$

Treatments: $F_0 = \dfrac{(455.72 - 38.57)/2}{38.57/11} = \dfrac{208.58}{3.506} = 59.5 > F_{0.05,2,11} = 3.98$: reject; promotions differ (R: 59.42).

Slope: reduced model error SS $= E_{yy} = 307.6$ (12 df); $F_0 = \dfrac{(307.6 - 38.57)/1}{3.506} = 76.7 > F_{0.05,1,11} = 4.84$: reject $H_0 : \beta = 0$; the covariate matters. $se(\hat\beta) = \sqrt{3.506/333.2} = 0.103$.
Estimates: $\bar{x}_{i\cdot} = 23.2,\ 26.4,\ 25.4$; $\bar{y}_{i\cdot} = 38.2,\ 36.0,\ 27.2$:

$$\begin{aligned}
\hat\tau_1 &= (38.2 - 33.8) - 0.8986(23.2 - 25) = 4.4 + 1.617 = 6.017,\\
\hat\tau_2 &= (36.0 - 33.8) - 0.8986(26.4 - 25) = 2.2 - 1.258 = 0.942,\\
\hat\tau_3 &= (27.2 - 33.8) - 0.8986(25.4 - 25) = -6.6 - 0.359 = -6.959.
\end{aligned}$$

Adjusted means: 39.82, 34.74, 26.84. Compare with the one-way ANOVA ignoring x: $MS_E = 307.6/12 = 25.63$ vs 3.51 with the covariate — the error variance is reduced by a factor of 7.

6 Testing for Parallel Slopes

Derivation

Model with treatment-specific slopes:

$$y_{ij} = \mu + \tau_i + \beta_i(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$$

$$\Rightarrow\ y_\ell = \mu + \tau_1 z_{1\ell} + \tau_2 z_{2\ell} + \beta_1(x_{1\ell} - \bar{x}) + \beta_2(x_{2\ell} - \bar{x}) + \beta_3(x_{3\ell} - \bar{x}) + \varepsilon_\ell,$$

where $x_{i\ell} - \bar{x}$ is the centered covariate if observation ℓ is in treatment i and 0 otherwise:

$$\mathbf{X} = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & x_{1j}-\bar{x} & 0 & 0 \\ \hline 1 & 0 & 1 & 0 & x_{2j}-\bar{x} & 0 \\ \hline 1 & -1 & -1 & 0 & 0 & x_{3j}-\bar{x} \end{array}\right],$$

each row block having n rows, and

$$\boldsymbol{\beta} = (\mu, \tau_1, \tau_2, \beta_1, \beta_2, \beta_3)^T.$$

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_a$ (parallel lines). Full model (separate slopes): each treatment is an ordinary simple regression, so

$$SS_E(\mathrm{F}) = \sum_{i=1}^{a} \Big[E_{yy}^{(i)} - \frac{(E_{xy}^{(i)})^2}{E_{xx}^{(i)}}\Big], \quad df_F = a(n-2),$$

with $E^{(i)}$ the within-treatment-i sums. Reduced model = common slope ANACOVA: $SS_E(\mathrm{R}) = E_{yy} - E_{xy}^2/E_{xx}$, $df_R = a(n-1) - 1$. Then

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})]/(a-1)}{SS_E(\mathrm{F})/[a(n-2)]} \overset{H_0}{\sim} F_{a-1,\,a(n-2)}.$$

Example 11.6: Parallelism for the cracker data

Within-treatment sums: trt 1: $E_{xx}^{(1)} = 54.8$, $E_{yy}^{(1)} = 78.8$, $E_{xy}^{(1)} = 59.8$; trt 2: 137.2, 142.0, 136.0; trt 3: 141.2, 86.8, 103.6. Separate slopes 1.091, 0.991, 0.734.

$$\begin{aligned} SS_E(\mathrm{F}) &= \left(78.8 - \tfrac{59.8^2}{54.8}\right) + \left(142.0 - \tfrac{136.0^2}{137.2}\right) + \left(86.8 - \tfrac{103.6^2}{141.2}\right) \\ &= 13.54 + 7.19 + 10.79 = 31.52 \ (9 \text{ df}), \\ F_0 &= \frac{(38.57 - 31.52)/2}{31.52/9} = \frac{3.53}{3.50} = 1.01 < F_{0.05,2,9} = 4.26. \end{aligned}$$

No evidence against parallel slopes: the common-slope ANACOVA is appropriate.

Example 11.7: Fiber strength by machine

Three machines, $n = 5$, y = strength, x = thickness:

	Machine 1		Machine 2		Machine 3	
	y	x	y	x	y	x
	36	20	40	22	35	21
	41	25	48	28	37	23
	39	24	39	22	42	26
	42	25	45	30	34	21
	49	32	44	28	32	15
Total	207	126	216	130	180	106

$S_{xx} = 261.73$, $S_{xy} = 282.60$, $S_{yy} = 346.40$; $T_{xx} = 66.13$, $T_{xy} = 96.00$, $T_{yy} = 140.40$; $E_{xx} = 195.60$, $E_{xy} = 186.60$, $E_{yy} = 206.00$.

$$\hat\beta = 186.6/195.6 = 0.954, \quad SS_E(\mathrm{F}) = 206.0 - \tfrac{186.6^2}{195.6} = 27.99 \text{ (11 df)}, \quad MS_E = 2.544,$$

$$SS_E(\mathrm{R}_\tau) = 346.4 - \tfrac{282.6^2}{261.73} = 41.27, \quad F_0 = \frac{(41.27 - 27.99)/2}{2.544} = 2.61 < 3.98:$$

no machine difference after adjusting for thickness. Slope: $F_0 = \frac{186.6^2/195.6}{2.544} = 69.97 > 4.84$: thickness is significant. $\hat\mu = 40.2$, $\hat\tau = (0.182,\ 1.219,\ -1.402)$. Parallelism: $SS_E(\mathrm{F}) = 25.25$ (9 df), $F_0 = \frac{(27.99-25.25)/2}{25.25/9} = 0.49$: parallel slopes accepted.

7 Assumption Checking

Residuals $e_{ij} = y_{ij} - \hat y_{ij}$: normal Q–Q plot; residuals vs fitted values; residuals vs treatment; residuals vs x (checks linearity in the covariate). For the cracker data all plots are satisfactory.

8 Selected exercises

Motivation. Precision of adjusted means and the gain over one-way ANOVA.

Exercise 11.1 (Adjusted treatment means; Montgomery 15.18)

In the model of this chapter, let $\hat\mu_i^{\mathrm{adj}} = \bar y_{i\cdot} - \hat\beta(\bar x_{i\cdot} - \bar x_{\cdot\cdot})$.
(a) Show $\mathrm{E}\hat\beta = \beta$ and $\mathrm{E}\hat\mu_i^{\mathrm{adj}} = \mu + \tau_i$.
(b) Show $\mathrm{Cov}(\bar y_{i\cdot}, \hat\beta) = 0$ and $\mathrm{Var}(\hat\mu_i^{\mathrm{adj}}) = \sigma^2\Big[\frac{1}{n} + \frac{(\bar x_{i\cdot} - \bar x_{\cdot\cdot})^2}{E_{xx}}\Big]$.
(c) Deduce the $100(1-\alpha)\%$ interval

$$\hat\mu_i^{\mathrm{adj}} \pm t_{\alpha/2,\,a(n-1)-1}\sqrt{MS_E\Big[\frac{1}{n} + \frac{(\bar x_{i\cdot} - \bar x_{\cdot\cdot})^2}{E_{xx}}\Big]}$$

and evaluate it for promotion 1 of the cracker data ($t_{0.025,11} = 2.201$).

Solution.

(a)

Step 1. $E_{xy} = \sum_i \sum_j (x_{ij} - \bar x_{i\cdot}) y_{ij}$ because $\sum_j (x_{ij} - \bar x_{i\cdot})\bar y_{i\cdot} = 0$. Taking expectations, the terms $\mu + \tau_i$ and $-\beta\bar x_{\cdot\cdot}$ are constant within treatment i and vanish against $\sum_j (x_{ij} - \bar x_{i\cdot}) = 0$, leaving

$$\mathrm{E}E_{xy} = \beta \sum_i \sum_j (x_{ij} - \bar x_{i\cdot}) x_{ij} = \beta E_{xx},$$

so $\mathrm{E}\hat\beta = \beta$.

Step 2. $\mathrm{E}\bar y_{i\cdot} = \mu + \tau_i + \beta(\bar x_{i\cdot} - \bar x_{\cdot\cdot})$, hence $\mathrm{E}\hat\mu_i^{\mathrm{adj}} = \mu + \tau_i$.

(b) By independence, $\mathrm{Cov}(\bar y_{i\cdot}, \hat\beta) = \frac{\sigma^2}{nE_{xx}} \sum_j (x_{ij} - \bar x_{i\cdot}) = 0$. Therefore

$$\mathrm{Var}(\hat\mu_i^{\mathrm{adj}}) = \mathrm{Var}(\bar y_{i\cdot}) + (\bar x_{i\cdot} - \bar x_{\cdot\cdot})^2 \mathrm{Var}(\hat\beta) = \frac{\sigma^2}{n} + \frac{\sigma^2 (\bar x_{i\cdot} - \bar x_{\cdot\cdot})^2}{E_{xx}}.$$

(c)

Step 1. $\hat\mu_i^{\mathrm{adj}}$ is a linear function of the least squares estimate, so it is normal with the mean and variance above, and it is independent of SS_E (as in Ch. 8, $\mathrm{Cov}(\hat{\boldsymbol\beta}, \mathbf{e}) = \mathbf{0}$), where $SS_E/\sigma^2 \sim \chi^2_{a(n-1)-1}$.

Step 2. Hence $\{\hat\mu_i^{\mathrm{adj}} - (\mu + \tau_i)\} / \sqrt{MS_E[\cdots]} \sim t_{a(n-1)-1}$; inverting gives the interval.

Step 3 (cracker, promotion 1). $\hat\mu_1^{\mathrm{adj}} = 39.82$, $\bar x_{1\cdot} - \bar x_{\cdot\cdot} = -1.8$, $MS_E = 3.506$, $E_{xx} = 333.2$:

$$\widehat{\mathrm{Var}} = 3.506\Big(\frac{1}{5} + \frac{3.24}{333.2}\Big) = 0.735, \qquad 39.82 \pm 2.201\sqrt{0.735} = 39.82 \pm 1.89 = [37.93,\ 41.71].$$

The farther a treatment's covariate mean lies from $\bar x_{\cdot\cdot}$, the less precise its adjusted mean. ■

Exercise 11.2 (What the covariate buys)

Let $r_w = E_{xy}/\sqrt{E_{xx}E_{yy}}$ be the pooled within-treatment correlation of x and y.
(a) Show $SS_E = (1 - r_w^2)E_{yy}$, so the ratio of the ANCOVA error mean square to the one-way ANOVA error mean square is $(1 - r_w^2)\, a(n-1)/\{a(n-1)-1\}$. Check this against the cracker data.
(b) If the covariate means are equal in all treatments ($T_{xx} = 0$), show that $\hat\tau_i = \bar y_{i\cdot} - \bar y_{\cdot\cdot}$ and that the adjusted treatment sum of squares equals T_{yy}.

Solution [Supplement].

(a) $SS_E = E_{yy} - E_{xy}^2/E_{xx} = E_{yy}\{1 - E_{xy}^2/(E_{xx}E_{yy})\} = (1 - r_w^2)E_{yy}$. The one-way ANOVA has error SS E_{yy} on $a(n-1)$ df and ANCOVA has SS_E on $a(n-1)-1$ df, which gives the ratio. Cracker data: $r_w^2 = 299.4^2/(333.2 \times 307.6) = 0.875$, and $(1 - 0.875) \times 12/11 = 0.137 = 3.506/25.63$: one degree of freedom spent on β removes about 87% of the error.

(b)

Step 1. $T_{xx} = n\sum_i (\bar x_{i\cdot} - \bar x_{\cdot\cdot})^2 = 0$ forces $\bar x_{i\cdot} = \bar x_{\cdot\cdot}$ for all i, so $\hat\tau_i = \bar y_{i\cdot} - \bar y_{\cdot\cdot}$ and also $T_{xy} = 0$.

Step 2. Then $S_{xx} = E_{xx}$ and $S_{xy} = E_{xy}$, so the adjusted treatment SS is

$$\Big(S_{yy} - \frac{S_{xy}^2}{S_{xx}}\Big) - \Big(E_{yy} - \frac{E_{xy}^2}{E_{xx}}\Big) = S_{yy} - E_{yy} = T_{yy}.$$

With balanced covariate means the covariate acts only on the error, like blocking; when the $\bar x_{i\cdot}$ differ it also corrects the means, at the extra variance of Exercise 11.1(b). ■